ALCANCE AVIATION

# CLEARED FOR THE INTERVIEW

The Complete Pilot Interview Preparation Guide

*Airline · Cargo · Fractional · Corporate*

**SECOND EDITION**

Expanded with 16 chapters · FAA chart practice · Cold temperature corrections



BY

**ALCANCE AVIATION**

# CLEARED FOR THE INTERVIEW

## The Complete Pilot Interview Preparation Guide

A comprehensive resource for pilots preparing for airline, cargo, fractional, and corporate aviation interviews worldwide.

Alcance Aviation




## How to Use This Book

This guide is organized into 16 chapters covering every aspect of a professional pilot interview — from personal background questions to advanced technical knowledge.

Start with Chapter 1 and work through each section. Do not skip ahead to the technical chapters without reviewing the HR and behavioral chapters first. Interviewers eliminate most candidates on the HR portion, not the technical.

Use the TMAAT chapter (Chapter 6) to prepare your stories BEFORE the interview. Write them down using the STAR method and practice them out loud.

The Technical Study Guide (Chapter 7) covers concepts asked at airlines, cargo operators, fractional companies, and corporate flight departments worldwide.

All questions in this guide are drawn from real interviews conducted at aviation companies around the world. No company names are used to protect the integrity of the material.

# ABOUT ALCANCE AVIATION

*Founded and led by Alessandro Ruppel Rolim*

Alcance Aviation was built on a simple idea: the pilots most qualified to prepare you for an interview are the ones who have lived them — on both sides of the table.

Alessandro Ruppel Rolim, the founder of Alcance Aviation, started flying at sixteen. By nineteen, he was in the right seat of a Cessna Citation. At twenty-four, he landed his first airline job on the Boeing 737-300 and -400. From there, he helped launch a low-fare carrier operating the 737-700 and -800, before expatriating to China, where he flew the Airbus A320 and A330 for seventeen years.

When COVID-19 brought global aviation to a halt, Ruppel relocated to the United States, where he currently flies for a U.S. carrier and serves as a simulator instructor on the Boeing 737-300, -400, and -700.

That path — from charter to airline, single-aisle to widebody, across three continents — is the foundation of the coaching methodology behind this book.

## Why Alcance exists

Alcance Aviation started the way most meaningful things do: out of frustration. After paying a premium for interview coaching that fell short, Ruppel set out to build something better — a preparation approach rooted in real cockpit experience, current hiring standards, and the specific patterns interviewers actually use.

Today, Alcance Aviation coaches pilots preparing for airline, cargo, fractional, and corporate interviews around the world. We have never run an advertisement. Our students come through referrals from other pilots who have sat in our sessions and, more importantly, landed the job. That pipeline has kept our online classes continuously booked.

## Why this book

Thousands of coaching hours made one thing clear: the questions repeat, the patterns are predictable, and preparation — not luck — is what separates the pilots who get hired from the ones who do not. This book distills those patterns into a single, focused resource, so that any pilot preparing for an interview has access to the same structured preparation our students do.

**Your uniform is earned the day you are hired. This book is here to help you earn it.**

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# INTRODUCTION

There is a myth in aviation that some pilots are simply “good at interviews.” That the charismatic ones walk in, say the right things, and walk out with an offer — while the rest grind through sim evaluations and technical questions, hoping their logbook speaks for itself.

*It is a comforting story. It is also wrong*.

After thousands of coaching hours across airline, cargo, fractional, and corporate interviews, one thing is clear: the pilots who get hired are not the ones with the best stories or the smoothest delivery. They are the ones who prepared for the pattern. Interviews repeat. The questions repeat. The way interviewers score answers repeats. Preparation — not personality — is what closes the gap between a qualified pilot and a hired pilot.

This book is built on that pattern. Sixteen chapters cover what every serious interview will test you on: personal background, company fit, background checks, CRM and safety, leadership, TMAAT behavioral questions, technical knowledge, weather, regulations and airspace, scenario-based problem solving, key words, chart reading practice, cold temperature altitude correction, and question banks. The questions are based on what pilots report being asked in real airline interviews. Study them out loud. Answer them until your answers feel natural, not rehearsed. Use the STAR method where it fits. And never lie — interviewers check.

**Your uniform is earned the day you are hired. This book is here to help you earn it.**

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# Chapter 1

## Personal & Background

**These questions open almost every interview. Know your story cold. Walk through your resume chronologically, keep it aviation-focused, and end with why you are here today.**

1. **Tell me about yourself / Who are you?**
2. **How did you get into aviation / flying?**
3. **Walk me through your resume / aviation career.**
4. **What are your strengths?**
5. **What is your biggest weakness?**
6. **What are your best and worst characteristics?**
7. **What are your strengths and weaknesses?**
8. **What do you do in your free time / outside of work?**
9. **Where do you see yourself in 5 years?**
10. **Tell me about your most memorable flight.**
11. **Tell me about a time you had an emergency or bad experience while flying.**
12. **Tell me about a flight you wish you hadn't gone on.**
13. **Tell me about a time you had to think quickly / make an instant decision.**
14. **Tell me about a time you had to take a leadership position.**
15. **Tell me about a time you went above and beyond.**
16. **Tell me about a time you received constructive criticism.**
17. **Tell me about a time you learned from a mistake or failure.**
18. **Tell me about a time you disagreed with a superior — what did you do?**
19. **Tell me about a time you had a conflict with a colleague or supervisor.**
20. **Tell me about a time you had to make a critical decision under pressure.**




21. Tell me about a time you dealt with a coworker who was not following SOP.

22. Tell me about a time you helped a customer directly.

23. Tell me about a time you helped a co-worker outside of flying duties.

24. Tell me about a challenging flight.

25. Tell me about a time you flew with a difficult Captain — how did you handle it?

26. Tell me about a time you flew with someone you didn't enjoy — how did you handle it?

27. Tell me about a time when you had a conflict with someone in the cockpit.

28. Tell me about a time when poor CRM led to a mistake or near miss. What did you learn?

29. Tell me about a time a breakdown in SOP occurred.

30. Tell me something others might have a misconception about you.

31. What was your worst day in aviation?

32. What was your best day in aviation?

33. What was the hardest decision you had to make in the cockpit?

34. What is your most valuable trait as a pilot / instructor?

35. What are your pet peeves in the cockpit?

36. What do you consider your most memorable experience in an airplane?

37. Tell me about the pilot you admire the most.

38. Have you had any mentors throughout your career?

39. Have you passed a student that you regretted?

40. What is your favorite topic to instruct/teach?

41. Tell me about your most influential instructor.

42. What song would best describe you?

43. What is the last book you read?

44. What are your hobbies?




45. **What is your favorite funny / comedy movie?**

46. **What makes you happy?**

47. **What makes you upset or angry?**

48. **5 adjectives that describe you.**

49. **Do you think of work as a performance?**

50. **What was your favorite job you've had so far?**

# Chapter 2

## Company Fit & Motivation

**Research the company thoroughly before your interview. Know their fleet, culture, schedule, domiciles, and what makes them unique. “Why our company?” is the most important question you will be asked.**

1. **Why do you want to fly for our company?**
2. **Why should we hire you?**
3. **Why should we hire you over other qualified applicants?**
4. **What will you bring to this company?**
5. **What do you know about us? Have you listened to our podcasts or talked with our pilots?**
6. **What do you like about your current job?**
7. **What frustrates you about your current company?**
8. **Why did you leave your previous company?**
9. **Have you applied elsewhere? Who have you applied to?**
10. **What would you do if another company called and offered you a job at the same time?**
11. **If they call you in 8 months, are you going to go with one of them?**
12. **Are you familiar with our schedules?**
13. **How will your family deal with our schedules?**
14. **How do you feel about being on the road for extended periods?**
15. **How will you prepare for training if you get hired?**
16. **Do you think you will have any issues with training on a new aircraft?**
17. **What would be the toughest / hardest part about transitioning here?**
18. **What is your biggest challenge coming here?**
19. **What will be challenging for you if you’re hired?**




20. Are you ok with sitting in the right seat for a while / extended time to upgrade?
21. Would you have a problem being a First Officer after being a Captain for years?
22. How would you feel about flying with a Captain younger than you?
23. Where do you see yourself in 5 years with this company?
24. Do you have any questions for us?
25. How do you think you did on the interview?
26. How do we know you're not going to leave after training?
27. What would you do if you didn't get the job?
28. Would the long trips we do be a problem for you?
29. Can you accept an immediate training date?
30. Would you need to give your current employer 2 weeks notice?
31. Did you leave your previous employers on good terms?
32. Have you thought about the financial challenges of the first year?
33. What equipment / aircraft would you like and why?
34. What are your long-term goals?
35. What is the option if not this company?
36. What is one thing about coming here that you have bad feelings about?
37. Do you like working for your current company and why are you leaving?
38. Tell me why you left a previous job.
39. Why don't you want to work for an airline under Part 121?
40. Why are you moving from Part 121 to Part 135?
41. Based on your resume it looks like you could go to a major — so why here?
42. What is the difference between good service and elite service?
43. How do you feel about going back to being a First Officer?
44. How do you feel about the negative aspects of this job?




**45.** **Would you be okay driving a rental car for several hours to get to or from a trip?**

**46.** **We try to stay in nice hotels, but occasionally only a smoking/handicap room may be available. Are you okay resting in that environment?**

**47.** **Are you okay with — lifting bags, driving rental cars 5-7 hours, cleaning the cabin, flying smokers/pets, fueling/servicing the jet, staying in a less-than-ideal hotel?**

**48.** **Do you have any problem with passengers smoking on the aircraft or exotic pets on board?**

**49.** **Do you realize that you'll have to clean a lav, load bags, hold the dog, vacuum the airplane, wipe fingerprints, etc.?**

**50.** **Do you think what you wear to work is a uniform or costume?**

**51.** **What would be any obstacles that could keep you from moving to your base?**

**52.** **Will you move to your base?**

**53.** **Are you okay with 8-day trips?**

**54.** **What crew base would you prefer?**

**55.** **How did you prepare for this interview?**

# Chapter 3

## Background Check & Records

**Always be honest. Interviewers have seen everything. A disclosed issue handled professionally is far better than one that is discovered later. Never lie on your application or in the interview.**

1. **Have you had any training failures or required extra sim lessons?**
2. **Have you failed any check rides?**
3. **Have you had any accidents, incidents, or violations?**
4. **Have you had any FAA / regulatory violations?**
5. **They asked about criminal history.**
6. **Have you ever been arrested?**
7. **They asked about driving violations.**
8. **Can you pass a drug test?**
9. **Can you pass a security / background check?**
10. **Do you have anything in your past that we might find in the background check?**
11. **Were you ever fired or forced to resign?**
12. **Were you ever taken off flying duties?**
13. **Did the tower ever ask you to call a number after you land?**
14. **Have you ever had to call in fatigue? What was the situation?**
15. **Are you currently employed with a flying job?**
16. **What are the gaps in your employment?**
17. **When was your last line check?**
18. **How did you accumulate all of your flight hours?**
19. **Where did you receive your flight training?**
20. **What are you currently flying?**




**21.** **How much flight time do you have in a crewed aircraft?**

**22.** **How soon are you available for training?**

**23.** **Will you resign from your current employer?**

**24.** **Have you ever been late to a flight, hotel van, or crew bus?**

**25.** **Tell me about a time you broke a FAR or company SOP.**

**26.** **We do an extensive background check — is there anything you want to tell us about?**

**27.** **Is there anything that would embarrass yourself or the company that we would find on Google?**

**28.** **Have you ever been grounded from flying?**

**29.** **Have you ever deviated from SOPs?**

**30.** **Have you had any moving violations in an automobile?**

# Chapter 4

## CRM, Safety & Airmanship

**Safety is non-negotiable. Never compromise it for schedule, passenger pressure, or a captain's request. Show interviewers that you know when and how to speak up.**

### Crew Resource Management (CRM)

**1. Can you define Crew Resource Management and explain its importance in aviation?**

**2. What does CRM mean to you?**

**3. What is the difference between CRM and TEM?**

**4. Describe a situation where effective communication helped prevent a potential issue during a flight.**

**5. How do you handle disagreements or conflicts in a multi-crew environment?**

**6. How do you ensure that all crew members feel comfortable speaking up, especially junior ones?**

**7. What role does assertiveness play in CRM, and how do you balance it with respect for authority?**

**8. How do you apply CRM principles during abnormal or emergency situations?**

**9. What strategies do you use to maintain situational awareness as a crew?**

**10. How do you debrief after a flight to improve CRM and overall performance?**

**11. Have you ever seen CRM break down in the cockpit?**

**12. Tell me about a time you used CRM to resolve a crew member conflict.**

**13. Tell me about a time you overcame a difficult situation using CRM.**

**14. How do you manage an emergency as it relates to CRM?**

**15. Who is responsible for safety of flight of the aircraft?**

### Threat and Error Management (TEM)

**1. Can you explain Threat and Error Management and how it applies to daily flight operations?**

2. **Describe a time when you identified a threat before it became a problem.**

3. **Have you ever made an error in flight? How did you recognize it and mitigate it?**

4. **What types of threats do you consider during pre-flight planning?**

5. **How do you manage errors made by other crew members without compromising safety or teamwork?**

6. **Can you give an example of an undesired aircraft state and how you corrected it?**

7. **How do you ensure minor errors don't escalate into major incidents?**

8. **What role does communication play in managing threats and errors?**

## Safety & Airmanship Scenarios

1. **What would you do if your Captain was violating a safety protocol?**

2. **What would you do if the Captain shows up drunk / intoxicated?**

3. **What would you do if the Captain wanted to take off below minimums without a takeoff alternate?**

4. **What would you do if the Captain continues below DA without required visual reference and refuses to go missed?**

5. **What would you do if the Captain wanted to delay writing up a maintenance item?**

6. **What would you do if the Captain wanted to fly with a broken armrest?**

7. **What would you do if the Captain wanted to pretend the NAV light was fine until reaching home base?**

8. **What would you do if you and your Captain notice the only lavatory is on MEL for a 4.5-hour flight?**

9. **What would you do if the Captain is flying an ILS and is approaching full-scale deflection low on the GS and does not respond?**

10. **What would you do if your FO falls asleep on an early morning departure?**

11. **What would you do if you're fatigued to the point of being unsafe — would you still fly?**

12. **Your captain tries to convince you to fly even though you called in fatigued. What do you say?**




**13.** **Captain throws the checklist in the back and says “To hell with the SOP” — what do you do?**

**14.** **Tell about a situation where you put safety first.**

**15.** **How do you ensure your decisions are aligned with safety protocols and operational guidelines?**

**16.** **How do you maintain the highest standards of safety and professionalism while on duty?**

**17.** **How do you handle stress and fatigue during long or demanding flights?**

**18.** **How do you manage stress and fatigue?**

**19.** **How do you deal with complacency?**

**20.** **Have you ever declared an emergency?**

**21.** **Have you ever taken the controls from another pilot? If so, explain the situation.**

# Chapter 5

## Leadership & Communication

**Leadership in the cockpit is not about rank — it's about creating an environment where the best decision always wins. Show that you lead by example.**

1. **What do you think makes a good First Officer?**
2. **What are 3 qualities of a good First Officer?**
3. **What qualities make a good Captain?**
4. **What attributes make a good Captain?**
5. **What makes a good leader in the cockpit?**
6. **Of those attributes, which is your strongest and weakest?**
7. **Tell me about a time when you had to take a leadership role unexpectedly.**
8. **What qualities make you a good leader in the cockpit?**
9. **Can you share an example of how you effectively collaborated with a diverse crew?**
10. **How do you adapt your communication style for different audiences?**
11. **How do you ensure clear and effective communication with ATC, cabin crew, and passengers?**
12. **How would you handle a passenger exhibiting disruptive behavior during a flight?**
13. **What would those you've flown with say about you?**
14. **If I call your references, what 3 words might they use to describe you?**
15. **Tell us three words that people who have flown with you would use to describe you.**
16. **If we brought your squadron mates / crewmates in and asked them about you, what would they say?**
17. **What would your Captains say about you?**
18. **What would your Captains say you needed to improve on?**




19. What would a previous captain have to say that is negative about you?

20. How would you handle a young, arrogant captain?

21. Have you ever had a captain you just didn't get along with? If so, how did you handle that?

22. What would you do if you had a really awful client/customer on your airplane?

23. How would you deal with a delay? Passengers are upset, already 30 min late, weather delay going longer.

24. What are you going to tell the customer who has a multi-million dollar deal and you have to cancel the flight because the aircraft is broken?

25. Customer has been delayed by the company for 4 hours and you show up just in time to get chewed out — how do you handle it?

26. WWYD if you were at a remote airport and corporate says "Tell the owner the catering company dropped the ball, not us." The owners are driving up right now.

27. How would you deal with an intoxicated passenger?

28. You have 8 passengers that show up with bags that clearly won't fit — walk us through how you would handle this situation.

29. How do you build rapport with people in the company?

30. What steps do you take to foster a positive and respectful working environment?

## Simulator Instructor Questions

1. Can you explain how you troubleshoot a simulator when it's not functioning correctly?

2. How do you ensure that the simulator software and hardware stay updated?

3. What protocols do you follow to maintain simulator accuracy and reliability?

4. How would you adapt your teaching style to match the needs of different learners?

5. Can you describe how you assess a trainee's performance during simulations?

6. How do you keep trainees engaged during lengthy simulation sessions?

7. Imagine a trainee is consistently struggling to grasp certain concepts. How would you handle this?

8. How would you respond if technical issues disrupted a training session with tight deadlines?

9. Have you ever encountered resistance from trainees in accepting feedback? How did you overcome it?

10. What do you consider your most valuable trait as a simulator instructor?

11. Can you share a time when you went above and beyond to ensure the success of a trainee?

12. How do you foster a positive and productive training environment?

13. Can you share a time when you had to manage conflicts between trainees?

14. How do you communicate complex technical concepts to trainees with limited technical backgrounds?

15. Have you ever faced an unexpected challenge during a simulation session? How did you resolve it?

16. How do you prioritize tasks when balancing simulator maintenance and conducting training sessions?

17. Can you give an example of how you modified or improved a simulation scenario for better training?

18. How do you stay updated on new simulation technologies and techniques?

19. Have you ever suggested or implemented improvements to the simulator systems? What was the outcome?

20. How do you incorporate emerging trends in simulation training into your sessions?

21. How do you ensure trainees understand the objectives of a simulation exercise?

22. How do you measure the success of your training sessions?

23. What is your approach to providing constructive feedback to trainees?

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# Chapter 5B

## The STAR Method — How to Answer Behavioral Questions

One of the most common reasons pilots fail airline interviews is not a lack of experience — it is a lack of storytelling. They have done incredible things in the cockpit but cannot communicate them clearly under pressure.

The STAR Method solves this problem. It is a simple four-step framework that transforms a rambling answer into a powerful, structured story that interviewers remember.

Every TMAAT question in Chapter 6 should be answered using this method. Learn it, practice it, and own it before you walk into any interview room.

### What Does STAR Stand For?

**S — Situation:** Set the scene. Where were you? What was happening? Give enough context so the interviewer understands the environment.

**T — Task:** What was your specific role or responsibility in that situation? What were you expected to do?

**A — Action:** What did YOU specifically do? This is the most important part. Do not say "we" — say "I." Be specific about your actions, your words, your decisions.

**R — Result:** What happened as a result of your actions? What did you learn? How did it affect the crew, the flight, or the company?

**Keep each STAR answer between 90 seconds and 2 minutes when spoken aloud. Practice out loud — not just in your head.**

### Why Interviewers Love the STAR Method

Airline and fractional interviewers are trained to evaluate behavioral competencies. They are not just listening to your answer — they are listening for evidence of specific traits: leadership, communication, safety awareness, customer service, and resilience.

A STAR answer gives them exactly what they need. It is organized, credible, and easy to evaluate. A rambling answer without structure leaves the interviewer guessing and creates doubt.

Think of it this way: the interviewer has already talked to 10 pilots that day. Most gave vague answers like "I handled it professionally" or "we worked it out as a team." Your STAR answer will stand out because it is specific, real, and memorable.

### The 5 Most Common STAR Mistakes

**Mistake 1 — Using "We" Instead of "I"**

Interviewers want to know what YOU did. Saying “we handled the emergency” tells them nothing. Say “I called out the malfunction, I took the controls, I declared the emergency.”

> **Every action in your story must be something YOU did. Save “we” only for the outcome.**

### Mistake 2 — Being Too Vague

> I had a conflict with a captain once and we resolved it professionally.

This answer is useless to an interviewer. It has no situation, no action, no result. It tells them nothing about how you actually behave.

> We were on final into KJFK, the captain was 200 ft below our briefed glidepath and not responding to my callouts. I said clearly: ‘Captain, we are low, I recommend go-around.’ He initiated the go-around immediately. After landing we debriefed and he thanked me for speaking up.

### Mistake 3 — Talking Too Long

Some pilots get nervous and over-explain. A STAR answer should be tight — 90 seconds to 2 minutes. If you are still talking after 3 minutes, you have lost the interviewer. Practice cutting your stories down to the essential details only.

> **If your story takes more than 2 minutes to tell out loud, cut the Situation shorter. The Action and Result are what matter most.**

### Mistake 4 — Choosing a Weak Story

Not all stories are equal. A story about resolving a minor scheduling conflict is far weaker than a story about a genuine in-flight emergency or a safety confrontation with a captain. Choose stories that demonstrate real competency under real pressure.

Before your interview, prepare at least one strong STAR story for each of these categories: conflict with a superior, safety decision, customer service, leadership, mistake you learned from, and going above and beyond.

### Mistake 5 — No Result

Many pilots give a great Situation and Action but forget the Result. The Result is what closes the loop. It shows the interviewer that your actions actually worked — or more importantly, that you learned something when they did not.

> **Even if the outcome was not perfect, a result that includes a lesson learned is still powerful. Interviewers respect self-awareness.**

## Full STAR Answer Examples

### Example 1 — TMAAT You Had a Conflict With a Captain

Question: “Tell me about a time you had a conflict with a captain and how you resolved it.”

**S — Situation:** We were operating a charter flight from Miami to Nassau. Weather at destination was deteriorating faster than forecast, and our filed alternate was already below minimums.

**T — Task:** I was the First Officer and responsible for monitoring fuel and weather. I noticed we were approaching a point where our options would become very limited.

**A — Action:** I brought my concern to the captain calmly and clearly: ‘Captain, Nassau is now reporting 400 overcast and our alternate is below minimums. I recommend we discuss our fuel options before we pass our last divert point.’ He initially wanted to continue, confident the weather would improve. I acknowledged his experience, but respectfully asked that we review the METAR one more time together. When he saw the trend, he agreed to divert.

**R — Result:** We diverted safely with comfortable fuel reserves. The weather at Nassau never improved. After the flight, the captain told me I had done exactly the right thing by speaking up when I did. It reinforced for me that respectful assertiveness is not a challenge to authority — it is part of good CRM.

**Notice how this answer uses “I” throughout the Action, includes the specific weather details, and ends with a lesson learned. This is what a strong STAR answer looks like.**

### Example 2 — TMAAT You Provided Exceptional Customer Service

Question: “Tell me about a time you went above and beyond for a customer.”

**S — Situation:** We had just landed at a small regional airport after a fractional owner’s trip. As we were completing our shutdown checklist, my FO noticed a travel bag left in the aft cabin. The owner had already departed in his car and was heading to a lake house several hours away.

**T — Task:** As Captain, I had to decide how to handle the situation. We had no scheduled flights for the rest of the day.

**A — Action:** I called the owner directly, informed him we had found his bag, and told him I would personally drive it to him. I got his address and made the two-hour drive to his lake house that evening.

**R — Result:** The owner was genuinely surprised and grateful. He later sent a letter to the company praising the crew. That interaction led to him referring two of his business partners to the program. For me, it confirmed that in private aviation, the experience we create off the aircraft is just as important as the flight itself.

### Example 3 — TMAAT You Made a Mistake and What You Learned

Question: “Tell me about a time you made a mistake. What did you learn?”




**S — Situation:** Early in my career, I was flying a Part 135 cargo run on a tight schedule. We were departing from a small airport and I was under self-induced pressure to make our slot time.

**T — Task:** I was the Captain and Pilot Flying. My responsibility was to complete all pre-departure checks before taxi.

**A — Action:** In my rush, I abbreviated my walkaround. I missed that the fuel cap on the right wing was not fully secured. As I was taxiing out, a line technician ran toward the aircraft waving his arms. I stopped immediately and shut down. The fuel cap was loose and fuel had been venting.

**R — Result:** We missed our slot time, but we did not have a serious incident. I wrote a safety report, briefed my chief pilot, and personally apologized to the customer. More importantly, I changed my behavior permanently. Since that day, I have never rushed a walkaround regardless of schedule pressure. The schedule exists to serve safety — not the other way around.

**When answering mistake questions, never blame others, never minimize what happened, and always include what you changed as a result. Interviewers are not looking for perfection — they are looking for self-awareness and growth.**

### Example 4 — TMAAT You Had to Make a Decision Under Pressure

Question: “Tell me about a time you had to make a critical decision under pressure.”

**S — Situation:** We were at FL350 on a transatlantic route when we received an ECAM alert indicating a hydraulic system pressure loss. The timing was difficult — we were two hours from our destination and three hours from our departure airport.

**T — Task:** As First Officer, my role was to support the Captain, run the appropriate checklists, and provide accurate information so we could make the best decision together.

**A — Action:** I called out the ECAM, identified the affected system, and ran the QRH procedure while the Captain maintained aircraft control. I assessed our nearest suitable airports, calculated fuel burn for each option, checked weather at alternates, and presented the Captain with a clear summary: three options, the tradeoffs of each, and my recommendation. I also contacted dispatch and maintenance control proactively.

**R — Result:** We diverted to our closest suitable airport, landed uneventfully, and the aircraft was inspected by maintenance. The hydraulic issue was a sensor fault — no actual system damage. But we made the decision based on the information we had, which was the right approach. The Captain debriefed me afterward and said the quality of the information I provided made the decision straightforward. That experience showed me how critical clear, organized communication is in high-pressure moments.

## How to Prepare Your Own STAR Stories

Before your interview, write down at least one strong STAR story for each of the following categories. Keep them in a notebook and review them the night before.

1. Conflict with a captain or superior
2. A time you made a mistake and what you learned
3. A time you went above and beyond for a customer / passenger
4. A time you had to make a critical decision under pressure
5. A time you showed leadership unexpectedly
6. A time you identified a safety threat before it became a problem
7. A time you had to work with someone you found difficult
8. A time you received constructive criticism and how you responded
9. A time you had an emergency in flight
10. A time you went above and beyond for a co-worker or trainee

**Write each story on a single index card or note. Include the key facts: aircraft type, airport, your role, the specific actions you took, and the result. Review them the day before your interview.**

## ⚡ Quick Reference — The STAR Checklist

Before you finish each STAR answer, ask yourself these four questions:

- ✓ Did I clearly set the scene? (Situation)
- ✓ Did I explain what MY role and responsibility was? (Task)
- ✓ Did I use “I” throughout my actions — not “we”? (Action)
- ✓ Did I close the loop with a concrete result or lesson learned? (Result)
- ✓ Did I keep the answer under 2 minutes?
- ✓ Was I specific? (Aircraft type, airport, weather, exact words I used?)
- ✓ Did I avoid blaming others?
- ✓ Did I show self-awareness if the story involved a mistake?

If you can answer yes to all eight questions, your STAR answer is ready.

**The single best way to prepare is to practice out loud with another person. Record yourself on your phone. Listen back. You will immediately hear where your answers are weak.**

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# Chapter 5C

## The 15 Most Common Interview Mistakes

Most pilots who fail aviation interviews are qualified. They have the hours, the ratings, and the experience. They fail because of avoidable mistakes that have nothing to do with flying ability.

This chapter covers the 15 most common mistakes seen in airline, fractional, cargo, and corporate aviation interviews. Read each one carefully. Many of them will feel obvious when you read them — but under the pressure of a real interview, they are surprisingly easy to make.

Awareness is the first step. Practice is the cure.

### Mistake #1 — Not Researching the Company

Walking into an interview without knowing the company is the fastest way to signal that you are not serious. Interviewers always ask “What do you know about us?” or “Why do you want to fly for our company?” If your answer is generic, you will be eliminated.

Before any interview you must know: the company’s history and founding year, the fleet they operate, their routes or service areas, their culture and core values, what makes them different from competitors, and any recent news or expansions.

> **Weak answer:** I know you are a great company with a good reputation and a modern fleet.
>
> **Strong answer:** You pioneered the fractional ownership model and your dedicated crew program assigns the same crew to specific owners, which creates a level of consistency and personalization that sets you apart from every other fractional operator. I have spoken with several of your pilots and done extensive research on your operation — everything I have seen confirms this is the culture I want to be part of.
>
> **The Fix:** Spend at least 2-3 hours researching the company before your interview. Visit their website, listen to their podcast if they have one, and talk to pilots who work there.

### Mistake #2 — Giving Vague Answers Without Specifics

Vague answers kill interviews. “I handled it professionally” and “we worked together as a team” are phrases that mean nothing to an interviewer. They have heard them a hundred times. What they want are specific details — aircraft type, airport, exact words you used, what the outcome was.

> **Vague:** I once had a difficult situation with a captain and we resolved it through communication.
>
> **Specific:** We were on a positioning flight into KORD in IMC. The captain wanted to continue the approach with RVR below our company minimums. I called it out clearly:




'Captain, RVR is 1400, our minimums are 1800. I recommend we go missed.' He agreed and we executed the missed approach.

**The Fix:** Every TMAAT answer needs at least: where it happened, what aircraft, what your specific role was, and what exact actions you took. Use the STAR method from Chapter 5B.

### Mistake #3 — Badmouthing a Previous Employer

This is one of the most damaging mistakes a candidate can make. No matter how bad your previous company or captain was, speaking negatively about them in an interview makes YOU look unprofessional, disloyal, and difficult to work with.

Interviewers think: "If he talks about his last company this way, he will talk about us the same way in a year."

**Never say this:** My last company was poorly managed, the chief pilot had no idea what he was doing, and the pay was terrible. That's why I left.

**Say this instead:** I learned a tremendous amount in my previous role and I'm grateful for the experience. I'm at a point in my career where I'm looking for a company that values long-term relationships with pilots, and everything I have seen about your culture tells me this is that place.

**The Fix:** Always frame your reason for leaving as moving TOWARD something better — never running away from something bad. Even if the previous situation was genuinely terrible, find a professional way to describe it.

### Mistake #4 — Lying or Exaggerating on the Application or in the Interview

Aviation companies do thorough background checks. They verify flight hours, check ride history, employment records, driving records, and criminal history. If you lie — even by omission — and they find out, you will be permanently disqualified. Not just from this company, but from others who share hiring data.

A disclosed issue handled with professionalism is almost always recoverable. A lie that is discovered is never recoverable.

**If you have a check ride failure, an incident, a violation, or any mark on your record — disclose it proactively. Prepare a brief, honest explanation and focus on what you learned. Interviewers respect honesty far more than a perfect record.**

**The Fix:** Before your interview, review your entire record. If anything could appear in a background check, prepare a concise, honest, professional explanation for it. Practice saying it out loud until it comes naturally.

### Mistake #5 — Not Preparing for the Simulator

Many candidates prepare extensively for the HR portion of the interview and neglect the simulator. This is a serious mistake. The simulator evaluation is a significant part of your score and can eliminate you even if your HR interview was excellent.

You do not need to be perfect in the sim. Interviewers know you have never flown that specific aircraft. What they are evaluating is: your instrument scan, your ability to brief and execute a procedure, how you handle the V1 cut, and your CRM and verbal callouts.

> **Common mistake:** Showing up to the sim with no preparation, unfamiliar with the cockpit layout, and no plan for the V1 cut procedure.
>
> **Right approach:** Watch YouTube videos of the specific aircraft sim (Challenger 601, Learjet 45, etc.) to familiarize yourself with the cockpit layout. Know the V-speeds they will give you. Practice verbalizing every deviation and correction. Brief the ILS approach out loud before you fly it.
>
> **The Fix:** The sim profile is almost always sent in advance. Study it. Know the profile cold. Practice your ILS brief and your V1 cut callouts out loud before the day of the interview.

### Mistake #6 — Poor Body Language and Lack of Eye Contact

Interviewers are evaluating you from the moment you walk in the door. Slouching, avoiding eye contact, crossing your arms, mumbling, or looking at the floor all send negative signals — even if your words are perfect.

In private and fractional aviation especially, you will be interacting with high-net-worth owners and executives on a daily basis. Companies want to see that you carry yourself with confidence and professionalism.

> **Sit up straight, lean slightly forward to show engagement, make steady eye contact with both interviewers, speak clearly and at a measured pace, and smile naturally. These are simple things that make a big difference.**
>
> **The Fix:** Practice your answers in front of a mirror or record yourself on video. Watch your posture, your eye contact, and your facial expressions. Most people are surprised by what they see.

### Mistake #7 — Talking Too Much — or Too Little

Both extremes are problems. Candidates who ramble for five minutes on every answer lose the interviewer's attention. Candidates who give one-sentence answers seem unprepared and disengaged.

The ideal answer to most HR questions is 60 to 90 seconds when spoken. TMAAT answers can be up to 2 minutes. Technical answers should be as concise as possible while being complete.

> **Too long:** Going on for 4-5 minutes with side stories, tangents, and unnecessary background details.

**Too short:** Answering “Why do you want to fly for us?” with “I like the company and the schedule.”

**Right length:** A clear, structured answer that covers the key points and stops when the point has been made.

**The Fix:** Time yourself when you practice. Record yourself and play it back. If you go over 2 minutes on any answer, cut it. If you are under 30 seconds, add more specific detail.

## Mistake #8 — Not Having Questions Ready for the Interviewers

At the end of almost every interview, you will be asked: “Do you have any questions for us?” Saying “No, I think you covered everything” is a missed opportunity. It signals low interest and low preparation.

Having thoughtful questions ready shows that you did your research, you are genuinely interested, and you are evaluating them as much as they are evaluating you. This changes the dynamic of the conversation in your favor.

**Strong questions to ask:** “What does the training pipeline look like for new hires and what can I do to set myself up for success?” / “What do the pilots who have been here the longest say they love most about the company?” / “How would you describe the culture between crews on long trips?”

**Never ask about salary, days off, or vacation in a first interview. Those conversations come after you have the offer.**

**The Fix:** Prepare 3-4 genuine questions before every interview. Write them down and bring them with you. It is perfectly acceptable to glance at your notes.

## Mistake #9 — Showing Arrogance or Overconfidence

There is a fine line between confidence and arrogance. Confidence is attractive in a pilot. Arrogance is a red flag. Companies want pilots who are self-assured but also coachable, humble, and aware of their limitations.

Arrogance shows up in subtle ways: dismissing the importance of the interview, claiming to already know everything, speaking negatively about less experienced pilots, or making it clear that you are doing the company a favor by applying.

**Arrogant:** I have 8,000 hours and I’ve pretty much seen everything. Training should be straightforward for me.

**Confident and humble:** I bring a lot of experience to this role and I’m excited to apply it here. At the same time, I know every operation has its own culture and procedures, and I’m genuinely looking forward to learning the right way to do things in your environment.

**The Fix:** Always pair your strengths with openness to learning. Show that you are confident in your abilities AND aware that there is always more to learn.

## Mistake #10 — Freezing on Technical Questions

Technical questions can come at any point in the interview — not just in a dedicated technical section. Some interviewers weave them naturally into the conversation. “You mentioned flying into JFK — what are the holding speed restrictions at 14,000 feet?”

The worst response is silence, followed by a panicked guess. A much better response is to show your thought process, even if you do not have the exact answer immediately.

> **Wrong approach:** Freezing, saying “I’m not sure” and moving on without attempting an answer.
>
> **Right approach:** “At 14,000 feet under ICAO rules, the max holding speed is 230 knots. Above 14,000 it increases to 265 knots. I believe under FAA rules it is similar but let me think through that —” and then reason through it.

**Interviewers are not always looking for the perfect answer. They are looking for how you handle uncertainty. Thinking out loud and reasoning through a problem is far better than silence or a random guess.**

> **The Fix:** Review Chapter 7 (Technical Study Guide), Chapter 8 (Weather), and Chapter 9 (Regulations) in the weeks before your interview. Do not try to memorize everything — understand the concepts so you can reason through them.

### Mistake #11 — Dressing Inappropriately

Aviation interviews are professional environments. You are applying to be the face of the company to high-value clients and passengers. How you dress signals how seriously you take the opportunity.

The standard for pilot interviews is business professional. For men: a dark suit, white or light blue dress shirt, conservative tie, polished dress shoes. For women: a dark business suit or professional dress, conservative and polished.

**Do not show up in business casual, khakis, or an open collar shirt. Even if the company culture is relaxed, the interview is formal. You can always dress down after you get hired.**

**Lay out your interview clothes the night before. Make sure everything is pressed, clean, and fits well. Shine your shoes. Get a haircut the week before — not the day of. These small details matter more than most people think.**

> **The Fix:** When in doubt, overdress. No interviewer has ever eliminated a candidate for being too well dressed. Many have eliminated candidates for appearing sloppy or unprepared.

### Mistake #12 — Not Being Ready to Discuss Weaknesses Honestly

“What are your weaknesses?” is one of the most feared questions in any interview. Most candidates try to dodge it with fake weaknesses: “I work too hard” or “I’m a perfectionist.” Interviewers have heard these a thousand times. They are not impressed.

A genuine, self-aware answer about a real weakness — paired with what you are doing to address it — is far more impressive than a clichéd non-answer.

> **Cliché answer:** My biggest weakness is that I care too much about my work and sometimes put in too many hours.
>
> **Honest and self-aware:** Early in my career I struggled with delegating. I wanted to control every aspect of the flight and sometimes that created unnecessary workload for me and reduced my FO's engagement. I recognized this pattern, started actively involving my FOs in decision-making, and I've seen a significant improvement in crew performance and my own situational awareness as a result.
>
> **The Fix:** Choose a real weakness that is not a core safety competency. Pair it with specific steps you have taken or are taking to improve. This shows self-awareness and a growth mindset — both of which interviewers value highly.

### Mistake #13 — Arriving Late or Unprepared Logistically

This one should be obvious — but it happens. Traffic, wrong building, parking problems, forgetting a document — all of these create stress before the interview even starts. A stressed candidate performs worse.

Most companies provide detailed instructions in their pre-arrival email. Read it thoroughly. Know exactly where you are going, how long it takes to get there, where to park, and what documents you need to bring.

> **Plan to arrive 20-30 minutes early. If you arrive too early, wait in your car or a nearby coffee shop. Walking in 30+ minutes early can also create awkwardness. Aim for 10-15 minutes early at the reception desk.**
>
> **The Fix:** The night before: confirm the address, plan your route, lay out all your documents (logbooks, certificates, medical, passport, licenses), and set two alarms. Treat the interview morning like an early departure — no rushing, no surprises.

### Mistake #14 — Failing to Follow Up After the Interview

Most candidates walk out of an interview and wait. The best candidates send a brief, professional thank-you email within 24 hours. This is a small gesture that reinforces your interest and professionalism, and it keeps your name in the interviewer's mind.

The email does not need to be long. Two or three sentences is enough. Thank them for their time, mention one specific thing from the conversation that excited you, and reaffirm your interest in the position.

> **Example thank-you email:** Subject: Thank You — [Your Name] — Pilot Interview [Date]. Dear [Interviewer Name], thank you for the time you invested in meeting with me today. The conversation about your dedicated crew program and the culture you have built confirmed everything that drew me to this opportunity. I am genuinely excited about the possibility of joining the team and I look forward to hearing from you. Best regards, [Your Name].

**The Fix:** Get the name and email address of your interviewers before you leave. If you cannot get their email, send the thank-you through the recruiter. Send it within 24 hours — not a week later.

### Mistake #15 — Not Practicing Out Loud

This is the most common and most costly mistake of all. Most candidates prepare by reading their answers in their head. This feels like preparation — but it is not. The interview is a spoken performance. Reading and speaking are completely different skills.

When you practice only in your head, your answers sound perfect. The moment you say them out loud for the first time — under pressure, in front of two strangers — you stumble, you lose your train of thought, you say "um" twenty times, and you realize your answer makes no sense.

The pilots who perform best in interviews are the ones who have said every answer out loud at least ten times before walking in the door.

**Thinking through an answer in your head does NOT count as practice. You must say it out loud, out of your chair, in a professional tone, as if you are actually in the interview.**

**Find a partner — a fellow pilot, a friend, your spouse — and do a full mock interview. If no one is available, record yourself on your phone and watch it back. You will immediately see where you need to improve.**

**The Fix:** In the two weeks before your interview, practice at least 5 STAR answers and your top 10 HR answers out loud every day. On the night before, do one final run-through of your most important answers. Go to bed early. Show up rested and ready.

## The 15 Mistakes — Quick Reference

Here is a quick summary of every mistake covered in this chapter. Review this list the night before your interview.

1. Not researching the company
2. Giving vague answers without specific details
3. Badmouthing a previous employer
4. Lying or exaggerating on your application or in the interview
5. Not preparing for the simulator
6. Poor body language and lack of eye contact
7. Talking too much — or too little
8. Not having questions ready for the interviewers
9. Showing arrogance or overconfidence




10. Freezing on technical questions
11. Dressing inappropriately
12. Not being ready to discuss weaknesses honestly
13. Arriving late or unprepared logistically
14. Failing to follow up after the interview
15. Not practicing out loud

**If you have read this chapter carefully and addressed each of these mistakes in your preparation, you are already better prepared than the majority of candidates who will walk into that interview room.**

# Chapter 6

## TMAAT — Behavioral Questions

**TMAAT = Tell Me About A Time. Use the STAR method: Situation, Task, Action, Result. Be specific: aircraft type, airport IATA code, your role (PF/PM), weather, and YOUR specific actions. Practice your stories out loud before the interview.**

### CRM, Crew & Conflict

1. **TMAAT you flew with a difficult crewmember.**

2. **TMAAT you had to resolve a conflict between crewmembers.**

3. **TMAAT you worked as a team to accomplish a goal with someone you did not get along with.**

4. **TMAAT you had a conflict with a captain and how did you resolve it.**

5. **TMAAT you had friction with another pilot in flight and how did you handle it.**

6. **TMAAT you had to change your approach to dealing with a coworker's behavior.**

7. **TMAAT you had a stressful flight and you helped the Captain make better decisions.**

8. **TMAAT you corrected a coworker who was not following standards/rules.**

9. **TMAAT someone broke a rule or SOP in front of you.**

10. **TMAAT you intentionally broke a rule or company SOP.**

11. **TMAAT you were asked to do something you didn't agree with.**

12. **TMAAT someone made a decision/policy you didn't agree with and how you dealt with it.**

13. **TMAAT you corrected either a leader or instructor.**

14. **TMAAT you led a team with diversified perspectives to reach a goal or accomplish a task.**

15. **TMAAT you had a disagreement with your captain and had to challenge him/her.**

16. **TMAAT you had to say no to a captain.**

17. **TMAAT you had to convince the other pilot.**




**18.** **TMAAT you had to pass important information to someone or a group of people.**

**19.** **TMAAT you had a communication problem / misunderstanding — how did you deal with it?**

## Safety, Emergencies & Decision-Making

**1.** **TMAAT you had an emergency.**

**2.** **TMAAT you handled a significant emergency in flight or helped back up the Captain during an emergency.**

**3.** **TMAAT you had a particularly stressful event and how you dealt with it.**

**4.** **TMAAT you made a decision too quickly.**

**5.** **TMAAT you made a mistake and what did you learn?**

**6.** **TMAAT you had to make a critical decision under pressure.**

**7.** **TMAAT you had to change or modify your plans.**

**8.** **TMAAT you had incomplete information but still had to make a decision.**

**9.** **TMAAT multiple threats occurred simultaneously — how did you prioritize and respond?**

**10.** **TMAAT you had to deal with factors outside of SOP.**

**11.** **TMAAT you had to think outside the box / deal with complicated information.**

**12.** **TMAAT you identified a threat before it became a problem.**

**13.** **TMAAT you had to deviate from SOPs or any normal procedure to solve a problem.**

## Leadership, Growth & Above & Beyond

**1.** **TMAAT you went above and beyond.**

**2.** **TMAAT you did the right thing because it was the right thing to do.**

**3.** **TMAAT you worked hard to achieve a professional goal.**

**4.** **TMAAT you had to deal with or create change.**

**5.** **TMAAT you made a positive change to company culture.**




6. TMAAT you were proactive in getting a professional job to which you aspired.

7. TMAAT you did something that gave you a sense of accomplishment.

8. TMAAT you experienced adversity and how you overcame it.

9. TMAAT you lost credibility and what did you do to restore it.

10. TMAAT you showed leadership.

11. TMAAT you needed to change something in your company.

12. TMAAT you demonstrated commitment to your work — above and beyond.

13. TMAAT you made a decision and stuck by it.

14. TMAAT you stayed back to make a decision.

15. TMAAT you got other team members involved to make a decision.

## Mentoring, Teaching & Encouraging Others

1. TMAAT you helped/encouraged someone to achieve or do more.

2. TMAAT you helped someone as an instructor.

3. TMAAT you offered encouragement either to a crew member or a passenger.

4. TMAAT you mentored a peer.

5. TMAAT you gave someone guidance and how it helped them.

6. TMAAT you gave praise to a coworker.

7. TMAAT you received praise from a peer or co-worker.

8. TMAAT a CA/leader that you admire and how have you incorporated their traits into your leadership skills.

9. TMAAT you requested feedback from others.

10. TMAAT you took inputs / feedback from other pilots or team members.

## Customer Service

1. TMAAT you provided excellent / exceptional customer service.




2. TMAAT you went above and beyond for a customer / went out of your way for a customer.

3. TMAAT you had to deal with an upset or difficult passenger.

4. TMAAT you had a really awful client/customer on your airplane — how did you resolve it?

5. TMAAT you helped a customer directly.

6. TMAAT you provided a positive experience for your passengers.

## Adaptability & Extra Duties

1. TMAAT you had to accomplish a task that was outside your duties or purview.

2. TMAAT you were asked to do something outside of your core responsibilities — what was the outcome?

3. TMAAT you had to take a leadership position unexpectedly.

4. TMAAT you had to quickly build a rapport with somebody.

5. TMAAT you had to quickly learn a new skill — how did you handle that?

6. TMAAT you had to deal with change in the industry.

# Chapter 6 — Part 2

## More TMAAT Questions with Model Answers

This is the second part of Chapter 6. TMAAT means Tell Me About A Time. These are the behavioral questions pilots report most often, grouped by type. They continue the list in Part 1, and similar questions were joined into one.

For each group you will find the questions and one model answer you can use for the whole type. Prepare at least one real story for each group, and use the STAR method: Situation, Task, Action, Result.

### Conflict & Interpersonal Friction

**Use for any question about a conflict, a disagreement between people, or a difficult colleague.**

**Q1. TMAAT you witnessed a conflict between other people and what you did about it.**
**Q2. TMAAT you were involved in a conflict that you could not fully resolve.**
**Q3. TMAAT you had to set aside your ego to work with someone.**
**Q4. TMAAT you calmed tension in the flight deck.**
**Q5. TMAAT you changed your attitude or behavior to work better with someone.**

**Model answer (STAR). Replace the story with your own.**

**Situation:** On a long trip, the First Officer and I had different ideas about how to split the radio and checklist work. The tension started to affect the quality of our briefings.

**Task:** My task was to fix the working relationship without letting it affect the safety of the flight.

**Action:** I asked for a private moment on the ground, described what I was seeing in neutral words, and asked for his view first. We agreed on a simple split of duties and on a rule that either of us could call a time-out. I stayed calm and did not take it personally.

**Result:** The rest of the trip was smooth, and we both said so in the debrief. The lesson: address friction early, talk about behaviors and not about people, and agree on a clear working method.

**Key points:** Early action, calm tone, listening first, focus on safety and the outcome. Never blame the other person.

### Disagreement, Advocating & Defending a Decision

**Use when someone disagreed with you, questioned your decision, or you had to push for a different course of action.**

**Q6. TMAAT someone disagreed with or challenged a decision you made (in the airplane or at work).**
**Q7. TMAAT you had to justify or explain a decision to a crew that did not agree.**
**Q8. TMAAT you had to defend a decision later, after the fact.**
**Q9. TMAAT you had to persuade or influence a crew to change their course of action.**

**Q10. TMAAT you had to advocate strongly for an action.**
**Q11. TMAAT you had to advocate for a different decision from your crew.**
**Q12. TMAAT someone questioned or doubted your judgment.**
**Q13. TMAAT you had to stand your ground when others disagreed.**
**Q14. TMAAT you had to change someone's mind by explaining your reasoning.**
**Q15. TMAAT you made a decision that others did not agree with, and how you reacted.**
**Q16. TMAAT you made an unpopular decision that was necessary.**
**Q17. TMAAT your authority as Pilot in Command was questioned.**
**Q18. TMAAT you stood up for a decision under pressure.**
**Q19. TMAAT you had to justify an operational decision.**
**Q20. TMAAT you had to advocate for a change within a team.**
**Q21. TMAAT you had to make a decision and justify it with limited data.**
**Q22. TMAAT you and another pilot had opposing views.**
**Q23. TMAAT you advocated for a crew member.**

**Model answer (STAR). Replace the story with your own.**

**Situation:** During a trip with a mixed crew, the First Officer disagreed with my plan to divert early because of a developing line of weather at the destination.

**Task:** I needed to explain my reasoning, hear his concerns, and still make a timely and safe decision as Captain.

**Action:** I asked for his concerns first and he raised a valid point about fuel. I showed him the weather trend and the fuel numbers for the alternate. I explained that I was choosing the margin that kept every option open, and I thanked him for speaking up.

**Result:** We diverted with comfortable fuel, landed uneventfully, and the destination closed shortly after. In the debrief he said he understood the decision. I learned that a decision is easier to defend when you show the data and invite the challenge.

**Key points:** Listen, show data, explain the 'why', stay open to being wrong, keep the final decision safe. Show you can advocate and also accept advocacy.

## Stress, Crisis & Pressure

**Use for 'most stressful flight', 'felt rushed', 'felt pressured' and 'solved a crisis'.**

**Q24. TMAAT you had your most challenging flight.**
**Q25. TMAAT you felt pressured or rushed in the flight deck.**
**Q26. TMAAT you felt overwhelmed or were negatively affected at work.**
**Q27. TMAAT you faced a challenging situation and how you overcame it.**
**Q28. TMAAT you had a high-pressure problem that you had not planned for.**
**Q29. TMAAT you had an unsafe situation in flight and how you recovered.**
**Q30. TMAAT you had to adapt under pressure.**
**Q31. TMAAT you had to deal with a crisis using CRM.**
**Q32. TMAAT you felt pressured to fly (and what you did).**
**Q33. TMAAT you were pressured to make a decision you were not comfortable with.**
**Q34. TMAAT you felt pressured to do something unsafe or questionable.**
**Q35. TMAAT you received pressure from someone senior to you.**
**Q36. TMAAT you were caught between competing priorities under pressure.**

**Q37. TMAAT you were forced to do something outside your comfort zone.**
**Q38. TMAAT you made the right decision despite pressure to do something else.**
**Q39. TMAAT you completed several tasks successfully under pressure.**
**Q40. TMAAT you were stressed and what you did about it.**

**Model answer (STAR). Replace the story with your own.**

**Situation:** After a late departure and a long duty day, we were rushed to make a tight turn when a maintenance item appeared before departure.

**Task:** I had to stay safe and correct while there was pressure to leave on time.

**Action:** I slowed the pace on purpose, said so to the crew, and called maintenance and dispatch to define the options. I wrote the plan down: what we know, what we need, and what time we would decide. We did not leave until the item was properly resolved.

**Result:** We departed later, safely, and the company supported the decision. I learned that the first tool against pressure is to slow down, name it, and use the checklists and resources.

**Key points:** Name the pressure, slow down, use resources (crew, dispatch, checklists), safety above schedule. For a real emergency use Aviate – Navigate – Communicate.

## SOP & Integrity

**Use when you saw someone break a rule, or when integrity was in question.**

**Q41. TMAAT a crew member did not want to follow instructions.**
**Q42. TMAAT you counseled someone for violating an SOP.**
**Q43. TMAAT you saw a captain break an SOP.**
**Q44. TMAAT someone you worked with had their integrity questioned and how you handled it.**
**Q45. TMAAT you witnessed a breach of integrity and what you did about it.**
**Q46. TMAAT you questioned another crew member's integrity.**

**Model answer (STAR). Replace the story with your own.**

**Situation:** On a trip, a crew member skipped a required check to save time and said it was 'not needed today'.

**Task:** I needed to bring the crew back to the SOP without damaging the relationship.

**Action:** I stopped the flow calmly, asked for the check, and explained that I wanted to follow the procedure every time. I described the reason in one sentence and did the check together. After the flight I talked privately and said I would report a repeated pattern to the proper channel.

**Result:** The crew member agreed and we kept standard procedures for the rest of the trip. I learned that you can protect the standard and the person at the same time: be direct, be respectful, and escalate if it continues.

**Key points:** Safety first, speak up in the moment, respect the person, escalate through the proper channel if needed. Show integrity, not judgment.

## Change & Improvement

**Use when you improved a process, advocated for a change or influenced your organization.**

**Q47. TMAAT you improved a process or SOP and how you implemented it.**
**Q48. TMAAT you successfully changed a policy in your organization.**
**Q49. TMAAT you led an initiative that resulted in a procedural change.**
**Q50. TMAAT you led a team to make a process better.**
**Q51. TMAAT you influenced change or helped to improve a culture.**
**Q52. TMAAT you implemented a policy that others resisted.**
**Q53. TMAAT you challenged the status quo.**
**Q54. TMAAT you led a change initiative.**
**Q55. TMAAT you had to adapt to change.**

**Model answer (STAR). Replace the story with your own.**

**Situation:** In my previous operation, a recurring delay happened because a briefing sheet was incomplete before departure.

**Task:** I wanted to improve the process without being asked and get the others to accept it.

**Action:** I collected examples of the delays, drew up a simpler one-page checklist, and tested it on my own flights for a month. I then presented the results to the chief pilot and offered to train the others.

**Result:** The new checklist was adopted, and the delays caused by missing information dropped. I learned that a change is accepted more easily when you bring data, a tested solution and an offer to help.

**Key points:** Problem, data, solution, buy-in, result. Show initiative and humility, and mention what you learned.

## Leadership

**Use for any leadership question: success, failure, challenge, or influence.**

**Q56. TMAAT you displayed your best or most impactful leadership.**
**Q57. TMAAT you faced your most difficult leadership situation.**
**Q58. TMAAT you failed as a leader and what you learned.**
**Q59. TMAAT your leadership was challenged.**
**Q60. TMAAT your leadership positively influenced someone.**
**Q61. TMAAT you led a team that failed to meet a deadline.**
**Q62. TMAAT you encountered obstacles as a leader.**
**Q63. TMAAT you had to lead a group of people to achieve a desired outcome.**
**Q64. TMAAT you did not have all the answers and still had to lead.**
**Q65. TMAAT you were given responsibility without authority.**
**Q66. TMAAT you had to take over in a leadership vacuum.**
**Q67. TMAAT you had to earn back trust as a leader.**
**Q68. TMAAT you took ownership of a team failure.**
**Q69. TMAAT you led under difficult circumstances.**
**Q70. TMAAT you gave harsh or negative feedback.**
**Q71. TMAAT your leadership style did not work and you changed it.**

**Model answer (STAR). Replace the story with your own.**

**Situation:** As a senior pilot in a small operation, I was asked to lead a group of pilots through a transition to a new procedure with a tight deadline.

**Task:** I had to align the group, keep the standard and deliver on time.

**Action:** I explained why the change mattered, divided the work according to each person's strengths, set short checkpoints, and made myself available for questions. When a pilot struggled, I paired him with a more experienced colleague.

**Result:** We finished on time, and the new procedure worked without incident. What I learned: a leader makes the goal clear, trusts the team, and checks progress early.

**Key points:** Style: clear goal, trust, support. Mention the team, not just you. For a failure story, own it and show the lesson.

## Mentorship & Influence

**Use when you coached, trained, guided or inspired someone.**

**Q72. TMAAT you mentored or coached someone and made a difference.**
**Q73. TMAAT someone asked you to be a mentor, and why.**
**Q74. TMAAT someone looked to you for guidance.**
**Q75. TMAAT you gave advice and the person did not follow it.**
**Q76. TMAAT you helped someone through self-doubt.**
**Q77. TMAAT someone came to you for personal support.**
**Q78. TMAAT you offered unsolicited but valuable advice.**
**Q79. TMAAT your feedback empowered someone.**
**Q80. TMAAT you mentored someone into a leadership role.**
**Q81. TMAAT you recognized someone's potential.**
**Q82. TMAAT you offered guidance during a crisis.**
**Q83. TMAAT your mentorship prevented a mistake.**
**Q84. TMAAT you mentored someone from a different background.**
**Q85. TMAAT you influenced others to do the right thing.**
**Q86. TMAAT you modeled professionalism.**
**Q87. TMAAT you inspired someone to persevere.**
**Q88. TMAAT you led by quiet example.**
**Q89. TMAAT you created a development opportunity for someone.**
**Q90. TMAAT you helped someone prepare for a major challenge.**
**Q91. TMAAT you were a role model for your organization.**
**Q92. TMAAT you mentored someone and what they are doing now.**

**Model answer (STAR). Replace the story with your own.**

**Situation:** A newly hired First Officer was uncomfortable with the arrival briefings and doubted himself.

**Task:** I wanted to help him build confidence without taking over.

**Action:** I shared how I prepare, let him brief the next legs and gave him feedback on one point at a time. I also told him about a mistake I made early in my career, so he could see that learning is normal.

**Result:** In a few weeks his briefings were clear and calm, and he later told me it changed how he flew. I learned that a good mentor listens first and lets the other person do the work.

**Key points:** Listen, one point at a time, share your own mistakes, let them own the result. Mention the measurable difference.

## Teamwork & Collaboration

**Use for 'worked as a team', 'used your crew' and 'brought a team together'.**

**Q93. TMAAT you worked as a team to accomplish a goal or solve a problem.**
**Q94. TMAAT you collaborated with a group to achieve a common goal.**
**Q95. TMAAT you used the entire crew's experience to make a decision.**
**Q96. TMAAT you brought a team together to accomplish a goal.**
**Q97. TMAAT you encouraged collaborative teamwork.**
**Q98. TMAAT you supported a struggling teammate.**
**Q99. TMAAT you handled team dysfunction.**
**Q100. TMAAT you built rapport in a difficult crew environment.**
**Q101. TMAAT you used teamwork for your crew and your customer.**
**Q102. TMAAT you coordinated with a crew in flight.**
**Q103. TMAAT a failure to brief caused an issue.**
**Q104. TMAAT you resolved a team conflict.**

**Model answer (STAR). Replace the story with your own.**

**Situation:** A maintenance issue appeared before a tight departure, and the best solution depended on information from several people: dispatch, maintenance, the cabin crew and my First Officer.

**Task:** I had to bring everyone's information together and make a safe decision quickly.

**Action:** I asked each person for the facts in turn, repeated the plan back to confirm, and asked the First Officer for his view before deciding. We chose the conservative option and told the cabin crew.

**Result:** The problem was solved with no safety impact and little delay. I learned that each person holds a piece of the picture, and a good decision uses all of them.

**Key points:** Use 'we', show you asked for input, and show you kept the decision and communication clear.

## CRM

**Use for any question about Crew Resource Management, breakdowns or recovering a crew.**

**Q105. TMAAT CRM broke down and what you did.**
**Q106. TMAAT you witnessed a CRM breakdown.**
**Q107. TMAAT you failed at CRM during a flight.**
**Q108. TMAAT you used CRM to solve a problem in the flight deck.**
**Q109. TMAAT you used CRM for collaborative decision making in flight.**
**Q110. TMAAT a mission did not go as planned and you used CRM to overcome it.**
**Q111. TMAAT someone was disengaged during a flight and you used CRM to re-engage them.**
**Q112. TMAAT you had to re-establish CRM in the flight deck.**
**Q113. TMAAT you had to intervene in a CRM breakdown.**
**Q114. TMAAT CRM was ineffective and how you course-corrected.**
**Q115. TMAAT you experienced bad CRM and how you overcame it.**
**Q116. TMAAT you were not performing well and how it affected CRM.**
**Q117. Which part of CRM is most important to you, other than communication?**

**Model answer (STAR). Replace the story with your own.**

**Situation:** During a busy approach, the First Officer went quiet and stopped calling deviations. I noticed the workload was high and our communication had broken down.

**Task:** My task was to restore the crew teamwork before it affected the approach.

**Action:** I stated the problem calmly, reduced the workload by simplifying the plan, and asked the First Officer to confirm the next steps. I invited him to speak up on any deviation and thanked him each time he did.

**Result:** The approach was stabilized and the crew worked normally again. In the debrief we talked about it openly. I learned that CRM is rebuilt by simple, calm communication and by reducing the workload.

**Key points:** Recognize early, state the problem, reduce workload, invite input, debrief. Mention the specific CRM skills: communication, workload management, situational awareness, leadership and followership.

## Criticism & Feedback

**Use when you received criticism or negative feedback.**

**Q118. TMAAT you were criticized by a supervisor, peer or leader and how you handled it.**
**Q119. TMAAT you were criticized for your flying and how you reacted.**
**Q120. TMAAT someone criticized a decision you made.**
**Q121. TMAAT you were corrected by a peer.**
**Q122. TMAAT your coworkers were disappointed in you.**
**Q123. TMAAT you received difficult feedback from a colleague.**
**Q124. TMAAT you received negative feedback and how you reacted to it.**

**Model answer (STAR). Replace the story with your own.**

**Situation:** After a flight, a check airman told me that my approach briefings were too long and unfocused.

**Task:** I wanted to take the feedback, understand it and improve.

**Action:** I thanked him, asked for a specific example, and did not defend myself. I rewrote my briefing into a short structure of threats, plan and backup, and I asked him to observe my next flight.

**Result:** On the next flight he said the briefing was clear. I now ask for feedback on purpose. I learned that criticism is information, and the way you react decides whether people keep giving it.

**Key points:** Listen without defending, ask for details, act, follow up. Never blame the person who criticized you.

## Decision Making: Time Critical & Incomplete Information

**Use for quick decisions, decisions without all the information, and time-critical team decisions.**

**Q125. TMAAT you made a decision without all the information.**
**Q126. TMAAT you took off or started a flight without all the information.**
**Q127. TMAAT you made a decision with low situational awareness.**
**Q128. TMAAT you had to think on your feet with missing information.**
**Q129. TMAAT you made a decision with an incomplete briefing or missing data.**
**Q130. TMAAT you made a mistake because you lacked full information.**
**Q131. TMAAT you had to make a decision outside your comfort zone without full information.**
**Q132. TMAAT you operated unprepared or felt underprepared.**
**Q133. TMAAT you used your team to make a quick decision with limited information.**
**Q134. TMAAT you made a time-critical decision.**
**Q135. TMAAT you and your crew solved a time-critical problem.**
**Q136. TMAAT you made a quick decision as a crew without sacrificing quality.**
**Q137. TMAAT you had to make a decision when time was a factor.**
**Q138. TMAAT you used all available resources to make a time-sensitive decision.**
**Q139. TMAAT you used CRM to make a time-critical decision.**
**Q140. TMAAT you had to act quickly in a time-critical situation.**

**Model answer (STAR). Replace the story with your own.**

**Situation:** On departure, an indication appeared that did not match the other instruments. I did not have the full picture and the time to decide was short.

**Task:** I needed a safe decision with what I knew, and I needed to use the crew.

**Action:** I confirmed who was flying, asked for the facts from the First Officer, applied the checklist, and chose the most conservative option that kept all choices open. I gave a short briefing to the crew and told ATC.

**Result:** We stabilized the airplane and landed without problems. Later the cause was found, and the decision proved correct. I learned a simple model: collect the facts, list the options, pick the safest, act, and review.

**Key points:** Name your decision process (for example: Aviate – Navigate – Communicate, then facts, options, decision, review). Show you ask for help and accept an imperfect picture.

## Accomplishment & Impact

**Use for 'proudest moment' and 'made an impact on your organization'.**

**Q141. TMAAT you were proud of an accomplishment in your career.**
**Q142. TMAAT you accomplished something significant and what you learned.**
**Q143. TMAAT you made an impact on your organization.**
**Q144. TMAAT you helped your organization grow.**
**Q145. TMAAT you achieved excellence within your organization.**
**Q146. TMAAT your words or actions inspired someone.**
**Q147. TMAAT you made a difference to a customer or team member with your words.**
**Q148. TMAAT you overcame a major obstacle and what it meant to you.**
**Q149. TMAAT someone came to you because of something you had done well.**
**Q150. TMAAT you took the initiative to make an improvement.**
**Q151. TMAAT you used all your resources and are proud of the outcome.**

**Model answer (STAR). Replace the story with your own.**

**Situation:** When I joined a flight department, the training program for new pilots had no structure and new hires took too long to be ready.

**Task:** I wanted to build a better program and measure the results.

**Action:** I created a step-by-step training plan, trained the instructors, and tracked the time to qualification for each new pilot. I asked the crews for feedback and adjusted twice.

**Result:** The time to qualification dropped and the new hires gave better feedback. I am proud of it because it helped the whole team. I learned that a small system, well explained, makes a large difference.

**Key points:** Choose a result you can measure, share the credit, explain why it mattered to you.

## Communication & Adapting Your Style

**Use when you had to change how you communicate to be understood.**

**Q152. TMAAT you had to change your communication style.**

**Q153. TMAAT you changed your communication because someone was not receptive or not understanding you.**
**Q154. TMAAT your message was not landing and you tailored it.**
**Q155. TMAAT you had to change your teaching method to help someone understand.**
**Q156. TMAAT you had to break down barriers to communicate.**
**Q157. TMAAT you communicated unsuccessfully and how you improved.**
**Q158. TMAAT you had to motivate a team by adjusting how you communicated.**
**Q159. TMAAT you had to modify how you communicated with someone who was disengaged.**
**Q160. TMAAT poor communication caused confusion.**
**Q161. TMAAT you led through a miscommunication.**
**Q162. TMAAT your communication made a difference.**

**Model answer (STAR). Replace the story with your own.**

**Situation:** I was training a new First Officer who was not responding to my explanations of an approach procedure.

**Task:** I needed him to understand the procedure well, so I had to change my method.

**Action:** I noticed that long explanations were not working. I switched to a short drawing, asked him to explain the procedure back to me, and used one example from his own experience.

**Result:** He understood it and flew the procedure well in the simulator the next day. I learned that if the message is not landing, the problem is the delivery, and I should change my approach, not blame the listener.

**Key points:** Notice, adapt, check understanding. Show you take responsibility for the communication.

## Failure, Mistakes & Adversity

**Use for failure, setback, mistake or 'did not meet standards'.**

**Q163. TMAAT you failed and what you learned from it.**
**Q164. TMAAT you failed to meet a deadline or a goal and how you handled it.**
**Q165. TMAAT you had a struggle in training.**
**Q166. TMAAT you had to own up to a mistake.**
**Q167. TMAAT you made a mistake that caused a project to fail.**
**Q168. TMAAT you handled something poorly and what you did.**
**Q169. TMAAT you felt inadequate and what you did to overcome it.**
**Q170. TMAAT you were told that you did not meet standards.**
**Q171. TMAAT you had your biggest failure.**
**Q172. TMAAT you failed to make the right decision as a leader.**
**Q173. TMAAT your team failed to meet a deadline.**

**Model answer (STAR). Replace the story with your own.**

**Situation:** Early in a new aircraft type, I struggled in a part of the training and did not meet the standard on one event.

**Task:** I had to understand what went wrong and reach the standard.

**Action:** I told my instructor honestly, asked for a plan, studied the weak area every day, and practiced with a colleague on the ground. I asked for a re-check when I was ready.

**Result:** I passed the re-check and became more careful in my preparation. I learned to own a mistake quickly, ask for help early and turn the setback into a habit of study.

**Key points:** Pick a real example, own it without excuses, show the fix and the lesson. Never choose a mistake that shows poor judgment about safety.

## Customer Service

**Use for passengers, customers, and 'above and beyond' questions.**

**Q174. TMAAT you demonstrated excellent customer service.**
**Q175. TMAAT you had a stressful or difficult interaction with a customer.**
**Q176. TMAAT you helped solve a stressful passenger situation.**
**Q177. TMAAT you had to deliver bad news to a customer.**
**Q178. TMAAT you made a decision that caused inconvenience and had to explain it to the customer.**
**Q179. TMAAT you showed empathy to a customer under stress.**
**Q180. TMAAT you could have provided better customer service.**
**Q181. TMAAT you did something to improve customer service.**
**Q182. TMAAT you provided customer service in a time-critical or stressful environment.**
**Q183. TMAAT you resolved a dispute or customer service challenge.**

**Model answer (STAR). Replace the story with your own.**

**Situation:** Before departure, a passenger was upset because a connection was at risk and the delay was growing.

**Task:** I wanted to help the passenger feel informed and cared for, without breaking any rule.

**Action:** I went to the cabin, introduced myself, listened first, and explained the facts: what we knew, what we were doing and when I would update them. I asked the agent to help with the connection and made a short announcement for everyone.

**Result:** The passenger calmed down and thanked the crew. Later we received a compliment. I learned that people accept bad news better when they feel heard and informed.

**Key points:** Empathy, honesty, action. Show that safety always comes first and that service is part of the professional job of a pilot.

# Chapter 7

## Technical Knowledge — General Study Guide

**This chapter covers technical topics commonly asked at airline, cargo, fractional, and corporate aviation interviews worldwide. Review these concepts thoroughly and be prepared to explain them clearly and concisely.**

### RVSM — Reduced Vertical Separation Minima

Applies from FL290 to FL410.

Rate of climb/descent: 500 ft/min to 1,000 ft/min.

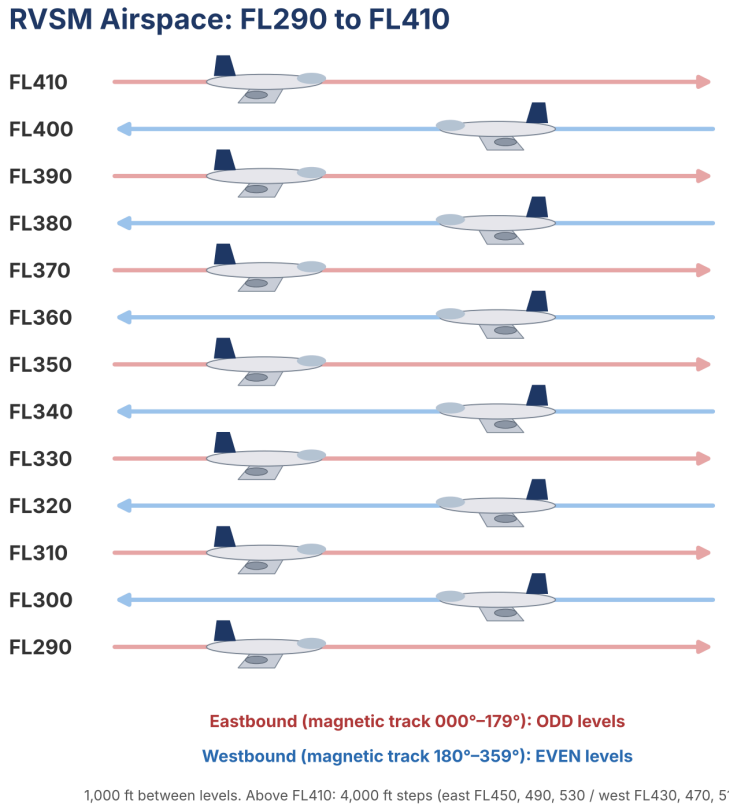
Cannot overshoot assigned level by more than +150 ft.

Altimeters: within 75 ft of field elevation on the ground; max 200 ft apart in flight.

Flight plan: include letter “W” in equipment field.

Weather deviation (no clearance possible): if less than 5 NM from the route, maintain the flight level. SLOP: fly 0, 1 or 2 NM to the right of the route for wake turbulence.

Deviation of 5 NM or more: Eastbound — climb 300 ft if right, descend 300 ft if left. Westbound — descend 300 ft if right, climb 300 ft if left.



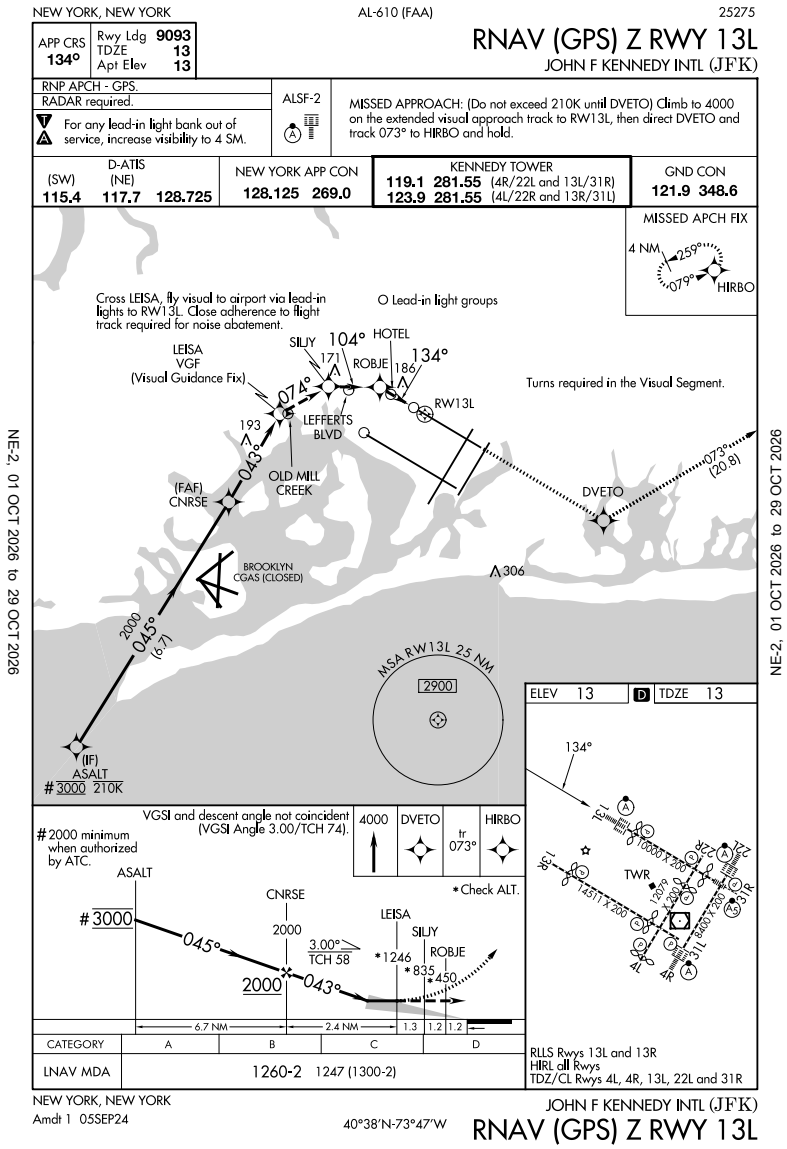
## RNAV — Area Navigation

Flight between 2 points without any radio station.

Advantages: time and fuel saving, reduces dependence on radar vectoring, more efficient use of airspace.

Waypoints defined by geographical position (latitude and longitude).

Two waypoint procedures: Fly-By (turn starts prior to reaching waypoint) and Fly-Over (fly exactly over the waypoint).




John F. Kennedy Intl (KJFK) - RNAV (GPS) Z Rwy 13L. FAA chart (public domain), cycle 01 OCT to 29 OCT 2026.
For study only; not for navigation.

## ETOPS — Extended Range Twin Engine Operations

Two-engine aircraft: in case of engine failure, must be able to fly and land at an adequate airport within the authorized time (air calm).

Standard ETOPS: 60 minutes. Can be extended to 207 minutes.

ETOPS up to 180 min: fire category 4 required.

ETOPS up to 207 min: fire category 7 required.

Three generators required for dispatch.

Required holding fuel: 15 min at 1,500 ft.

Adequate airport: satisfies aircraft performance (landing weight, equipment); must be available and anticipated 2 hours before use.

Suitable airport: adequate airport PLUS required ceiling and visibility, and safe landing with one engine inoperative.

## Three Single-Engine Descent Strategies

Standard strategy: 300 kts / M.78.

Obstacle strategy: Green dot speed, MCT.

Fixed speed strategy (ETOPS): 320 kts / M.78 / MMO M.80 / 350 kts MCT.

## Holding Speeds

FAA: Up to 6,000 ft — 200 kts | 6,001 to 14,000 ft — 230 kts | Above 14,000 ft — 265 kts.

ICAO (PANS-OPS): Up to 14,000 ft — 230 kts | 14,000-20,000 ft — 240 kts | 20,000-34,000 ft — 265 kts | Above 34,000 ft — M.83.

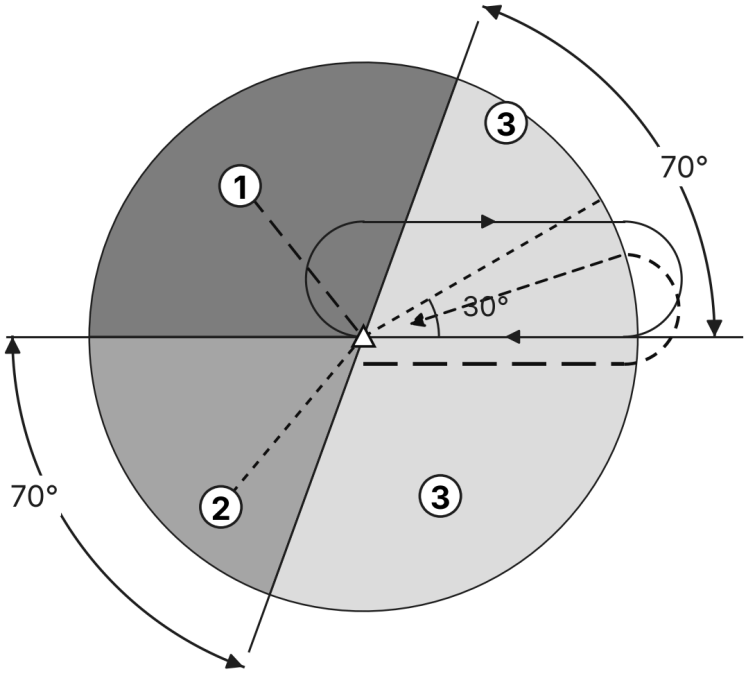
Speed reduction for holding: 3 min before entering hold when ATC does not request.

Turn: 3 degrees per second | 25 degrees bank or follow flight director | 30 degrees bank maximum.

Timing: Up to FL140 — 1 min | Above FL140 — 1 min 30 sec.

Entries: Parallel (110°), Teardrop (70°), Direct.

Standard holding: all turns to the right. Non-standard holding: all turns to the left.




1 Parallel entry (110° sector)
2 Teardrop entry (70° sector) 3 Direct entry (180° sector)

## Procedure Turn 45/180 Degrees

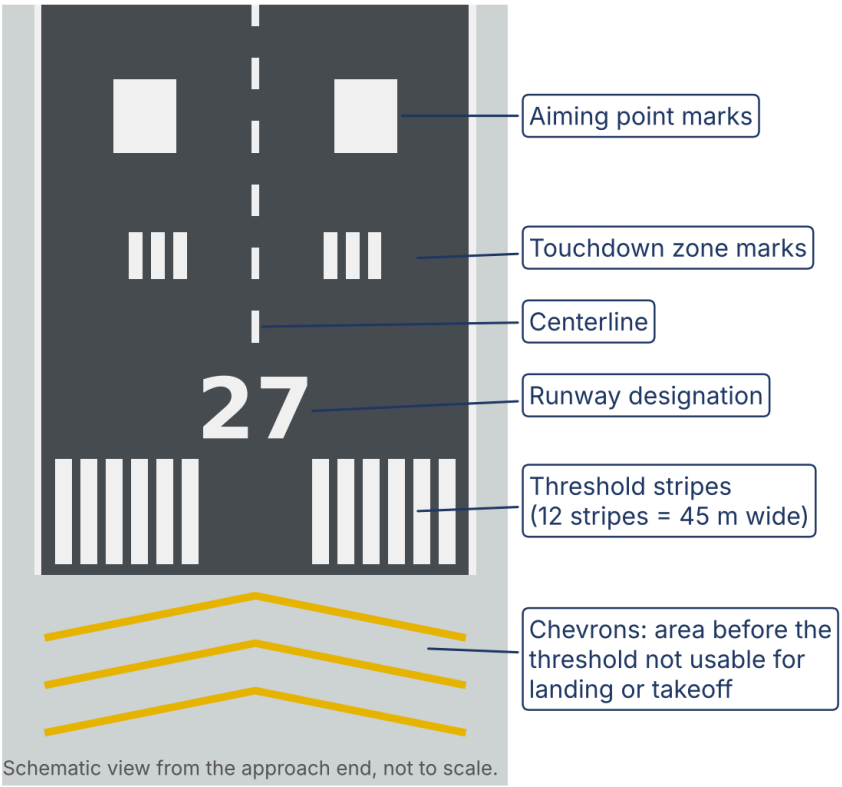
First entry: outbound 45°, time 1 min or 45 sec with wings level, re-intercept inbound.

Second entry: outbound 80°, as soon as heading reached start turn 260° to re-intercept inbound.

Used when no suitable fix permits a direct entry onto an instrument approach.

## Runway Threshold Stripes

18m wide — 4 stripes | 23m wide — 6 stripes | 30m wide — 8 stripes | 45m wide — 12 stripes | 60m wide — 16 stripes.



Schematic view from the approach end, not to scale.

## Take-off Path Segments

1st segment: starts at 35 ft (15 ft wet), until gear fully retracted. Gradient: 2 engines 2.4% | 3 engines 2.7% | 4 engines 3.0%.

2nd segment: starts after gear fully retracted until 400 ft (1,000 ft). Speed V2 or V2+10 kts, takeoff power. Gradient: 2 eng 2.4% | 3 eng 2.7% | 4 eng 3.0%.

3rd segment: starts after 400 ft (1,000 ft), speed V2, finishes when flaps fully retracted. Gradient: 2 eng 1.2% | 3 eng 1.4% | 4 eng 1.5%.

4th segment: starts after flaps fully retracted until 1,500 ft. Gradient: 2 eng 1.2% | 3 eng 1.4% | 4 eng 1.5%.

## Approach BAN

If weather drops below minimums before the outer marker: must go around before passing the outer marker.

If already past the outer marker: can continue the approach until MDA.

## Wake Turbulence Categories

Light (L): below 7,000 kg. Medium (M): 7,000 to 136,000 kg. Heavy (H): above 136,000 kg.

Note: 757 is classified as Heavy.

Departure separation: Light/Medium behind Heavy — 5 NM or 2 min.

Arrival separation: Medium behind Heavy — 2 min or 5 NM | Light behind Heavy — 3 min | Light/Medium behind Heavy — 5 NM.

## ATC Light Signals

Steady Green: cleared for takeoff / cleared to land.

Flashing Green: cleared to taxi / return to land.

Steady Red: stop / give way to other aircraft and continue circling.

Flashing Red: taxi clear of landing area or runway in use / airport unsafe, do not land.

Flashing White: return to starting point on airport / land at this airport and proceed to apron.

Red Pyrotechnic: do not land for the time being.

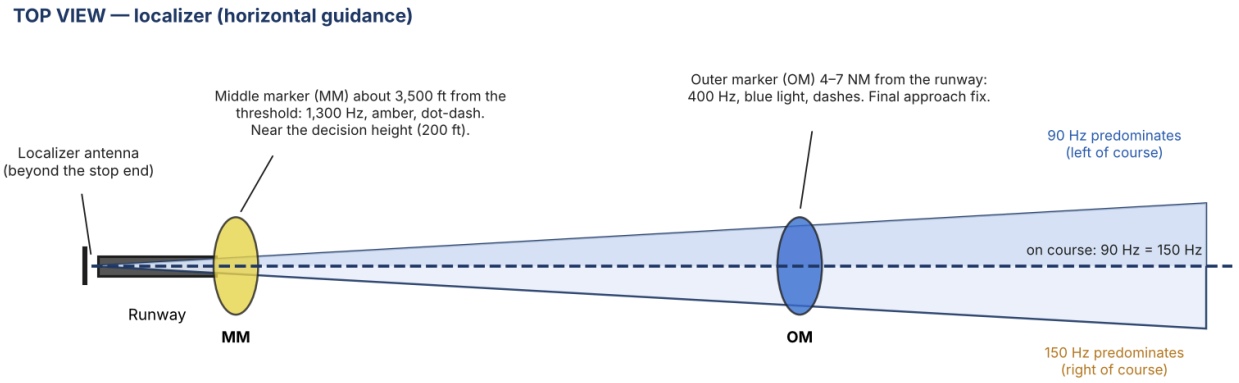
## Instrument Landing System (ILS)

Localizer range: ±2.5° each side (full width 5°). Range: 18 NM. Beyond 18 NM: ±10° at 18 NM, ±35° at 10 NM each side. Frequency: 108.10 to 111.95 MHz.

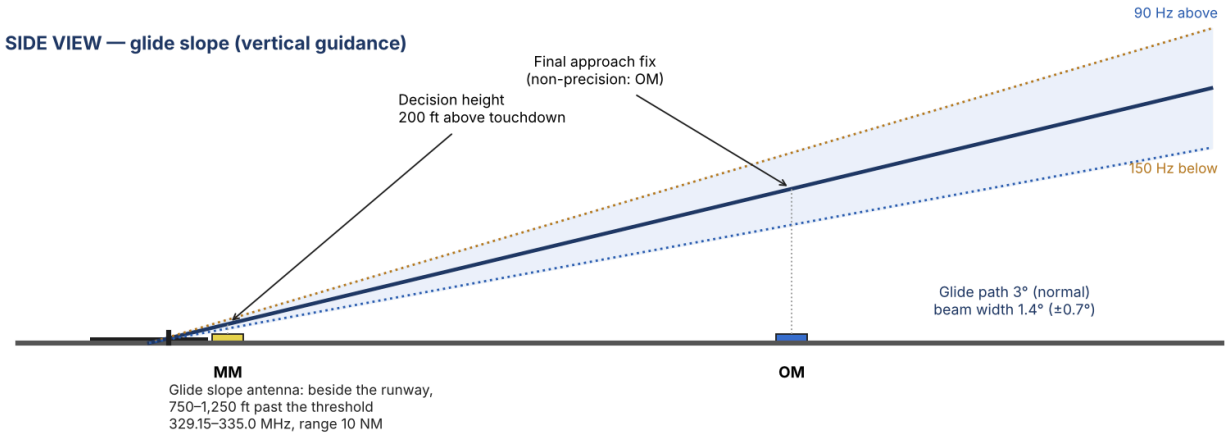
Glide Slope: range 10 NM. Frequencies: 329.15 to 335.0 MHz. Normal glide path: 3°. Beam width: normally 1.4°. Loss of GS: approach can continue as non-precision approach.

PRM (Precision Runway Monitor): required for simultaneous close parallel approaches. Close parallel = less than 4,300 ft between runway centerlines.

**ILS — Instrument Landing System: how the beams and markers fit together**




*Full-scale deflection ±2.5° (5° total). Range 18 NM (±10°), 10 NM (±35°).*




*Vertical scale exaggerated. Without the markers, DME or GPS fixes can replace them.*

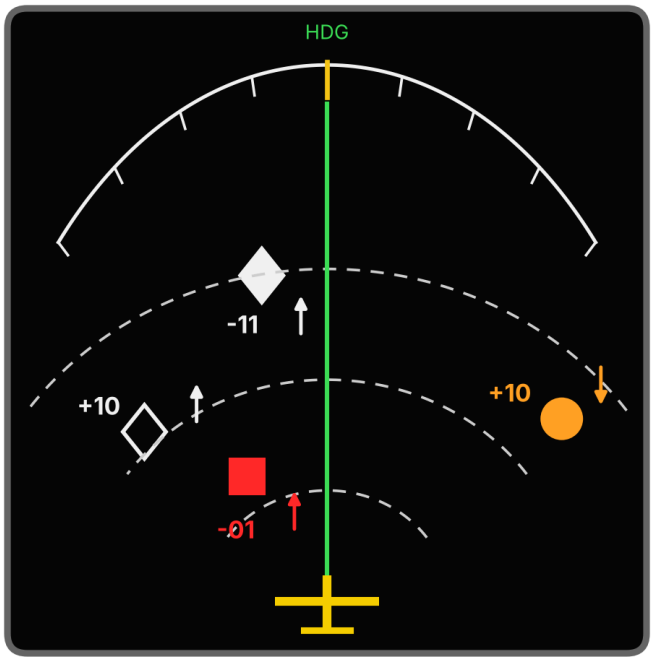
## TCAS — Traffic Alert and Collision Avoidance System

Detects any aircraft with a transponder. Range: 30-40 NM. Altitude range: 9,900 ft above or below.

White diamond: proximate traffic. Amber circle: traffic advisory (TA) — aural “Traffic Traffic,” about 40 sec.

Red square: real collision threat (RA) — aural voice, resolution advisory (vertical orders), about 25 sec.

RA procedure: AP OFF / FD OFF — fly orders — clear of conflict: FD ON / AP ON. Call out “TCAS, I HAVE CONTROL.”




**Other traffic**
open diamond, white or cyan: no threat

**Proximate traffic**
solid diamond: within 6 NM and 1,200 ft, no threat

**Traffic advisory (TA)**
solid amber circle: "Traffic, traffic"

**Resolution advisory (RA)**
solid red square: follow the vertical orders

The number is the relative altitude in hundreds of feet:
+10 = 1,000 ft above; -01 = 100 ft below. Arrow: climbing or descending.
*Schematic of a navigation display, not to scale.*

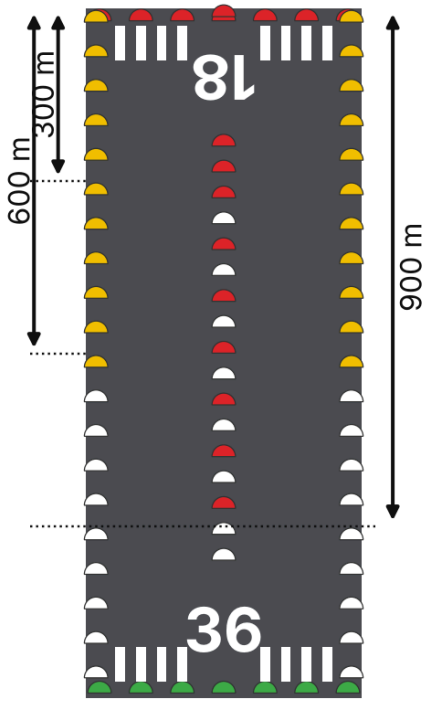
## Runway Edge Lights

Classified by intensity: HIRL (High), MIRL (Medium), LIRL (Low).

Color: white; yellow in the last 600m (2,000 ft), the caution zone; the lights at the runway end are red.

Spacing between edge lights: 60m.

Center lights: white until last 900m (3,000 ft), then alternating red/white until last 300m, then red only.
CAT II/III spacing: longitudinal spacing 7.5m, 15m, or 30m. CAT II requires at least 30m.
CAT III: 7.5m and 15m.




**End**
Red. Required when edge lights are present.

**Edge**
White; the last 600 m yellow.
Required for night operations or precision approaches.

**Centerline**
White; from 900 m to 300 m from the end, alternating red and white; the last 300 m red.
Required for CAT II/III approaches.

**Threshold**
Green. Required when edge lights are present.

Colors follow the ICAO standard. In the United States the yellow zone of the edge lights is the last 2,000 ft (or half the runway, whichever is less). Schematic view, not to scale.

## Touchdown Zone Lights

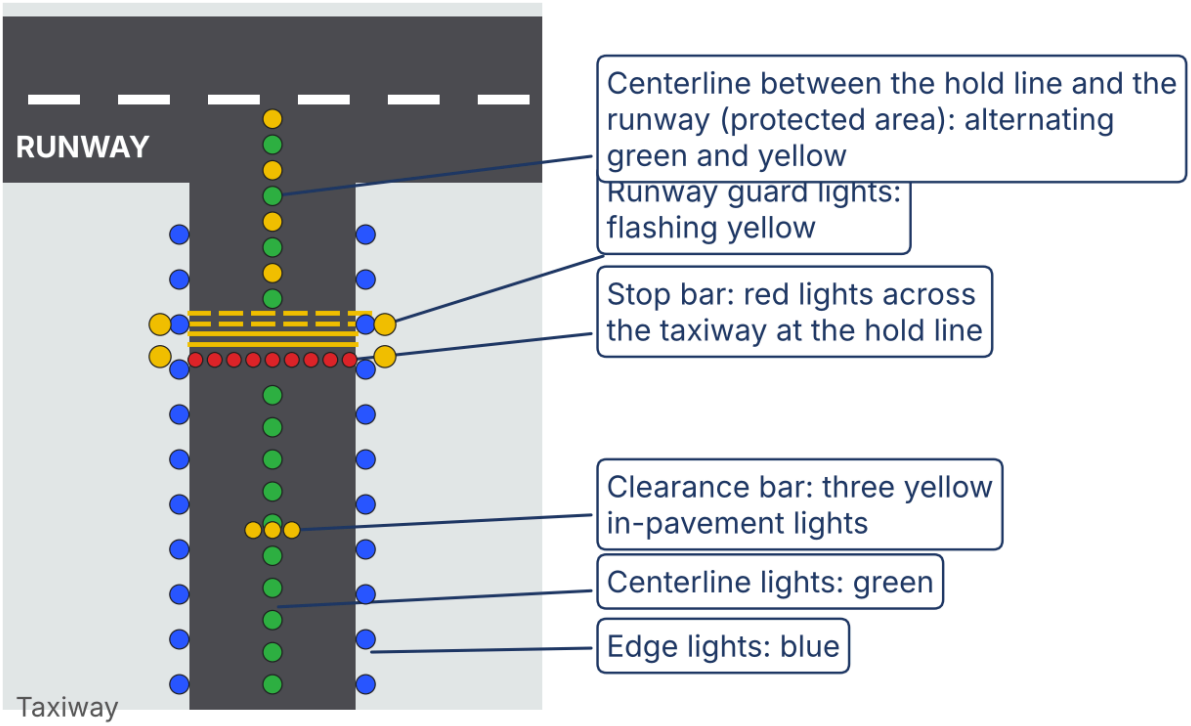
Required for CAT II and III.
Extend from threshold for 900m longitudinal distance (not to exceed midpoint).
Three row lights, color: white.

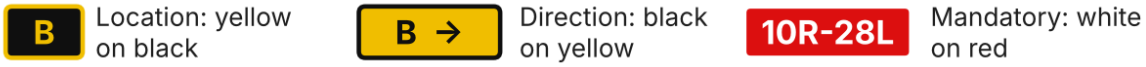
## Taxiway Lighting

Edge lights: blue. Used for LVO but not specific for CAT II/III.

Centerline: green. At RVR below 400m (1,312 ft), lateral spacing not to exceed 15m (7.5m in curves). In sensitive areas: alternating green and yellow.

Stop bars: red. Intervals of 3m. Used at RVR below 400m (1,312 ft). Required for all CAT III.




*Schematic, not to scale. Equipment varies by airport.*

## LVO — Low Visibility Operations

Visibility below 400m (1,312 ft) but not less than 150m (492 ft).

If no RVR available: pilots can make the assessment.

Minimum visibility for takeoff: 125m (410 ft).

Runway cannot be contaminated. Crosswind: less than 10 kts.

Takeoff from the beginning of runway. Static takeoff. TOGA power. Lights off. Must be in force.

## RVR — Runway Visual Range

High-accuracy instrument. Measured at three points: touchdown, midpoint, and endpoint. Reported up to three points. Starts detecting when below 1,500m.




METAR and TAF show RVR with “R” prefix before visibility value.

## Approach Minimums (CAT I / II / III)

CAT I: DH 200 ft — RVR 550m at touchdown. Visibility 800m. Midpoint RVR not required.
CAT II: DH between 200-100 ft — RVR at least 300m at touchdown. Midpoint not required.
CAT IIIa: DH between 100-50 ft — RVR 200m touchdown, 150m midpoint, 50m rollout.
CAT IIIb: DH below 50 ft — RVR 50m at touchdown, midpoint, and rollout.
CAT IIIc: DH 0 ft — RVR 0m.

## Wind Limitations — CAT II and CAT III

Headwind: 30 kts managed / 20 kts selected.
Max tailwind: 10 kts.
Max crosswind: 20 kts.

## Flight Parameter Deviation Call-Outs

IAS: +10 kts / -5 kts — callout: SPEED.
Rate of descent 1,000 ft/min or more — callout: SINKRATE.
Pitch attitude: 10° up / 2.5° down — callout: PITCH.
Bank angle: 7° or more — callout: BANK ANGLE.
Localizer: 1/4 dot — callout: LOCALIZER.
Glide Slope: 1 dot — callout: GLIDE SLOPE.

## CAT II Autoland

Automatic landing until 2,500 ft. Autobrake medium. Auto callout. Single engine approved. Flaps full or 3.
Fail Passive: pilot assumes control, computer cannot continue. Landing not completed automatically.
Fail Operational: automatic landing continues with remaining system; flare and landing completed automatically.

## MEL — Minimum Equipment List

Cat A: rectified at 00:01 next day. Cat B: rectified in 3 days. Cat C: rectified in 10 days. Cat D: rectified in 120 days.

* = placard INOP | O = operation procedure required | M = maintenance procedure required.

CDL (Configuration Deviation List): part missing, operation restriction, performance penalties, maintenance actions.

Maintenance schedule types: Transit (each transit), Pre-flight (prior first flight), Daily check (every 36 hrs), Weekly check (every 8 days).

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL AVIATION ADMINISTRATION

**MASTER MINIMUM EQUIPMENT LIST**

AIRCRAFT: Sample twin turboprop | REVISION NO: 1 | PAGE NO: 30-5

| 1. SYSTEM, SEQUENCE NO. & ITEM | REPAIR CATEG. | 2. NO. INSTALLED | 3. NO. REQ. FOR DISPATCH | 4. REMARKS AND EXCEPTIONS |
|---|---|---|---|---|
| 30 ICE AND RAIN PROTECTION | C | 2 | 3 | 4 |
| 30-1 Windshield wipers | C | 2 | 1 | (O) One may be inoperative provided the airplane is not operated in known or forecast precipitation. |
| | C | 2 | 0 | (O) May be inoperative provided:<br>a) the flight is not planned in known or forecast precipitation, and<br>b) the crew can maintain the required visibility. |

### How to read the page

1. System (ATA chapter 30 = ice and rain protection), sequence number and item.

C. Repair category: C = repair within 10 calendar days
(A: as stated in the remarks, B: 3 days, D: 120 days).

2. Number installed on the aircraft.

3. Number required for dispatch. First row: 1 of 2 required. Second row: 0 required, so both may be inoperative if the conditions in column 4 are met.

4. Remarks and exceptions. (O) = operations procedure required, (M) = maintenance procedure required.

*Illustrative example with the layout of an FAA MMEL page. Not an actual MMEL.*

## Types of Aquaplaning

Dynamic: due to standing water.

Viscous: surface is damp, thin film of fluid the tire cannot penetrate.

Reverted rubber (rubber reversion): long skid, rudder reversion, wet surface.

Control: release brakes if skid is sensed. On wet runway, reverse thrust is most effective.

## Weather Terms

CAVOK: visibility equal or greater than 10 km, no clouds below 5,000 ft, no precipitation, no thunderstorm.

BECMG (Becoming): permanent change, phenomena starts after the specified time.

TEMPO: temporary variation.

TAF: issued for 9 hours, updated 4 times daily, international TAFs valid for 24 hours.

## GPS — Global Positioning System

Needs 4 satellites: 2 establish lat/long, 1 confirms the fix, 1 gives altitude information.

## Takeoff Distance Definitions

TORA: Take-Off Run Available — usable length of runway.

TORR: Take-Off Run Required.

Clearway: area at end of runway, 75m each side of extended centerline (can be water). Not defined surface.

TODA: Take-Off Distance Available = TORA + Clearway.

Stopway: unprepared surface at end of runway supporting the aircraft.

ASDA: Accelerate-Stop Distance Available = TORA + Stopway.

Balanced field: TODA = ASDA.

MEA (Minimum En-Route Altitude): lowest published altitude between radio fixes that meets obstacle clearance requirements.

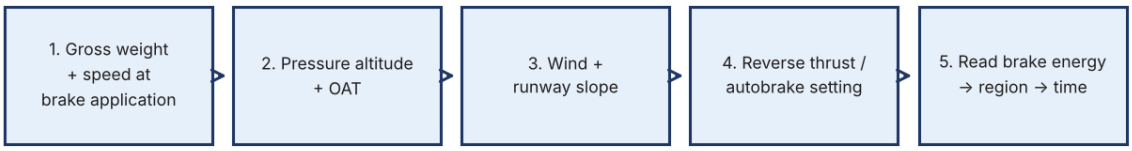
MOCA (Minimum Obstacle Clearance Altitude): lowest altitude in effect between radio fixes on VOR.

MORA (Minimum Off-Route Altitude): derived by Jeppesen; provides obstacle clearance within 10 NM of route centerline.

## Brake Cooling Schedule

Brake cooling schedule: a chart in the AFM/QRH that tells how long to wait after a high-energy stop (rejected takeoff, heavy or fast landing, repeated stops) before brakes are safe for a new takeoff. Fuse plugs in the wheels melt when hot and deflate the tire safely instead of letting it burst.

### BRAKE COOLING SCHEDULE — how it is used




*Inputs from the AFM / QRH chart of the aircraft. The values in the table below are illustrative only.*

| REGION | TYPICAL ACTION | WHAT TO EXPECT |
|---|---|---|
| **1 Normal** | Continue normal operation. No cooling needed. | Brakes stay inside normal limits. |
| **2 Caution** | Wait for the cooling time in the chart before the next takeoff. | Hot brakes: reduced energy margin for a rejected takeoff. |
| **3 Fuse plug zone** | Do not take off. Stay clear of the wheels (approach only from the front or rear of the gear). | Fuse plugs may release air and deflate tires 10–15 min after the stop. |
| **4 Emergency** | Park without the parking brake. Call fire service. Keep people clear for about 1 hour. | Possible brake or wheel fire; wheel/tire failure risk. |

*Higher brake energy → longer cooling time. Heavy weight, high speed, high altitude, high temperature, tailwind and no reverse all raise the brake energy. Illustrative table: always use the cooling schedule of your aircraft.*

## BRAKE COOLING TABLE (illustrative)

Brake cooling time in minutes, by gross weight and ground speed at brake application

| GROSS WT (lb/1000) | GROUND SPEED (knots) | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 80 | 90 | 100 | 110 | 120 | 130 | 140 | 150 | 160 | 170 | 180 |
| 240 | | | 4 | 11 | 18 | 26 | 35 | 44 | 54 | 64 | MZ |
| 260 | | | 7 | 14 | 22 | 31 | 40 | 50 | 61 | 72 | MZ |
| 280 | | 2 | 9 | 17 | 26 | 35 | 45 | 56 | 68 | MZ | MZ |
| 300 | | 4 | 12 | 20 | 30 | 40 | 50 | 62 | 74 | MZ | MZ |
| 320 | | 7 | 15 | 24 | 33 | 44 | 56 | 68 | MZ | MZ | MZ |
| 340 | 1 | 9 | 17 | 27 | 37 | 49 | 61 | 74 | MZ | MZ | MZ |
| 360 | 3 | 11 | 20 | 30 | 41 | 53 | 66 | MZ | MZ | MZ | MZ |
| 380 | 4 | 13 | 23 | 33 | 45 | 58 | 71 | MZ | MZ | MZ | MZ |
| 400 | 6 | 15 | 25 | 37 | 49 | 62 | MZ | MZ | MZ | MZ | MZ |

Gray zone: wheel fuse plugs may melt. Delay the takeoff and inspect after one hour.

Black zone (MZ, melt zone): clear the runway, no parking brake, do not approach the gear or taxi for one hour.

*Use after a rejected takeoff or a landing where brake overheat is suspected. Illustrative values only: use the brake cooling table of your own aircraft.*

## RNP — Required Navigation Performance

Improved navigation capability. Depends on aircraft equipment and navigation infrastructure.

ICAO defines RNP values for four navigation phases: Oceanic, En-route, Terminal, Approach.

## Other Navigation & Approach Systems

ALS (Approach Light System): basic means to transition from instrument to visual flight for landing.

VASI (Visual Glide Slope Indications): 3-5 NM during day, up to 20 NM at night.

ATIS (Automatic Terminal Information Service): continuous broadcast of recorded non-control information at high-activity terminals.

## Braking Action Reports

Good: deceleration and directional control feel normal for the braking applied.

Medium (formerly “Fair”): deceleration or directional control is noticeably reduced.

Poor: deceleration or directional control is significantly reduced; expect a long roll.

Nil: minimal or no braking, and directional control is uncertain.

Who reports: the pilot who has just landed. He or she reports what was felt on the runway: how well the aircraft slowed down (deceleration) and how well it stayed straight (directional control), using the words Good, Medium, Poor or Nil. Tell ATC the runway and the time. Report Poor or Nil at once, so other crews and the airport are warned. A report is a feeling, so it changes with aircraft type, weight, tires and braking used: use it together with the runway condition codes of the airport.

### BRAKING ACTION REPORT — Runway Condition Assessment Matrix (RCAM)

| CODE | RUNWAY SURFACE (examples) | WHAT THE PILOT FEELS | REPORTED AS |
|---|---|---|---|
| 6 | Dry; frost; wet (water 1/8 in or less) | Not applicable | — |
| 5 | Slush, dry snow or wet snow, 1/8 in or less | Normal deceleration and directional control | GOOD |
| 4 | Compacted snow (−15 °C and colder); slippery when wet | Between good and medium | GOOD TO MEDIUM |
| 3 | Dry or wet snow deeper than 1/8 in; compacted snow warmer than −15 °C | Noticeably reduced | MEDIUM |
| 2 | Water or slush deeper than 1/8 in | Between medium and poor | MEDIUM TO POOR |
| 1 | Ice | Significantly reduced | POOR |
| 0 | Wet ice; slush or water over ice; water over compacted snow | Minimal or none; control uncertain | NIL |

*How it is reported: the airport gives a code for each third of the runway (touchdown / midpoint / rollout), for example 5/5/3, on the ATIS or the NOTAM. Pilots who just landed can report the braking action they felt (Good, Medium, Poor, Nil). The word “Fair” was replaced by “Medium”. Any report of Poor or Nil should be passed on at once. Summary of the FAA matrix: always check the current FAA documents.*

## Runway Contamination Levels

### RUNWAY CONTAMINATION LEVELS

| | |
|---|---|
| **DRY** | No visible moisture. |
| **DAMP** | Surface changes color because of moisture, but it is not shiny. |
| **WET** | Shiny surface or visible water, but 1/8 in (3 mm) or less of water. |
| **CONTAMINATED** | More than 25% of the runway length and width (or the cleared width) is covered by water or slush deeper than 1/8 in (3 mm), snow, compacted snow, ice or frost. |

*Braking performance gets worse from top to bottom: use the contaminated runway performance in the AFM.*

### Contaminants

| CONTAMINANT | RELATIVE DENSITY (water = 1.0) | EQUIVALENT WATER DEPTH | TYPICAL MAXIMUM DEPTH FOR TAKEOFF* |
|---|---|---|---|
| **Standing water** | 1.0 | Depth as measured | 1/2 in (13 mm) |
| **Slush** | about 0.85 | Depth × 0.85 | 1/2 in (13 mm) |
| **Wet snow** | about 0.5 | Depth × 0.5 | 1 in (25 mm) |
| **Dry snow** | about 0.2 | Depth × 0.2 | 4 in (100 mm) |
| **Compacted snow** | — | Contaminated at any depth | See AFM / operator limits |
| **Ice** | — | Contaminated at any depth | See AFM; often no takeoff |

** Typical values; the real limit is in the AFM or the operator's manual. Example: 1/2 in of slush ≈ 0.4 in (about 11 mm) of water. A wet or contaminated runway needs a longer takeoff and landing distance and a lower crosswind limit. Check ATIS, NOTAMs and the runway condition codes. Summary of FAA and ICAO definitions; always check the current documents.*

## VREF & CRM

VREF: reference speed = 1.3 x VSO (stall speed in landing configuration).

5 Focus Points of CRM Training: Situation Awareness, Communication Skills, Teamwork, Task Allocation, Decision-Making.

## Engine Out SID

Must be performed with maximum bank of 15° at 1,500 ft, at green dot speed.

## Landing Distance

Actual Landing Distance (ALD): measured from 50 ft above runway threshold until complete stop. Uses approach speed VLS (1.23 x VS configuration), maximum braking,




ground spoiler operative.

VLS+5 for CAT II/III.

Required landing distance on wet runway: 115% of dry runway distance.

## Failure Levels (ECAM)

Level 3: Safety / Red / Immediate action.

Level 2: Abnormal / Amber / Awareness.

Level 1: Degradation / Amber or None / Monitoring.

## ATC Flight Plan Equipment Codes

Wake turbulence: H (Heavy, above 136,000 kg) | M (Medium, 7,000-136,000 kg) | L (Low, below 7,000 kg).

Equipment: S=Standard (VHF/VOR/ILS) | D=DME | W=RVSM | X=MNPS | R=RNP.

Other: SEL=SELCAL | OPR=Operator | STS=Hospital or engine inoperative.

## Special Operations — Flying in China: QFE & Metric Altimetry

Flying in China is one of the most unique operational experiences a professional pilot can have. Unlike the rest of the world, China uses a combination of QFE altimeter settings and metric altitudes that is unlike any other airspace system. If you have never operated there before, the learning curve is steep. If you have, this knowledge is a powerful differentiator in any international aviation interview.

This section explains exactly how the Chinese altimetry system works, how to use the conversion tables, and the critical mistakes that pilots make when they fly in China for the first time.

### The Three Altimeter Settings — QNH, QFE, and QNE

QNH: Altimeter set to local sea level pressure. The altimeter reads altitude above mean sea level (MSL). This is the standard setting used worldwide for en-route and approach operations.

QFE: Altimeter set to the atmospheric pressure at the airport elevation. When set to QFE, the altimeter reads ZERO on the runway. Height above the airport is displayed directly — not altitude above sea level.

QNE: Standard pressure setting (1013.25 hPa / 29.92 inHg). Used above the transition altitude. The altimeter reads Flight Level (FL).

**The key difference: with QNH the altimeter reads your altitude above sea level. With QFE it reads your height above the airport. On the runway with QFE set correctly, your altimeter reads zero.**

### Why China Uses QFE

China's civil aviation system was historically influenced by Soviet aviation standards, which used QFE as the primary altimeter setting for approach and landing.

The logic behind QFE: during approach and landing, the pilot reads height directly above the runway. There is no need to mentally subtract the airport elevation from the altitude readout to calculate height above terrain.

This simplifies the approach — the pilot knows at 300 meters on the altimeter, they are 300 meters above the runway. At zero, they are on the runway.

Despite global standardization efforts, China continues to use QFE below the transition level for domestic operations. International operations follow ICAO standards but crews must be fully familiar with QFE procedures.

### Metric Altitudes — Meters Instead of Feet

China uses METERS for all altitude and height references below the transition level, and FLIGHT LEVELS above the transition altitude.

ATC clearances, approach plates, minimums, and traffic advisories are all given in meters.

A clearance to "descend to 900 meters" means 900 meters QFE — 900 meters above the airport elevation.

This requires the crew to have a metric conversion table readily available and to think in meters, not feet, for the entire approach phase.

## Feet to Meters Conversion — The Basics

1 foot = 0.3048 meters. 1 meter = 3.281 feet.

Quick mental conversion: divide feet by 3.3 to get approximate meters. Multiply meters by 3.3 to get approximate feet.

For precision operations, always use the official conversion table. Mental approximations are for situational awareness only.

### Key Altitude Conversion Reference:

| Feet | Meters | Common Use |
|---|---|---|
| 100 ft | 30 m | Decision Height CAT I reference |
| 200 ft | 61 m | Typical CAT I DH |
| 300 ft | 91 m | Stabilized approach check |
| 500 ft | 152 m | Config/speed check gate |
| 600 ft | 183 m | |
| 700 ft | 213 m | |
| 1,000 ft | 305 m | Stabilized approach gate |
| 1,500 ft | 457 m | |
| 2,000 ft | 610 m | Traffic pattern altitude reference |
| 3,000 ft | 914 m | Typical initial approach altitude |
| 5,000 ft | 1,524 m | |
| 6,000 ft | 1,829 m | Holding speed change level |
| 10,000 ft | 3,048 m | Speed restriction reference |
| 14,000 ft | 4,267 m | Holding speed change level |
| 18,000 ft | 5,486 m | |
| 20,000 ft | 6,096 m | |
| 29,000 ft | 8,839 m | China transition level reference |

## How the System Works in Practice — The Two Tables

Chinese operations require two separate reference tables that the crew uses during descent and approach.

TABLE 1 — En-route descent table (meters): Used from cruise level down to the transition level. ATC assigns altitudes in meters. The crew cross-checks their altimeter (set to QNE / 1013) against the metric conversion table to confirm they are at the correct flight level.

TABLE 2 — QFE approach table (meters): After passing the transition level and setting QFE, all altitude references switch to height above the airport in meters. The crew uses this table to cross-check approach gates, minimums, and traffic pattern heights.

The transition between the two tables happens at the transition level, which varies by airport in China. ATC will instruct you when to set QFE and the QFE value will be provided.

## Transition Level and Transition Altitude in China

China uses a standardized transition altitude system. The transition altitude varies by region and airport elevation.

In most of eastern China (low elevation airports): transition altitude is typically 3,000 meters (approximately 9,843 ft) MSL.

At high elevation airports (western China, Tibet, Yunnan): the transition altitude is higher to account for terrain.

Above the transition level: standard pressure (QNE / 1013.25 hPa) is used and altitude is expressed in metric flight levels.

Below the transition level: QFE is set and altitude is expressed in meters above the airport.

## QFE Setting Procedure

ATC will provide the QFE value for the destination airport, typically on the ATIS or during the approach clearance.

The QFE value is given in hectopascals (hPa). Both pilots set QFE on their altimeters simultaneously and cross-check.

After setting QFE: the altimeter should read approximately the airport elevation in meters above sea level MINUS the airport elevation — which should be close to zero if you are already on the ground, or should read your current height above the airport if airborne.

Cross-check: if you are at 3,000 meters QNH and the airport elevation is 50 meters, after setting QFE your altimeter should read approximately 2,950 meters.

Always verbally confirm the QFE setting with your crew: “QFE set, cross-checked.”

## Practical Approach Example — What It Looks Like in the Cockpit

Cruising at FL350 (metric FL equivalent). ATC begins descent: “Descend to 8,400 meters.” Crew consults metric table: 8,400 m = approximately FL276.

Continuing descent. At transition level ATC instructs: “Set QFE 1008, cleared ILS approach runway 36.”

Crew sets QFE 1008 on both altimeters, cross-checks, and confirms verbally.

From this point, all altitude callouts are in meters QFE — height above the airport.

Approach gates: 900m (approximately 3,000 ft) — configuration check. 600m — gear down, flaps approach. 300m — final config, stable.

Decision Height: typically 60 meters (approximately 200 ft) for a standard CAT I ILS.

At touchdown: altimeter reads zero (or very close to zero). This confirms QFE was set correctly.

## Common Mistakes Western Pilots Make in China

Forgetting to switch from QNH to QFE at the transition level. The altimeter continues reading altitude above sea level, not height above the airport. Minimums will appear much higher than they should be — extremely dangerous at airports with high field elevation.

Mixing feet and meters mentally. A clearance to “maintain 900” in China means 900 METERS, not 900 feet. 900 meters is approximately 2,953 feet — a very different altitude.

Using the wrong table. Using the en-route metric table when you should be on the QFE approach table — or vice versa — creates dangerous altitude confusion.

Not briefing the QFE value before beginning the approach. Both pilots must set the same QFE value and cross-check before descending below the transition level.

Assuming the missed approach is flown in QFE. Some operators revert to QNH for the missed approach. Know your operator’s procedure before the flight.

**The most dangerous mistake is descending below the transition level without setting QFE. If the airport is at 50 meters elevation, your altimeter with QNH set will read 50 meters when you are on the ground — not zero. You may believe you have 50 meters of height remaining when you have none. This error has contributed to serious incidents.**

## Why This Matters in an Interview

Questions about special operations — QFE, metric altimetry, high-altitude airports, RVSM, ETOPS — are common at airlines that operate internationally, especially in Asia, the Middle East, and regions with high-terrain airports.

Being able to explain the difference between QNH, QFE, and QNE clearly and accurately signals operational maturity and international experience.

Knowing how the Chinese two-table system works in practice — from personal experience or thorough study — demonstrates preparation that most candidates simply will not have.

If asked about China operations or metric altimetry, walk the interviewer through the system step by step: what QFE is, why China uses it, how the transition works, and what the common errors are. This level of detail is rare and impressive.

**If you have personally flown in China, prepare a specific STAR story about that experience. It is a powerful differentiator in any international airline interview.**

## China Operations — Quick Reference

Altimeter setting below transition level: QFE (hPa).

- Altitude units: METERS (not feet) below transition level.
- Altimeter reads: height above airport (zero on the runway).
- Transition level: varies by airport/region, typically 3,000 m in eastern China.
- Above transition level: standard pressure (1013.25 hPa), metric flight levels.
- Two tables required: (1) en-route metric descent table, (2) QFE approach table in meters.
- Always cross-check QFE setting with both crew members before descending below transition level.
- Conversion: 1 ft = 0.3048 m / 1 m = 3.281 ft. Quick rule: divide feet by 3.3 for meters.

## Ground De-Icing & Anti-Icing Procedures

Many professional pilots come from warm-weather countries — Brazil, the Middle East, Africa, Southeast Asia, the Caribbean — where snow and ice are simply not part of daily life. They build thousands of hours of solid flying experience without ever once standing next to an airplane in a snowstorm wondering what to do next.

This section exists because of that reality.

Early in a career that began in warm climates, the first exposure to cold-weather operations can be genuinely intimidating. The aircraft looks different covered in frost. The ground crew is rushing. Everyone around you seems to know exactly what is happening — and you are trying not to show that you do not.

One of the most important lessons learned early in cold-weather flying came not from a manual, but from a simple question. When the uncertainty of de-icing procedures became apparent, an instructor offered advice that proved invaluable:

> **“Go find the operators who fly in cold weather every day. Ask them: what type of de-icing are you doing today? That one question will teach you more than any manual. And it will save your life until you build your own experience.”**

That advice proved correct. Asking experienced cold-weather crews — “What type of de-ice are you doing? What fluid? What are the holdover times today?” — became a habit that bridged the gap between theory and experience many times over.

This section gives you the foundation. The rest comes from asking the right questions and building your own experience in the field.

## De-Icing vs. Anti-Icing — Understanding the Difference

### De-Icing

- De-icing is the process of REMOVING existing contamination (ice, snow, frost, slush) from the aircraft surfaces.
- Uses heated fluid sprayed under pressure to melt and wash away the contamination.
- De-icing alone does not provide protection against new contamination forming after treatment.

Type I fluid is most commonly used for de-icing due to its low viscosity and heating properties.

### Anti-Icing

Anti-icing is the process of PREVENTING new contamination from forming on clean aircraft surfaces.

Applied after de-icing or directly to a clean aircraft when precipitation is expected.

Thicker, more viscous fluids (Type II, III, IV) are used for anti-icing because they stay on the surface longer.

Anti-icing provides a protection window called the HOLDOVER TIME — the estimated time the fluid remains effective.

### The Two-Step Procedure

Step 1 — DE-ICE: Apply heated Type I fluid to remove all existing contamination. Aircraft must be clean — critical surfaces free of all ice, snow, and frost — before takeoff.

Step 2 — ANTI-ICE: Apply Type II, III, or IV fluid to protect clean surfaces during taxi and before takeoff.

The two-step procedure is standard when active precipitation is falling or expected during taxi.

Single-step procedure (anti-icing only) may be used when the aircraft is already clean and precipitation is light.

**NEVER take off with frost, ice, or snow on critical surfaces — wings, control surfaces, engine inlets, sensors. Even a thin layer of frost on the wing can reduce lift by up to 30% and increase stall speed significantly. There are no exceptions to this rule.**

## The Complete De-Ice Procedure Flow

The following is the standard decision flow for ground de-icing operations. Follow these steps in sequence every time cold-weather conditions are present.

**Step 1 — Complete All Normal Checklists:** Complete all normal pre-departure checklists as you would on any other day. De-icing is an addition to your normal flow — not a replacement for it. Engines, systems, and avionics must be configured correctly before de-icing begins.

**Step 2 — Assess Current Conditions:** Evaluate conditions visually and using available weather data. Step outside and look at the aircraft. Check the METAR for current precipitation type, visibility, and temperature. Check the TAF for trends. The combination of visual assessment and weather data gives you the full picture.

**Step 3 — Identify the Type of Precipitation:** Using visual observation, METAR, and the FAA Weather Intensities Table, determine what you are dealing with: freezing rain,

freezing drizzle, snow, light snow, wet snow, sleet, frost, or freezing fog. Each type of precipitation has different characteristics and requires different fluid considerations.

**Step 4 — Determine Precipitation Intensity:** Using the FAA Weather Intensities Table, classify the precipitation as: HEAVY, MODERATE, LIGHT, or VERY LIGHT. Intensity affects both your fluid type selection and your holdover time. Heavier precipitation degrades fluid protection faster.

**Step 5 — Check Temperature:** Obtain the current Outside Air Temperature (OAT) from the METAR or the aircraft instruments. Temperature is critical — it determines which fluids are effective and which holdover times apply. Some fluids lose effectiveness below certain temperatures.

**Step 6 — Check Available Fluid Types at the Airport:** Contact ground operations or the FBO and ask: what de-ice fluid types are available at this airport? Not every airport stocks all four fluid types. Knowing what is available before you need it avoids surprises. This is also where the golden rule applies — ask the operators who fly here every day what they are using.

**Step 7 — Select the Appropriate Fluid and Procedure:** Based on precipitation type, intensity, temperature, and available fluids, select the appropriate procedure: single-step or two-step, and which fluid types to use. Consult the holdover time tables to determine how much time you have after treatment.

**Step 8 — Coordinate with Ground Crew:** Brief the de-ice crew on what you need: single or two-step, fluid types, areas to be treated. Confirm they will treat all critical surfaces: wings (upper and lower surfaces), horizontal stabilizer, vertical stabilizer, fuselage if required, engine inlets, and sensors (pitot tubes, AOA vanes, static ports).

**Step 9 — Verify Clean Aircraft and Start Holdover Time Clock:** After de-icing is complete, perform the clean aircraft check: visually confirm all critical surfaces are free of contamination. The holdover time clock starts when the LAST fluid application ends — not when de-icing began. Record the start time.

**Step 10 — Takeoff Before Holdover Time Expires:** Monitor conditions during taxi. If precipitation intensifies or holdover time is about to expire before takeoff, return for re-treatment. Never take off after holdover time has expired. When in doubt — get treated again.

**The pre-takeoff contamination check is your last line of defense. Before lining up on the runway, confirm visually (or have the FO confirm) that critical surfaces are still clean. If you cannot see the wings from the flight deck, ask ground crew or ATC for a visual check.**

## De-Ice Fluid Types — Type I, II, III & IV

### Type I — De-Icing Fluid

Color: Orange.

Composition: High concentration of propylene or ethylene glycol mixed with water, heated before application.

Primary use: DE-ICING — removing existing ice, frost, and snow from aircraft surfaces.

Characteristics: Low viscosity, flows off the aircraft quickly, provides minimal holdover time.

Effective temperature range: down to approximately -25°C (-13°F) depending on dilution.

Holdover time: very short — typically 3 to 45 minutes depending on conditions. Essentially provides no anti-ice protection in active precipitation.

When to use: first step of a two-step procedure when contamination is present.

## Type II — Anti-Icing Fluid

Color: Clear or slightly straw/pale yellow.

Composition: Lower concentration of glycol with thickening agents (pseudoplastic fluid).

Primary use: ANTI-ICING — protecting clean surfaces from new contamination.

Characteristics: Thicker than Type I, stays on surfaces during taxi, shears off during takeoff roll due to aerodynamic forces (above approximately 100 kts).

Effective temperature range: down to approximately -25°C (-13°F).

Holdover time: significantly longer than Type I — up to 65 minutes in light snow conditions.

Important: Type II requires a minimum rotation speed to ensure the fluid shears off properly. Check the Aircraft Flight Manual (AFM) for minimum rotation speed requirements when Type II is used.

When to use: second step of a two-step procedure, applied after Type I has cleaned the aircraft.

## Type III — Anti-Icing Fluid

Color: Yellow-green.

Composition: Similar to Type II but with different viscosity properties.

Primary use: ANTI-ICING for commuter/regional aircraft with lower rotation speeds.

Characteristics: Intermediate viscosity between Type I and Type II. Designed for aircraft that rotate at speeds too low for Type II to shear off properly.

Holdover time: similar to Type II.

When to use: anti-icing on commuter turboprops and regional jets with lower V-speeds.

## Type IV — Anti-Icing Fluid

Color: Green.

Composition: Similar to Type II but with improved thickening agents for longer holdover times.

Primary use: ANTI-ICING — provides the longest holdover time of all fluid types.

Characteristics: Very thick, excellent adhesion to surfaces, longest protection window available.

Effective temperature range: down to approximately -25°C (-13°F).




Holdover time: the longest of all types — up to 80+ minutes in light snow conditions.

When to use: second step of two-step procedure when maximum protection time is needed, such as long taxi delays or heavy precipitation expected.

Most commonly used at major hubs and busy airports where taxi times can be long.

**Fluid Type Quick Reference:**

| Type | Color | Primary Use |
|---|---|---|
| Type I | Orange | De-icing (remove contamination) |
| Type II | Clear / Pale Yellow | Anti-icing (longer holdover) |
| Type III | Yellow-Green | Anti-icing (low V-speed aircraft) |
| Type IV | Green | Anti-icing (maximum holdover time) |

## Holdover Time — What It Is and How to Use It

### Definition and Importance

Holdover Time (HOT) is the ESTIMATED time that de-icing/anti-icing fluid will prevent the formation of frost or ice and the accumulation of snow on treated aircraft surfaces.

It is an ESTIMATE — not a guarantee. Actual protection time depends on precipitation rate, temperature, humidity, wind, and the specific fluid used.

Holdover time begins when the LAST fluid application ends.

If holdover time expires before takeoff: the aircraft must be re-inspected. If contamination is found, re-treatment is required.

Holdover time tables are published by the FAA, Transport Canada, and EASA. Always use the most current approved tables.

### Factors That Reduce Holdover Time

Higher precipitation intensity (heavy snow reduces HOT significantly vs. light snow).

Higher precipitation rate (freezing rain degrades fluid faster than light drizzle).

Wind (accelerates fluid degradation).

Higher OAT (paradoxically, warmer temperatures near freezing can reduce HOT in some conditions).

Fluid dilution (Type I applied with too much water loses effectiveness faster).

### Approximate Holdover Times — General Reference

Type I fluid: 1-45 minutes (very condition-dependent, essentially minimal protection in active precip).

Type II fluid: 15-65 minutes (light freezing drizzle to light snow).

Type IV fluid: 20-80+ minutes (light freezing drizzle to light snow).

In FREEZING RAIN: all holdover times are dramatically shorter. Type IV in light freezing rain: approximately 15-30 minutes.

In HEAVY SNOW: Type IV may provide only 20-30 minutes of protection.

Always consult the current official HOT tables for precise values — these are approximate guides only.

**Holdover time tables are GUIDELINES, not guarantees. You are the final authority. If the aircraft looks contaminated before takeoff — regardless of whether the HOT has expired — you do not depart. Period.**

## FAA Weather Intensities Table — How to Use It

The FAA publishes a Weather Intensities Table that helps crews classify precipitation intensity for holdover time selection. It uses visibility and precipitation rate to classify conditions as Very Light, Light, Moderate, or Heavy.

### Precipitation Classification — General Guidelines

VERY LIGHT: Scattered snowflakes or very light drizzle. Visibility greater than 1 mile. Minimal accumulation rate.

LIGHT: Visibility 5/8 to 1 mile in snow. Accumulation less than 1 inch per hour. Light freezing drizzle.

MODERATE: Visibility 5/16 to 5/8 mile in snow. Accumulation 1-3 inches per hour. Moderate freezing rain.

HEAVY: Visibility below 5/16 mile in snow. Accumulation more than 3 inches per hour. Heavy freezing rain.

Always cross-reference the METAR (visibility, precipitation type and intensity codes) with visual observation to classify conditions accurately.

**When in doubt about precipitation intensity — always classify UP. If you cannot decide between light and moderate, use the moderate holdover time. The cost of re-treatment is always less than the cost of taking off with degraded protection.**

## The Golden Rule for Inexperienced Cold-Weather Pilots

If you are new to cold-weather operations — whether it is your first winter season or your first time flying into a snow-covered airport — there is one habit that will serve you better than any manual:

**Find the operators who fly in cold weather every day. Ask them: “What type of de-icing are you doing today? What fluid? What are your holdover times in these conditions?”**

## No-Tower Operations (NONTOWERED Airports)

Operating at airports without an operating control tower is a fundamental skill for professional pilots. Whether you are flying cargo into a rural freight hub, positioning to a maintenance facility, or operating internationally at smaller airports, you must know the procedures, communication requirements, and lighting systems involved.

■ *No-tower operations are frequently asked in cargo and regional airline interviews. Know your CTAF procedures, pilot-controlled lighting steps, and what to do when you cannot obtain an IFR clearance on the ground.*

### Weather Requirements

Before initiating any approach or departure at a nontowered airport, the crew must obtain current official weather and NOTAMs. Acceptable sources include:

- ACARS (Aircraft Communications Addressing and Reporting System)
- FSS — Flight Service Station
- AWOS — Automated Weather Observing System
- ASOS — Automated Surface Observing System

Never assume conditions are suitable based on inflight visual assessment alone. Always obtain official weather before committing to an approach.

### Pilot-Controlled Lighting (PCL)

Many nontowered airports use radio-controlled lighting systems that allow pilots to activate and adjust runway lights from the cockpit by keying the aircraft microphone on the published frequency. All lighting systems at a given airport — whether it has one runway or multiple — operate on the same radio frequency.

**Three intensity levels are available:**

| MICROPHONE CLICKS (within 5 seconds) | LIGHTING FUNCTION |
|---|---|
| 7 clicks | Highest intensity |
| 5 clicks | Medium intensity (or lower REIL / REIL off) |
| 3 clicks | Lowest intensity (or lower REIL / REIL off) |

Lighting remains illuminated for **15 minutes** from the most recent activation. It cannot be extinguished manually before the 15-minute period expires.

■ *Even if the runway lights are already on when you arrive, key the microphone as directed anyway. This resets the 15-minute clock and ensures full lighting duration for your approach and landing.*

Jeppesen approach charts note the type of lighting system associated with each runway, including the frequency used to activate it (e.g., "Activate on 122.8" or "Activate on 122.9 when Tower Not Available").

### Communications at Nontowered Airports — CTAF

At airports without an operating control tower, radio communications between aircraft and visual separation are the primary methods of avoiding traffic conflicts. All pilots must **continuously monitor the CTAF (Common Traffic Advisory Frequency) when within 10 NM of a nontowered airport**.

When approaching or departing a nontowered airport, broadcast your position and intentions as follows:

| FACILITY | FREQUENCY | OUTBOUND (Departing) | INBOUND (Arriving) |
|---|---|---|---|
| No tower / FSS open | CTAF | Before taxi; taking runway; clear of area | 10 NM out; downwind, base, final; clear of runway |
| No tower / FSS closed | CTAF (blind broadcast) | Before taxi; taking runway; clear of area | 10 NM out; downwind, base, final; clear of runway |
| UNICOM (no tower) | Published CTAF (or CTAF blind if unable) | Before taxi; taking runway; clear of area | 10 NM out; downwind, base, final; clear of runway |
| No tower / no FSS / no UNICOM | 122.9 (blind broadcast) | Before taxi; taking runway; clear of area | 10 NM out; downwind, base, final; clear of runway |

**Example inbound position call:**

> *"Cedar Rapids Muni Traffic, [Callsign] 10 miles east, 3,000 feet, VFR arrival straight-in Runway 27, Cedar Rapids Muni."*

### Obtaining a Departure Clearance (IFR)

When departing IFR from an airport without an operating control tower, make every effort to obtain a departure clearance before takeoff. Options include:

- Contact the FSS on the published frequency
- Use a Remote Communication Outlet (RCO) — an FSS with extended range
- Contact Center or Clearance Delivery directly by phone prior to departure

When operating from an airport without a control tower, the clearance should include a **release time** — specifying the earliest time the aircraft may depart. If a void time is received, the crew must be airborne before that time or cancel the clearance and contact ATC before attempting departure again.

### Departing Without an IFR Clearance

If it is not possible to obtain an IFR departure clearance on the ground, a VFR takeoff is permissible **only if ALL of the following conditions are met:**

- The airport does not have operating ATC facilities and it is not otherwise possible to obtain an IFR clearance on the ground
- VFR weather conditions exist at the time of takeoff
- The flight remains in VFR conditions at all times while operating under VFR
- Weather conditions allow sufficient visibility to identify and avoid obstacles, maneuver using external visual references, and maintain minimum altitudes
- An IFR clearance is obtained as soon as practicable after takeoff — but under no circumstances farther than **50 NM** from the departure airport

⚠ **Departing VFR without a clearance is a last resort, not a convenience. Contact Flight Control or Dispatch before departure to ensure they advise ATC of the no-clearance departure. You must get IFR clearance within 50 NM — no exceptions.**

### Postflight — Closing the Flight Plan

When no FAA communication facility exists at the destination airport, it is the **Captain's responsibility** to close the IFR flight plan upon arrival. This can be done by:

- Contacting ATC directly on the appropriate frequency
- Contacting an FAA FSS by radio or phone
- Using Flight Control or Dispatch as a relay to ATC

Failure to close a flight plan will trigger a search and rescue response. This is a crew responsibility — do not assume it was handled automatically.

■■ **Never leave a destination airport without confirming the IFR flight plan is closed. An open flight plan at an uncontrolled airport will result in search and rescue activation.**

### Common Interview Questions — No-Tower Operations

Be ready to answer these in any cargo, regional, or corporate interview:

- What is the CTAF and when do you begin monitoring it?
- How do you activate pilot-controlled lighting? What are the three intensity levels?
- How long does PCL stay on after activation?
- What are your options for obtaining an IFR clearance at a nontowered airport?
- Can you depart IFR without a clearance? Under what conditions?
- What is a void time and what happens if you miss it?
- Who is responsible for closing the IFR flight plan at a nontowered destination?

■ *Interviewers at cargo airlines (UPS, FedEx, Atlas, ATSG) ask about nontowered operations frequently — cargo routes often serve smaller airports without towers. Knowing this cold shows operational awareness beyond the airline environment.*

Experienced cold-weather crews at your destination or departure airport have current, real-world knowledge that no book can fully replicate. They know what is actually working in the conditions on the ground right now. They know which fluid the airport has in stock, which truck is available, and how long the taxi to the runway actually takes.

Asking this question is not a sign of weakness or inexperience. It is a sign of professionalism and good airmanship. Every experienced cold-weather pilot knows that conditions change constantly and that local knowledge is invaluable.

As your cold-weather experience grows, you will develop your own instincts and judgment. Until then — ask. The answer to that one question has saved many flights and many pilots.

### Why This Matters in an Interview

### Common Interview Questions on De-Icing

- What are the types of anti-icing/de-icing fluids and when do you use each one?
- What is holdover time and what factors affect it?
- What is the difference between de-icing and anti-icing?
- Walk me through your de-ice decision flow before departure.
- What would you do if your holdover time expired during a long taxi?
- What precipitation types have NO holdover time? (Answer: volcanic ash, sand/dust — these are not precipitation types that fluids protect against. Also: some very heavy freezing rain conditions exceed fluid capability entirely.)
- What is the clean aircraft concept?
- Where do you find holdover time tables?

**If you come from a warm-weather country and the interviewer asks about cold-weather operations, be honest about your experience level — but show that you know the procedures thoroughly. Knowledge and awareness of your own limitations is a strength, not a weakness.**

# Chapter 8

## Weather & Meteorology

**Weather questions appear in almost every technical interview. Know your METARs and TAFs cold. Be prepared to decode and brief them in real time.**

1. **Read and decode a METAR and TAF.**
2. **What does T00890072 / T01610061 at the end of a METAR mean?**
3. **What does -RA mean on a METAR?**

4. What does RAE mean on a METAR?

5. How do you convert Sea Level Pressure from a METAR?

6. What does it mean when the temperature and dew point are the same?

7. What does P stand for in the remarks section of a METAR? (e.g., P0005)

8. What do the terms BR and HZ mean in a METAR?

9. What does TSE051 mean in a METAR?

10. What does ALQDS mean on a METAR?

11. What does the $ sign at the end of a METAR mean?

12. How many times a day are TAFs issued?

13. What is the difference between TEMPO and BECMG on a TAF?

14. What is the forecast area for a TAF? What is its valid time?

15. What is LLWS? What is your windshear escape maneuver?

16. What does the D in D-ATIS stand for?

17. What side of a thunderstorm do you want to fly around? How far away should you remain?

18. What is the difference between a windshear caution vs. a warning?

19. What would you take into consideration with windshear advisories?

20. What is the jet stream and how do you tell its velocity on a weather chart?

21. What is an occlusion front and what are the three types?

22. What is the ITCZ?

23. What is the Coriolis force and how does it affect wind in each hemisphere?

24. In the northern hemisphere, how does wind circulate around a high-pressure system?

25. What is an isobar / isotherm / isotach? What are isopleths?

26. What weather will you expect after the TEMPO ends in a TAF?

27. What are the types of precipitation that have no holdover time?

28. What are the two types of anti-ice fluid?
29. How long after a pre-takeoff contamination check do you have?
30. Define icing conditions.
31. What indications would you see if you flew through volcanic ash?
32. What is the number in a rectangular box on a high-level weather chart?
33. What is the dark line with an arrow on a high-level weather chart?
34. Decode a METAR with low visibility and precipitation.
35. What is the difference between a SIGMET and an AIRMET? Give examples of each.

### ❄■ De-Icing & Anti-Icing

36. What are the four types of de-ice/anti-ice fluids? What color is each one and when do you use them?
37. What is holdover time? What factors affect it?
38. Walk me through your complete de-ice decision flow before departure.
39. What is the pre-takeoff contamination check? When must it be performed and who is responsible?
40. What is the clean aircraft concept and why is it critical?
41. What precipitation types have no holdover time protection?
42. What is the two-step de-ice procedure and when is it required?
43. How do you classify precipitation intensity for de-ice purposes? What are the four categories?
44. Why does Type II fluid require a minimum rotation speed? What happens if you rotate too slowly?
45. Where do you find the holdover time tables and what authority publishes them?
46. Can you take off with frost on the wings if it is very thin and smooth?
47. What critical surfaces must be treated during a de-ice procedure?
48. When does the holdover time clock start — when de-icing begins or when it ends?

# Chapter 9

## Maps, Plates, Regulations & Airspace

**Know your plates inside and out. Be ready to brief an ILS approach, a departure, and a STAR. Regulatory knowledge shows you are professional and current.**

### Maps, Plates & Airport Markings

1. **Brief an ILS approach plate.**
2. **Brief a departure plate and its symbology.**
3. **What weather do we need to fly to this airport?**
4. **What is the symbol for the highest obstacle on a chart?**
5. **What is the MSA on this approach plate? What range does it cover?**
6. **What is TCH on a chart?**
7. **When do you go missed on an ILS approach?**
8. **What happens when the glide slope goes out on an ILS approach?**
9. **What does the D next to the localizer frequency on a chart indicate?**
10. **How is the final approach fix defined on an ILS?**
11. **Brief an ILS or LOC approach when glide slope is out of service.**
12. **How would you fly a PRM approach differently?**
13. **When circling, what are your circling radius limitations for Cat D minimums?**
14. **What is the difference between Minimum Safe Altitude and Minimum Sector Altitude?**
15. **What are the hollow arrows containing numbers at the beginning of an approach?**
16. **What speed/altitude do you set when cleared for a STAR?**
17. **What does the Maltese cross mean on an approach plate?**
18. **DME fixes — how do we identify them?**




19. When can you descend below your DA?

20. What are the pink dots / pink circle on the taxi chart / airport diagram?

21. What does the star with a circle on an airport diagram mean?

22. What does ARP mean on an airport diagram?

23. What does a hot spot look like on a Jepp taxi chart?

24. What does a displaced threshold look like on a Jepp chart?

25. Fly-over vs. fly-by waypoints — what are the symbols on a SID?

26. Where can you find the lost comm procedure on the arrival plate?

27. What is the distance between runway centerline markings and edge light spacing?

28. What are lines of latitude / longitude?

29. What is the prime meridian?

30. Are METAR winds magnetic or true?

## Regulations & Fuel Requirements

1. What weather is required at an alternate?

2. When are you required to file a destination alternate airport?

3. What are the requirements for a departure alternate?

4. When would you need a takeoff alternate airport?

5. What is CDL? What is the difference between a CDL and MEL?

6. What difference does it make if visibility goes below minimums inside/outside the FAF?

7. Can you descend below DA with only the approach lights visible?

8. What conditions are required to be met to descend below MDA/DA?

9. What are standard takeoff minimums?

10. When doing an approach if you see the approach lights at DA, how much lower can you go?




11. What is the maximum holding airspeed at 6,000 ft? At 14,000 ft? Above 14,000 ft?

12. A TAF shows exactly 2,000 ft ceilings — would you need an alternate?

13. Can you shoot an approach if it goes below minimums before and after the FAF?

14. You are holding at 16,000 ft — what is the max holding airspeed and leg length?

15. If ATC says “descend via,” what does that mean?

16. When do you need a second alternate?

17. What is your airspeed as you pass below 10,000 feet?

18. What are company-derived minimums?

19. What is required reporting to ATC when entering a hold?

20. If RVR and visibility are both reported, which is controlling for commencing an approach?

21. What are the three types of entries for holding? What are the three types of holding exits?

22. What is RVR? Where is it measured?

23. What are Part 121 international and domestic fuel requirements?

24. What are Part 91 and Part 121 VFR minimums for takeoff?

25. At what altitude do you switch from indicated airspeed to Mach? Why?

26. What is ETOPS? What does it mean operationally?

27. What is RVSM airspace and what equipment is required?

28. What are the oxygen requirements for crew and passengers?

29. What altitude must one pilot wear an O2 mask at all times?

30. What are lost comm procedures?

31. What are the three types of holding entries?

32. DME Arc — if you fly a 15-mile arc for 60 degrees, how many miles have you traveled?




**33.** **Are Cat II or III minimums used in determining whether you need a takeoff alternate?**

**34.** **What RVR do low visibility takeoffs begin?**

**35.** **Are we ever allowed to take off below charted takeoff minimums?**

**36.** **What is the additional safety margin factored into landing data for a clean and contaminated runway?**

# Chapter 10

## Scenario-Based & Problem Solving

**These questions test your decision-making under pressure. Always prioritize: Aviate, Navigate, Communicate. Think out loud. Show your reasoning process — they want to see how you think, not just the answer.**

### Aerodynamics & Performance

1. **List your airplane's V-speeds.**
2. **What is TALPA?**
3. **What is the difference between Vx and Vy?**
4. **How do you calculate BINGO fuel?**
5. **What is V1? How does it change with runway condition?**
6. **What is V2? Can VMC and V2 be lower than V1? Can V1 be higher than VMCA?**
7. **What is Vmcg?**
8. **What is the touchdown zone defined as?**
9. **Does a contaminated runway affect your V speeds?**
10. **What conditions would reduce the maximum takeoff weight?**
11. **How would a removed landing gear door affect performance before the flight?**
12. **How far out do you need to start your descent from 30,000 ft to 10,000 ft on a 3-degree glideslope?**
13. **Can you calculate top of descent from 33,000 ft? (3:1 rule)**
14. **What are the segments of climb? Define 1st through 4th segments and their gradients.**
15. **What is drift down speed?**
16. **What is wake turbulence?**
17. **How could you burn extra fuel en route to avoid max landing weight?**
18. **What factors affect our landing distances?**




19. How do you calculate actual landing distance (ALD)? How are factored landings calculated?

20. Does density altitude change V1?

21. What is a balanced airfield?

22. What is screen height with engine failure at V1?

23. What is the minimum height you need to clear the runway if you have a V1 cut?

24. What is the additional safety margin for a wet/contaminated runway?

## Emergency Scenarios

1. What would you do if you experienced an engine failure on short final?

2. What would you do if you receive a TCAS RA to climb but you're at your max altitude?

3. What are your actions if you lost an engine right at V1?

4. What would you do if you experienced windshear right after departure? On landing?

5. What would you do if after takeoff the FO and CA attitude indicators disagree?

6. What is the first thing you would do if the airplane depressurized at 30,000 ft?

7. Walk me through the procedure for rapid decompression.

8. What would you do if you experience cabin depressurization at 34,000 ft?

9. What would you do if you lost GPS signal after the last fix on a STAR?

10. What would you do if flying at FL370 and the autopilot disengages and won't reengage?

11. What would you do if at 800 ft on an ILS your engine starts on fire?

12. What would you do if the flight deck starts smoking during the takeoff roll?

13. What is the go-around profile?

14. If your mains touch down during a go-around, can you continue to execute one?

15. What would you do if you had a depressurization between two cities?

16. **Walk me through a rejected takeoff. What are RTO criteria?**

17. **What would you do if you experienced windshear — escape procedure?**

18. **What do you do for a compressor stall?**

19. **What would you do if the METAR is reporting freezing rain?**

20. **Walk me through what happens after you turn the battery switch on.**

21. **What would you do if you got a cabin door unlocked fault over the Rockies?**

22. **What would you do if flying over the Rockies and your plane pitches up violently?**

23. **What would you do if tower asked you to depart from an intersection?**

24. **What would you do if fire detection says there is a fire but you have no other signs?**

25. **What would you do if weather deteriorates 45 min out from landing and you don't have an alternate filed?**

26. **What would you do if you experienced an engine failure 3 kts before V1?**

27. **What would you do if your FO calls runway in sight but you don't see it?**

28. **What is your CFIT escape procedure?**

29. **What is your windshear escape maneuver? When do you initiate it?**

30. **Explain what you would do with an engine failure on a two-engine aircraft.**

31. **What would you do if you lost an engine at 500 ft on a visual approach?**

32. **Coming in on approach below 200 ft — tower says go around. What do you do?**

33. **If you experience an engine failure at 1,500 ft with good weather — continue or go around?**

34. **What are your considerations when operating out of high altitude airports?**

35. **How does high density altitude, short runways, or high obstacles affect how much fuel you take?**

36. **How would you handle an emergency descent over the Rockies?**

37. **How would you manage an emergency as it relates to CRM?**

38. What would you do if en-route requesting weather deviation but ATC wasn't responding?

39. What would you do if you noticed missing static wicks on your walk-around?

40. What would you do if the weather was below minimums and the airplane was on fire?

## Operational Scenarios

1. You and your captain show up and the only lavatory is on MEL. What do you do?

2. What would you do if the Captain is flying an ILS and approaching full-scale deflection low on the GS and does not respond?

3. What would you do if hitting minimums on the approach and the Captain says continue?

4. What would you do if the glide slope goes out on a visual approach?

5. When can you descend if you're cleared for the approach but still 100 nm out?

6. What would you do if the FA told you a passenger stated the lavatory was broken?

7. What is the first climb segment?

8. What would you do if the CA wanted to fly the plane with a broken armrest?

9. How would you be an asset for the company?

10. What is the captain's role in customer service?

11. Would you accept a clearance to maintain 180 kts to a 4 nm final?

12. What is the max speed for a procedure turn?

13. What is the runway environment?

14. What is the minimum fuel requirement for an IFR flight?

15. How does an ILS work?

16. Walk me through a runway change after pushing back from the gate.

17. What is the captain's procedure during an RTO?

18. Hydroplaning formula?




19. What do you need to see to legally land once at 100 ft above TDZE?

20. How do engine anti-ice / wing anti-ice systems work?

21. How do you start an engine with an inoperative APU?

22. What anti/de-icing systems are on your current aircraft?

23. What is required equipment for RVSM airspace?

24. What is the runway lighting from the last 2,000 ft remaining?

25. What color are taxiway lights?

26. What is the difference between green and blue taxiway lights?

27. What does CTAF stand for?

28. What does ATIS stand for?

29. What is the reciprocal of a 067 course?

31. What are NATs? What is SLOP?

32. For an oceanic flight, besides an ATC clearance are any other clearances required?

33. What does LAHSO mean? What is the landing distance for LAHSO? Why would you decline a LAHSO?

34. What would you do if you were holding short and realized you didn't have enough fuel?

35. What speeds are associated with thrust reverser usage? When do you use thrust reversers?

36. What is the definition of transition altitude and transition level?

37. What is the difference between runway centerline markings and edge light spacing?

38. What are Category 1 and Category 2 airspace?

39. How many degrees of runway difference constitutes a circling procedure?

40. If the FO and CA airspeed indicators were off by 20 kts, how would you decide which to follow?

41. **Is VV a ceiling?**

42. **Give a generic takeoff briefing.**

43. **Brief an ILS approach.**

44. **Give an example position report over the NATs / on HF.**

45. **What is spatial disorientation?**

46. **What threats would you discuss before taking off from a high altitude airport?**

47. **What are your crew rest requirements?**

48. **What is a stable / stabilized approach?**

49. **Describe windshear escape procedures.**

50. **What is the critical engine?**

52. **What is SLOP/NAT — when do you have to be back on course? Do we need permission?**

53. **How would you brief a taxi?**

54. **How is an FMS loaded differently for a transoceanic flight vs. a domestic flight?**

55. **What are your company minimums for lower than standard takeoff minimums?**

56. **Can we accept an RNAV STAR? How do we load it?**

57. **What vertical mode do you use to descend to MDA?**

58. **What altitude do you set in the box on an LNAV approach?**

59. **When would you descend to the runway on an LNAV approach?**

60. **What are the procedures for loading an ETOPS flight plan?**

61. **As PM, where do you keep your hands and feet on takeoff and landing?**

62. **What do you do if your holdover time ends?**

# Chapter 11

## Key Words

The words airline interviewers listen for.

Interviewers listen for these words, but they want real examples, not a memorized list. For each key word, prepare a short story from your own career: the situation, what you did, and the result. Use the words naturally, in your own voice.

### Competencies (CRM / Soft Skills)

| Key word | What it means and how to show it |
|---|---|
| **Crew Resource Management (CRM)** | Using every resource (people, information, equipment) to operate safely. Show it with a story where the team made a better decision together. |
| **Threat and Error Management (TEM)** | Identifying threats, preventing and trapping errors, and managing undesired states before they become unsafe. |
| **Situational awareness** | Knowing what is happening around you and the aircraft, and anticipating what comes next. |
| **Decision-making / ADM (Aeronautical Decision Making)** | A structured process: gather information, assess the risk, decide, act, and then evaluate the result. |
| **Workload management** | Prioritizing tasks, delegating, and avoiding task saturation, especially in high-workload phases of flight. |
| **Communication / Assertiveness** | Clear, concise, closed-loop communication, and speaking up respectfully when something is not right. |
| **Leadership / Followership** | Setting the tone as captain, and supporting the captain and the plan as first officer. |
| **Conflict resolution** | Handling disagreements calmly, focusing on facts and safety instead of personalities. |
| **Professionalism / Integrity** | Doing the right thing consistently, even when nobody is watching. |
| **Accountability / Ownership** | Owning your mistakes, reporting them, and sharing what you learned. |




*Figure: the ten competencies, around one goal.*

## Safety & Culture

| Key word | What it means and how to show it |
|---|---|
| **Safety culture** | Shared attitudes and values where safety comes first, reporting is encouraged and everyone owns safety. |
| **Risk management / Risk mitigation** | Identifying hazards, assessing likelihood and severity, and applying controls to reduce the risk. |
| **SMS (Safety Management System)** | A company-wide, proactive system built on four components: safety policy, safety risk management, safety assurance and safety promotion. |
| **Just culture** | People are not punished for honest mistakes, but willful violations and reckless behavior are not tolerated. |
| **Standard Operating Procedures (SOPs)** | Standardized procedures that make the crew predictable and reduce errors. |
| **Compliance / Adherence to procedures** | Following procedures, regulations and checklists as written, and communicating any deviation and why. |
| **Go-around / Stabilized approach** | Meeting the stabilized approach criteria, and going around without hesitation when they are not met. A go-around is a normal, safe maneuver. |
| **Aeronautical judgment** | Sound decisions that combine knowledge, experience and risk assessment. |

## Technical (Technical Interview)

| Key word | What it means and how to show it |
|---|---|
| **Systems knowledge** | Hydraulic, electrical, pneumatic, fuel, flight controls, FMS and automation. Know how each system works, its normal operation and its failures. |

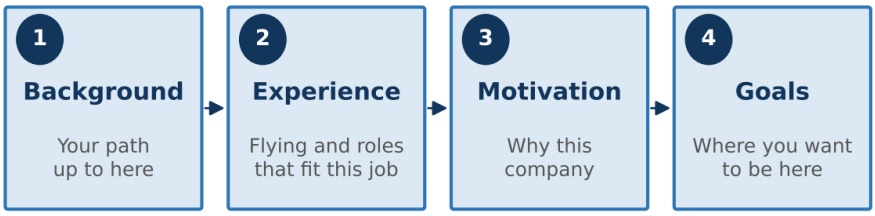
## Character & Personal Qualities

| Key word | What it means and how to show it |
|---|---|
| **Integrity / Honesty** | Telling the truth, including about your own mistakes. Never exaggerate your experience. |
| **Humility** | Recognizing what you do not know, learning from others and giving credit to the team. |
| **Resilience** | Recovering from setbacks, such as a failed checkride or a career interruption, and growing from them. |
| **Adaptability** | Adjusting to new aircraft, new crews, new procedures and changing conditions. |
| **Discipline** | Following the plan, the procedures and the standards, day after day. |
| **Curiosity / Continuous learning** | Always studying, asking questions and keeping your knowledge current. |
| **Teamwork / Collaboration** | Putting the crew and the operation before your own ego. |
| **Leadership by example** | Showing the standards through your own behavior, in the cockpit and outside it. |
| **Calm under pressure** | Staying composed, thinking clearly and communicating well in stressful situations. |
| **Self-awareness** | Knowing your strengths and limits, including fatigue and stress, and acting on them. |
| **Respect / Empathy** | Treating every person, from crew to ground staff to passengers, with respect and understanding. |
| **Reliability / Dependability** | Showing up prepared and on time, and doing what you said you would do. |
| **Work ethic** | Putting in the effort and preparation, with a track record that proves it. |
| **Passion for aviation** | A genuine, specific love of flying and of the profession, shown through your story. |
| **Commitment / Loyalty** | Wanting to build a long career with the company and the people in it. |
| **Positive attitude** | Constructive, solution-oriented and respectful, even in difficult moments. |

## "Tell Me About Yourself"

Build the answer in four parts, in about one to two minutes:

1. Background: your path up to here.
2. Relevant experience: the flying and the roles that matter for this job.
3. Motivation: why this company.
4. Long-term goals: where you want to be in the company.




*About 1 to 2 minutes in total*

*Figure: the four parts of the answer.*

## "Why This Airline?"

Use specific facts about the company, not generic praise. Cover these four points:

1. Company culture and values.
2. Safety record.
3. Growth opportunities.
4. Fleet and type of operation.

# Chapter 12

## Chart Reading Easy

In this section you see the chart and then every question an examiner could ask about it, with the answers. Read the chart first, then cover the answers and test yourself.

The charts in this section are FAA charts for Los Angeles International (KLAX):

- Airport Diagram
- KYLOW ONE RNAV Departure
- ILS or LOC Rwy 24L

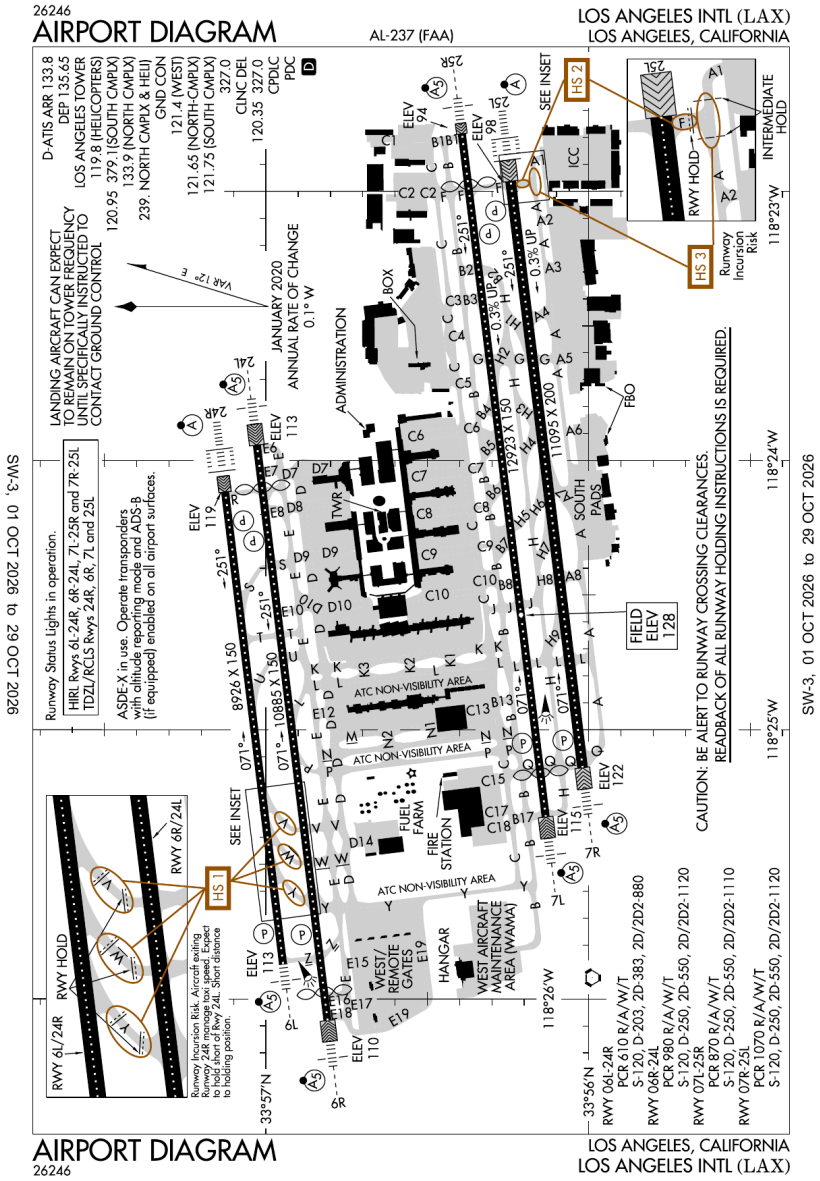
**Profile-view symbols you will meet on the charts**

| | | |
|---|---|---|
| 4000 | **Underlined altitude** | At or above this altitude (minimum) |
| 4000 | **Overlined altitude** | At or below this altitude (maximum) |
| 4000 | **Over + underlined** | Exactly this altitude (mandatory) |
| ✖ | **Maltese cross** | FAF on a precision approach |
| V | **V on the profile** | Visual Descent Point (VDP) |
|  | **Glide slope line** | Intercept altitude, e.g. GS 2200' |

*Figure: symbols on the profile view.*

*Charts are for study and interview preparation only. The charts are FAA charts and are public domain. Always use the current chart for real operations.*

# Los Angeles (KLAX) – Airport Diagram




*FAA chart (public domain), cycle 01 OCT to 29 OCT 2026. For study only; not for navigation.*

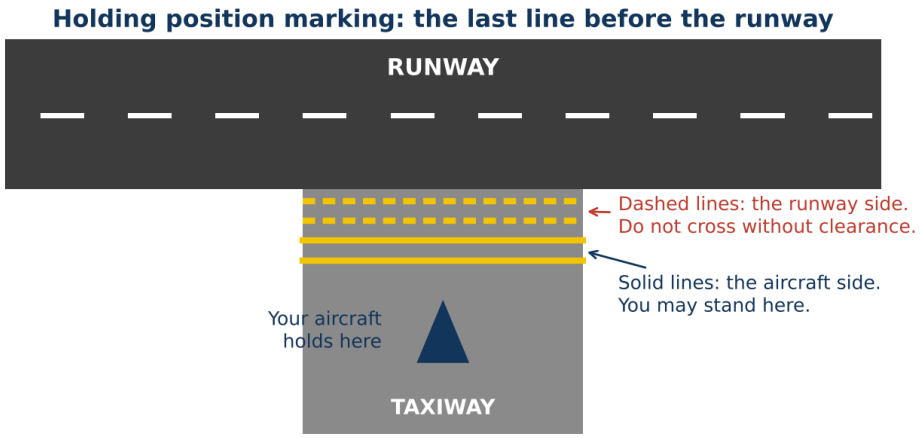



*Figure: the holding position marking.*

## Section 1 – Airport Identification & Basics

### Q1. What airport is this chart for, and what are its identifiers?

Los Angeles International. ICAO is KLAX and IATA is LAX. The title is "Airport Diagram", FAA chart ID 26246.

### Q2. What is the airport elevation?

128 ft. The chart shows it in the box "FIELD ELEV 128" on the middle of the diagram.

### Q3. What is the magnetic variation?

12° East, with an annual rate of change of 0.1° West (January 2020).

### Q4. What do the numbers on the border of the chart show?

The latitude and longitude grid: 33°57'N and 33°56'N, and 118°23'W, 118°24'W, 118°25'W and 118°26'W. They help you place a position on the airport.

### Q5. What do the cycle dates "SW-3, 01 OCT 2026 to 29 OCT 2026" mean?

The period when this printed chart is current. Always confirm that you have the current chart in your EFB or database before the flight.

## Section 2 – Runways

### Q6. How many runways does LAX have, and how are they arranged?

Four parallel runways, in two complexes:

- North complex: 6L/24R and 6R/24L
- South complex: 7L/25R and 7R/25L

### Q7. What are the runway lengths and widths?

All four runways are 150 ft wide:

- 6L/24R: 8,926 ft (2,721 m)
- 6R/24L: 10,885 ft (3,318 m)
- 7L/25R: 12,923 ft (3,939 m)
- 7R/25L: 11,095 ft (3,382 m)

### Q8. What are the runway headings?

All four runways have a magnetic heading of 071° / 251°.

**Q9. Which is the longest runway, and why does that matter?**

7L/25R at 12,923 ft. It is the best choice for heavy aircraft at high takeoff weights or for contaminated or reduced-braking conditions.

**Q10. Which is the shortest runway?**

6L/24R at 8,926 ft. It is still long, but check performance for heavy international departures.

**Q11. Convert each runway length to meters and confirm.**

8,926 ft = 2,721 m, 10,885 ft = 3,318 m, 11,095 ft = 3,382 m, 12,923 ft = 3,939 m.

**Q12. What is the difference between 6L and 6R, and between 7L and 7R?**

They are parallel runways, left and right as seen on the runway heading. The two 6s are the north complex (6L/24R, 6R/24L), and the two 7s are the south complex (7L/25R, 7R/25L).

**Q13. What is the true heading of Runway 7L/25R?**

Magnetic 071° plus variation 12°E gives true 083°. Rule: "East is least, West is best". Going from true to magnetic you subtract East variation, so from magnetic to true you add it.

**Q14. What is the magnetic heading when landing on 24R? What is the reciprocal?**

251° magnetic. The reciprocal is 071°.

**Q15. What is the length ratio between the longest and shortest runway?**

12,923 ÷ 8,926 ≈ 1.45, so the longest is about 45% longer. The difference is 3,997 ft (about 1,218 m).

**Q16. Which runway would you expect for a heavy international departure at maximum takeoff weight?**

The longest: 7L/25R at 12,923 ft. The final choice always depends on ATC, wind, noise-abatement procedures and performance calculations.

**Q17. How does the length of 6L/24R affect performance planning for a widebody?**

At 8,926 ft it has less margin. Compute takeoff distance, accelerate-stop and landing distance for the day's weight, temperature and runway condition before accepting it.

**Q18. What does the table with PCR numbers at the bottom right show?**

The pavement strength of each runway. PCR is the Pavement Classification Rating, followed by the pavement type and the allowable gear loads for single (S), dual (D), dual tandem (2D) and double dual tandem (2D/2D2) gear. The chart lists PCR 610 for 6L-24R, 980 for 6R-24L, 870 for 7L-25R and 1070 for 7R-25L.

**Q19. Why does the PCR matter to a pilot?**

Your aircraft's pavement rating (ACR) must not exceed the runway PCR, or the operator must have approval to use the runway. Check it for heavy aircraft and for runways with weight limits in the chart or NOTAMs.

## Section 3 – Communications

**Q20. What are the D-ATIS frequencies?**

Arrival 133.8 and departure 135.65.

**Q21. What is the Clearance Delivery frequency, and is there a data comm option?**

120.35 (UHF 327.0). The chart also shows PDC and CPDLC, so you can request the departure clearance by data link.

**Q22. What are the Ground frequencies?**

UHF 327.0 for all:

- West: 121.4
- North Complex: 121.65
- South Complex: 121.75

**Q23. What are the Tower frequencies?**

- North Complex: 133.9
- South Complex: 120.95 (UHF 379.1)
- Helicopters: 119.8

**Q24. What does "Landing aircraft can expect to remain on Tower frequency until specifically instructed to contact Ground Control" mean?**

After landing, do not switch to Ground by yourself. Stay on the Tower frequency until ATC tells you to contact Ground.

**Q25. Which Tower do you call if you land on 24R, and which if you land on 25L?**

24R is in the North Complex, so Tower 133.9. 25L is in the South Complex, so Tower 120.95.

**Q26. Which Ground do you call if you are parked in the West/Remote area?**

Ground West, 121.4. Confirm on the day, since ATC can assign a different one.

**Q27. What is PDC? What is DCL?**

PDC is Pre-Departure Clearance, delivered by text. DCL is Departure Clearance via Data Link (CPDLC), delivered through Data Comm.

**Q28. When would you use Data Comm instead of voice for the clearance?**

When the operator is equipped and the airport supports it. It reduces frequency congestion and read-back errors, and it is standard in most airline operations.

**Q29. What is D-ATIS, and what do you do with it?**

Digital ATIS, delivered by text or voice. Copy the information letter, check the runway in use, NOTAMs and weather, and tell ATC the current letter on initial contact.

**Q30. Why does LAX have separate North and South frequencies?**

The airport has two parallel runway complexes working independently. Splitting the frequencies keeps the workload manageable for ATC and reduces frequency congestion.

## Section 4 – Hot Spots (Runway Incursion Risk)

**Q31. What is a hot spot?**

A location on an aerodrome movement area with a history of potential risk of collision or runway incursion, where heightened attention by pilots and drivers is necessary.

**Q32. How many hot spots are shown on the chart, and how are they marked?**

Three: HS 1, HS 2 and HS 3. Each is a brown box with a brown line to an oval on the diagram, and an inset gives the detail.

**Q33. What does HS 1 say?**

"Runway Incursion Risk. Aircraft exiting Runway 24R manage taxi speed. Expect to hold short of Rwy 24L. Short distance to holding position." The ovals mark Taxiways V, W and Y between 6L/24R and 6R/24L.

**Q34. What is the hazard at HS 1?**

The two north-complex runways are close together. After landing 24R, the exit taxiways V, W and Y lead straight to the hold short of 24L, with little distance in between.

**Q35. What do you do when you exit Runway 24R at V, W or Y?**

Slow down on the exit, expect to hold short of Runway 24L, and do not cross without a clear instruction. Read back the hold short.

**Q36. Where are HS 2 and HS 3?**

At the east end of the south complex near Runway 25L, around Taxiways F, A, A1 and A2. The inset for HS 2 shows Runway 25L, Taxiway F and the runway holding position, and an intermediate hold on Taxiway A. HS 3 is marked "Runway Incursion Risk".

**Q37. What does the HS 2 inset tell you about Taxiway F?**

Taxiway F meets Runway 25L, and the inset marks the runway holding position (RWY HOLD) on it. You cannot cross it without ATC clearance.

**Q38. What does "For HS detail see inset" mean?**

The area is too complicated for the main diagram, so an enlarged inset shows the details.

**Q39. What do the "Rwy Hold" and "Intermediate Hold" markings represent?**

- Rwy Hold: the runway holding position, which you cannot cross without ATC clearance.
- Intermediate Hold: a holding position on a taxiway, used to protect a crossing taxiway or aircraft. It is also not to be crossed without ATC clearance.

**Q40. What does the caution at the bottom of the chart say?**

"Caution: be alert to runway crossing clearances. Readback of all runway holding instructions is required."

**Q41. Why is the readback of runway holding instructions required?**

A missed or misunderstood hold short is a common cause of runway incursions. The readback lets ATC confirm that you understood.

## Section 5 – Notes & Airport Features

**Q42. What does "Runway Status Lights in operation" mean?**

Red lights in the pavement, on the runways and taxiways (such as the runway entrance lights), warn you when it is unsafe to enter, cross or take off. Do not enter, cross or take off when you see red lights, and tell ATC.

**Q43. What lighting is listed for the runways?**

HIRL (high intensity runway lights) on 6L-24R, 6R-24L, 7L-25R and 7R-25L. TDZL (touchdown zone lights) and RCLS (runway centerline lights) on Runways 24R, 6R, 7L and 25L.

**Q44. What does the ASDE-X note say?**

"ASDE-X in use. Operate transponders with altitude reporting mode and ADS-B (if equipped) enabled on all airport surfaces." ATC surface radar uses your transponder to see you, so keep it on while taxiing.

**Q45. What are the "ATC Non-Visibility Areas"?**

Areas of the airport that cannot be seen from the control tower. ATC may not see you there, so ask or report if you are unsure, and do not rely on ATC to see you.

**Q46. What are the "Fire Station", "Fuel Farm", "FBO" and "Hangar" labels for?**

Fire Station is the airport rescue and firefighting base. Fuel Farm is the fuel storage area. FBO is the fixed-base operator for general aviation services. Hangar is a maintenance hangar.

**Q47. What is WAMA?**

West Aircraft Maintenance Area, the maintenance region on the west side, next to the West/Remote Gates.

**Q48. What are the "West/Remote Gates"?**

The gates on the west side of the field, near Taxiways E15 to E19, served by Ground West (121.4).

**Q49. What is the tower symbol (TWR) on the diagram?**

The control tower, in the middle of the terminals. The ATC non-visibility areas are the places the tower cannot see.

**Q50. What do the hatched chevron areas at the runway ends mean?**

They are paved areas before the runway end that are not usable as runway (such as blast pad or stopway areas). They are not included in the runway length.

## Section 6 – Taxiways & Surface Navigation

**Q51. How are the taxiways named at LAX, and why does it matter?**

They use letters with numbers for connectors, for example A1–A8, B1–B8, C1–C18, D7–D14, E6–E19, H1–H9 and K1–K3. Know the naming before you taxi, since the airport is complex and mistakes lead to incursions.

**Q52. What should you do when you are unsure about the taxi route?**

Stop, tell ATC you are unsure, and request progressive taxi or clarification. Do not guess.

**Q53. What is "progressive taxi" and when would you ask for it?**

ATC gives step-by-step instructions to the destination. Ask for it at a complex or unfamiliar airport, which describes LAX for many crews.

**Q54. What is the movement area vs the ramp area?**

The movement area is controlled by ATC (runways, taxiways), and the ramp is controlled by the airline or airport operations. At LAX some ramp areas have separate procedures.

**Q55. What is the difference between a taxiway and a taxilane?**

A taxiway is a defined path on the movement area, controlled by ATC. A taxilane is a path within a ramp or parking area, used for access to gates or parking positions, and it is usually not controlled by ATC.

**Q56. Where are Taxiways V, W and Y?**

They are the three exits between the north-complex runways 6L/24R and 6R/24L, at HS 1, near the west end of the runways.

## Section 7 – Scenarios & Phraseology

**Q57. You land Runway 24R and ATC gives no further instructions. What do you do?**

Slow down on the exit and expect to hold short of Runway 24L. Do not cross without a clear instruction, and read back every hold short.

**Q58. You are on Twy A and ATC says "turn left on Foxtrot." Can you cross Runway 25L?**

No. The chart shows a runway holding position on Taxiway F at Runway 25L (HS 2). Do not cross it without an explicit clearance.

**Q59. You are cleared onto a taxiway but see the intermediate holding position markings. What do you do?**

Hold at the intermediate markings until ATC clears you to continue.

**Q60. ATC says: "Taxi to runway 25L via A, F." What is your read-back?**

Read back the route and the hold short: "Taxi to runway 25L via Alpha, Foxtrot, hold short of runway 25L, [callsign]." Add the hold-short even if ATC did not say it, if you are not cleared to cross.

**Q61. ATC says "Cross runway 24L." What do you do before crossing?**

Check both directions for traffic (visually and with the airport moving map if equipped), confirm the clearance with the other pilot, read back "crossing 24L," then cross promptly.

**Q62. You vacate 24R and ATC is silent. What is the correct action?**

Stop before 24L unless cleared. Do not assume the next clearance. Call ATC if needed.

**Q63. You get a conflicting clearance from ATC and the chart. What do you follow?**

ATC clearance. The hot spot information on the chart is advisory. If something is unsafe or unclear, stop and clarify.

**Q64. How do you brief LAX taxi before pushback?**

- Review the diagram and hot spots HS 1, HS 2 and HS 3.
- Identify the departure runway and the likely taxi route.
- Note hold shorts and crossings.
- Check the NOTAMs for construction and closures.
- Agree on who is heads-down and who is heads-up.

**Q65. How would you handle a runway incursion alert?**

Stop immediately if you are on the ground and tell ATC. If airborne on approach, be ready for a go-around. Follow company procedures and report afterwards.

**Q66. What are the key tools for situational awareness on the ground?**

The airport diagram, the moving map (if equipped), both pilots outside, ATC instructions written down and read back, and a sterile cockpit during taxi.

**Q67. Why is taxiing at LAX considered high risk?**

Parallel runways close together, short distances between exits and hold lines, many taxiways, heavy traffic, construction and large aircraft.

**Q68. What is a "sterile cockpit" during taxi?**

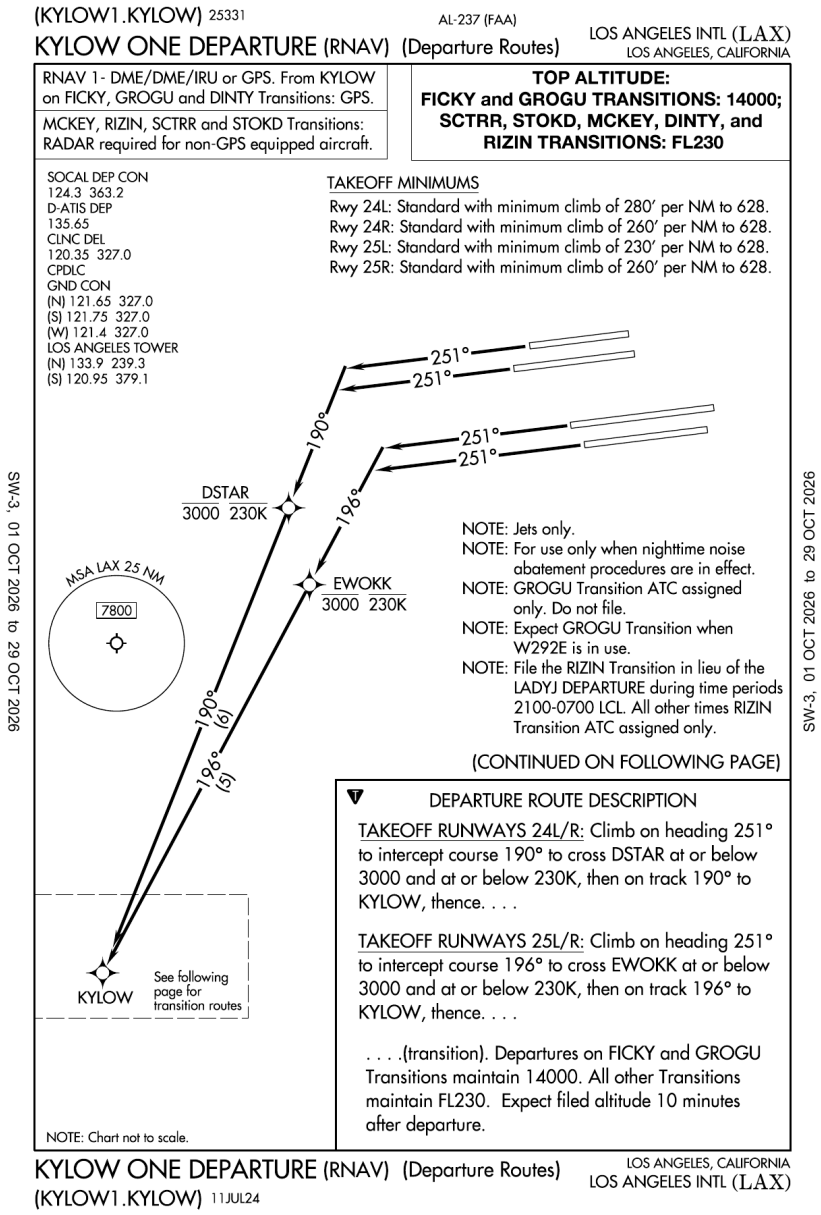
A rule limiting non-essential conversation and tasks during critical phases of flight, which includes taxi and flight below 10,000 ft. It reduces distractions.

**Q69. What is your best defense against runway incursions at LAX?**

- Brief the hot spots before taxi.
- Keep both pilots' eyes outside.
- Read back every hold short and runway crossing.
- Use the airport diagram during the entire taxi.

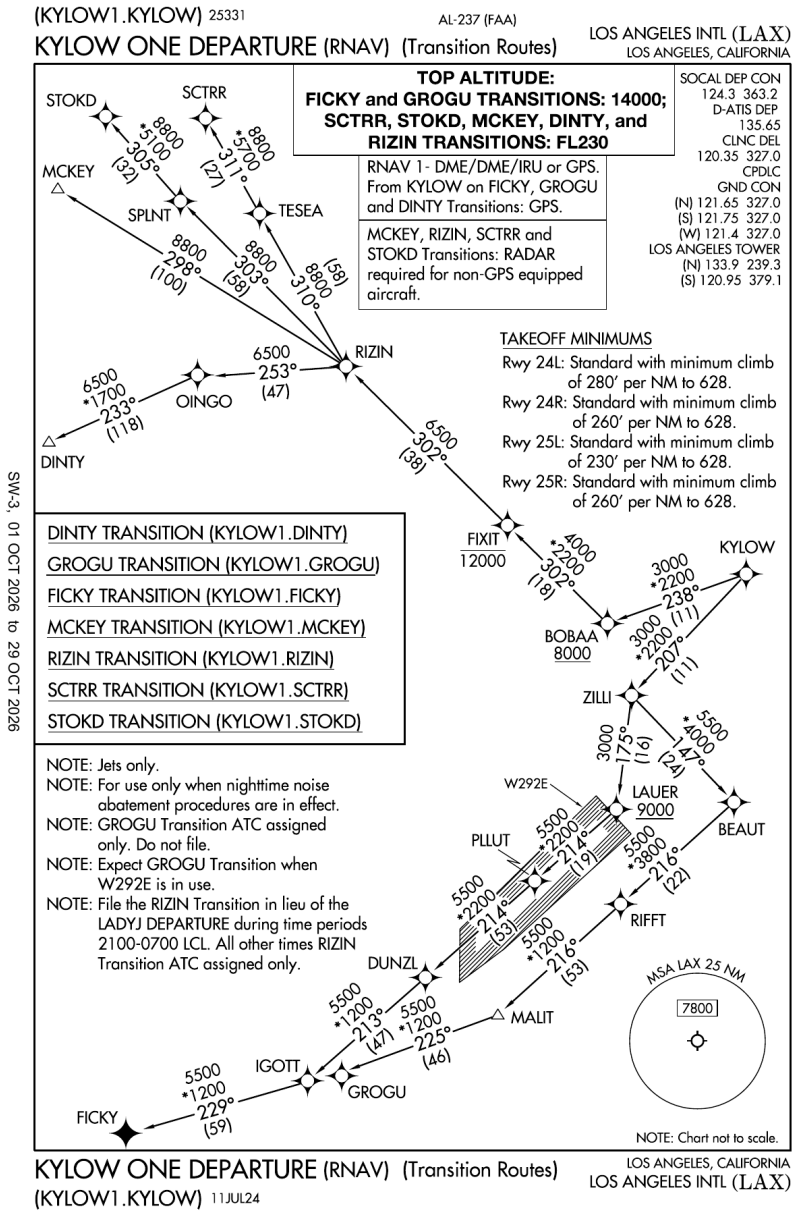
- If in doubt, stop and ask.

# Los Angeles (KLAX) – KYLOW ONE RNAV Departure (Departure Routes)




*FAA chart (public domain), cycle 01 OCT to 29 OCT 2026. For study only; not for navigation.*

# Los Angeles (KLAX) – KYLOW ONE RNAV Departure (Transition Routes)




*FAA chart (public domain), cycle 01 OCT to 29 OCT 2026. For study only; not for navigation.*

## At a Glance – Initial Climb and Gradients

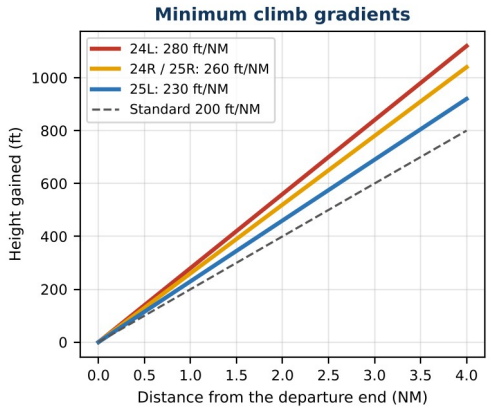
As printed on the chart.

| Runways | Min. gradient to 628 ft | Initial climb |
|---|---|---|
| 24L | 280 ft/NM | Heading 251°, intercept 190°, DSTAR at or below 3000 and 230K, track 190° to KYLOW |
| 24R | 260 ft/NM | Heading 251°, intercept 190°, DSTAR at or below 3000 and 230K, track 190° to KYLOW |
| 25L | 230 ft/NM | Heading 251°, intercept 196°, EWOKK at or below 3000 and 230K, track 196° to KYLOW |
| 25R | 260 ft/NM | Heading 251°, intercept 196°, EWOKK at or below 3000 and 230K, track 196° to KYLOW |

## At a Glance – The Seven Transitions

Distances are the sums of the leg distances printed on the chart (approximate).

| Transition | Route after KYLOW | Approx. NM | Top altitude | Navigation / filing |
|---|---|---|---|---|
| RIZIN | BOBAA – FIXIT – RIZIN | 67 | FL230 | Radar if non-GPS. File only 2100–0700 LCL |
| DINTY | BOBAA – FIXIT – RIZIN – OINGO – DINTY | 232 | FL230 | GPS required |
| MCKEY | BOBAA – FIXIT – RIZIN – MCKEY | 167 | FL230 | Radar if non-GPS |
| STOKD | BOBAA – FIXIT – RIZIN – SPLNT – STOKD | 157 | FL230 | Radar if non-GPS |
| SCTRR | BOBAA – FIXIT – RIZIN – TESEA – SCTRR | 152 | FL230 | Radar if non-GPS |
| FICKY | ZILLI – LAUER – PLLUT – DUNZL – IGOTT – FICKY | 205 | 14,000 | GPS required |
| GROGU | ZILLI – BEAUT – RIFFT – MALIT – GROGU | 156 | 14,000 | GPS required. ATC assigned only |




**Rate of climb needed (ft/min)**

| | 150 kt | 200 kt | 250 kt |
|---|---|---|---|
| 230 ft/NM | 575 | 767 | 958 |
| 260 ft/NM | 650 | 867 | 1083 |
| 280 ft/NM | 700 | 933 | 1167 |

*Rate (ft/min) = gradient × groundspeed ÷ 60*

*Figure: climb gradients and the rate of climb they need.*

## Section 1 – Chart Identification & Header

### Q1. What procedure is this chart for?

The KYLOW ONE RNAV Departure (KYLOW1.KYLOW) at Los Angeles Intl (KLAX/LAX), Departure Routes page, for runways 24L, 24R, 25L and 25R. It is an FAA chart, procedure ID 25331.

### Q2. What does "RNAV" in the title mean?

The departure is flown with area navigation (FMS/GPS) instead of ground-based navaids and vectors.

### Q3. What do the identifiers in parentheses, like (KYLOW1.KYLOW), mean?

They are the names used in the FMS navigation database: the SID name, a dot, then the transition. KYLOW1.KYLOW is the base procedure that ends at KYLOW. A transition appears after the dot, for example KYLOW1.RIZIN.

### Q4. Which runways is the departure for?

The text gives the routes for takeoff runways 24L/R and 25L/R. It is a west-flow departure and cannot be used from runways 6L/R or 7L/R.

### Q5. What do the two pages of the chart show?

The first page is titled "(Departure Routes)" and shows the initial climb by runway, the take-off minimums and the notes. The second page is titled "(Transition Routes)" and shows the seven transitions from KYLOW to the exit fixes.

### Q6. What do the dates on the chart mean?

The chart carries the FAA cycle dates in the margin ("SW-3, 01 OCT 2026 to 29 OCT 2026") and the procedure date at the bottom (11JUL24). The cycle dates tell you the period when this printed chart is current. Always check that your EFB or database is current.

### Q7. What does "NOTE: Chart not to scale" mean?

The drawing is only a sketch. Do not measure distances or angles on it. Use the courses and the fixes.

### Q8. What does the circle with 7800 and "MSA LAX 25 NM" show?

The Minimum Sector Altitude within 25 NM of the LAX VOR: 7,800 ft. It gives obstacle clearance and is used for orientation and emergencies.

## Section 2 – Navigation Requirements

### Q9. What navigation capability does the chart require?

RNAV 1 – DME/DME/IRU or GPS.

### Q10. What does RNAV 1 mean?

Area navigation with a total system error of no more than 1 NM for at least 95% of the flight time.

### Q11. What does DME/DME/IRU mean?

Position computed from two or more DME stations, with an inertial reference unit (IRU) carrying the position when DME updates are lost.

### Q12. Which transitions require GPS?

FICKY, GROGU and DINTY. The chart says: "From KYLOW on FICKY, GROGU and DINTY Transitions: GPS."

**Q13. Which transitions require radar for non-GPS aircraft?**

MCKEY, RIZIN, SCTRR and STOKD. Aircraft without GPS can fly them only with radar.

**Q14. Why would the chart require radar for non-GPS aircraft?**

The chart does not say. It typically means DME/DME coverage is not assured on those legs, so ATC radar monitoring is required.

**Q15. What is RAIM and why does it matter here?**

Receiver Autonomous Integrity Monitoring, the GPS check of its own integrity. Before a departure that requires GPS, confirm that RAIM (or the equivalent integrity function of your system) is available for your departure time.

**Q16. What does "Jets only" mean?**

Only jet aircraft may fly this SID. Turboprop and piston aircraft must request another departure.

**Q17. You filed the DINTY transition and your GPS fails before departure. What do you do?**

You cannot fly a GPS-required transition without GPS. Tell ATC and your dispatcher and request a transition or departure that your equipment supports (for example, a radar-required transition for non-GPS aircraft, if ATC agrees).

**Q18. You lose RNAV capability after takeoff. What do you do?**

Tell ATC immediately ("unable RNAV"), request vectors or a conventional clearance, and fly the last assigned heading and altitude until ATC replies.

## Section 3 – The Notes

**Q19. List the notes printed on the chart.**

- Jets only.
- For use only when nighttime noise abatement procedures are in effect.
- GROGU Transition ATC assigned only. Do not file.
- Expect GROGU Transition when W292E is in use.
- File the RIZIN Transition in lieu of the LADYJ DEPARTURE during time periods 2100-0700 LCL. All other times RIZIN Transition ATC assigned only.

**Q20. What does "For use only when nighttime noise abatement procedures are in effect" mean?**

The KYLOW ONE departure is intended for use only when the nighttime noise abatement procedures at LAX are in effect. Check the ATIS and NOTAMs for what is active.

**Q21. What does "Expect GROGU Transition when W292E is in use" mean?**

W292E is a warning area over the water. When it is active, expect ATC to assign you the GROGU transition.

**Q22. What does "ATC assigned only. Do not file." mean?**

You cannot put the GROGU transition in your flight plan. Only ATC can assign it.

**Q23. What does the RIZIN note mean?**

Between 2100 and 0700 local time, file the RIZIN transition instead of the LADYJ departure. At all other times, the RIZIN transition is ATC assigned only.

**Q24. What is LADYJ?**

The name of another LAX departure. This chart only tells you to file RIZIN instead of it at night. See the LADYJ chart for its details.

**Q25. What does "LCL" mean?**

Local time.

**Q26. Which transitions can you file yourself?**

The chart puts no filing restriction on DINTY, FICKY, MCKEY, SCTRR and STOKD (subject to the navigation requirements). GROGU is ATC assigned only. RIZIN can be filed only between 2100 and 0700 local, and is ATC assigned only at other times.

## Section 4 – Take-off Minimums

**Q27. What are the take-off minimums?**

Standard for every runway, with a minimum climb gradient to 628 ft:

- Rwy 24L: 280 ft/NM to 628
- Rwys 24R and 25R: 260 ft/NM to 628
- Rwy 25L: 230 ft/NM to 628

**Q28. What does "Standard" mean?**

Standard take-off weather minimums apply. Under Part 91 that is 1 SM visibility for aircraft with two engines or fewer, and 1/2 SM for aircraft with more than two engines. Airlines follow their operations specifications.

**Q29. What does "to 628" mean?**

You must climb at least that gradient until you reach 628 ft MSL.

**Q30. How far above the airport is 628 ft MSL?**

500 ft above the airport (the airport elevation is 128 ft, and 628 − 128 = 500).

**Q31. Which runway has the steepest required gradient?**

Runway 24L, at 280 ft/NM.

**Q32. Which runway has the least demanding gradient?**

Runway 25L, at 230 ft/NM.

**Q33. Why do the gradients differ by runway?**

Each runway end faces different obstacles and terrain off the departure end, so the required climb to clear them is different.

**Q34. What happens after you pass 628 ft?**

The chart gives no further gradient. In general, a departure procedure assumes at least 200 ft/NM unless a higher gradient is published, and ATC or the SID altitudes may require more.

**Q35. Do the published gradients apply with one engine inoperable?**

The gradients are written for normal operation. For engine-out, use your company's engine-out departure procedure and obstacle analysis, and brief it before departure.

## Section 5 – Climb Gradient Calculations

**Q36. What is the formula to convert a climb gradient into a rate of climb?**

Rate of climb (ft/min) = gradient (ft/NM) × groundspeed (kt) ÷ 60.

**Q37. What rate of climb do you need for 280 ft/NM at 150 kt groundspeed?**

700 ft/min (280 × 150 ÷ 60).

**Q38. What rate of climb do you need for 280 ft/NM at 250 kt?**
1,167 ft/min.

**Q39. What rate of climb do you need for 230 ft/NM at 200 kt?**
767 ft/min.

**Q40. What rate of climb do you need for 260 ft/NM at 300 kt?**
1,300 ft/min.

**Q41. What rate of climb do you need for 280 ft/NM at 175 kt?**
About 817 ft/min (280 × 175 ÷ 60 = 816.7).

**Q42. What rate of climb do you need for 260 ft/NM at 180 kt?**
780 ft/min (260 × 180 ÷ 60).

**Q43. What gradient do you achieve climbing at 3,000 ft/min at 250 kt groundspeed?**
720 ft/NM (3,000 × 60 ÷ 250), well above the 280 ft/NM maximum required on this chart.

**Q44. Why do you use groundspeed and not airspeed?**
The gradient is measured over the ground, so a tailwind raises your groundspeed and requires a higher rate of climb for the same gradient.

## Section 6 – Initial Climb (Runway Specific)

**Q45. What is the initial climb for runways 24L/R?**
Climb on heading 251° to intercept course 190° to cross DSTAR at or below 3000 and at or below 230K, then on track 190° to KYLOW.

**Q46. What is the initial climb for runways 25L/R?**
Climb on heading 251° to intercept course 196° to cross EWOKK at or below 3000 and at or below 230K, then on track 196° to KYLOW.

**Q47. Which fix goes with which runway?**
DSTAR for runways 24L/R and EWOKK for runways 25L/R.

**Q48. Which way is the turn, and how large is it?**
Left in both cases. From heading 251° to course 190° is a 61° left turn. From heading 251° to course 196° is a 55° left turn.

**Q49. What does "3000 230K" at DSTAR and EWOKK mean?**
Two restrictions at the fix: at or below 3,000 ft and at or below 230 knots. The 3000 and the 230K both have a line over them, the chart symbol for a maximum.

**Q50. What is the difference between "climb on heading" and "track"?**
"Heading" means fly that heading with no drift correction until you intercept the course. "Track" means fly the course over the ground, correcting for wind.

**Q51. What are the distances from DSTAR and EWOKK to KYLOW?**
The chart prints (6) NM on the 190° leg and (5) NM on the 196° leg. The chart is not to scale.

**Q52. At what point do you start the turn to intercept the course?**
The chart does not give a turn altitude. Follow your company's procedures for when to engage LNAV and any minimum height above the airport.

**Q53. What runway heading do you fly on runways 24 and 25?**

251°, as printed on the chart for all four runways.

**Q54. Does the initial climb also apply to runways 6 and 7?**

No. This SID is only for departures from runways 24L/R and 25L/R.

**Q55. What does the dashed box around KYLOW mean?**

It marks the area at the bottom of the chart where the transitions begin: "See following page for transition routes". The routes themselves are on the second page.

## Section 7 – Top Altitudes & Climb

**Q56. What is the top altitude for each transition?**

- FICKY and GROGU Transitions: 14,000 ft
- SCTRR, STOKD, MCKEY, DINTY and RIZIN Transitions: FL230

**Q57. What does "Expect filed altitude 10 minutes after departure" mean?**

Your cruise altitude will normally be cleared about 10 minutes after takeoff. Until then you are limited to the top altitude on the chart. It is also the point used for lost-communications planning.

**Q58. Which transitions limit you to 14,000 ft?**

FICKY and GROGU.

**Q59. Which transitions limit you to FL230?**

SCTRR, STOKD, MCKEY, DINTY and RIZIN.

**Q60. What is the difference between 14000 and FL230?**

14,000 is an altitude set on the local altimeter setting. FL230 is a flight level set on 29.92.

**Q61. Can ATC give you a different altitude than the chart?**

Yes. The charted top altitude is the standard. ATC can assign a different altitude, and you follow the clearance.

**Q62. What does the lost communications rule say about this departure?**

Under 14 CFR 91.185, fly the route in your clearance and use the highest of the last assigned altitude, the minimum IFR altitude (MEA), or the altitude ATC told you to expect. The chart tells you to expect your filed altitude 10 minutes after departure. Squawk 7600.

**Q63. How many transitions does the departure have, and what are they?**

Seven: DINTY, GROGU, FICKY, MCKEY, RIZIN, SCTRR and STOKD. The second page lists them with their FMS names (KYLOW1.DINTY, KYLOW1.GROGU and so on).

## Section 8 – Transitions & Routes (Page 2)

**Q64. What is the common point for all transitions, and what is the first fix after it?**

KYLOW. After KYLOW, the RIZIN, DINTY, MCKEY, STOKD and SCTRR transitions go first to BOBAA, and the FICKY and GROGU transitions go first to ZILLI.

**Q65. What is the route for the RIZIN transition?**

KYLOW – BOBAA – FIXIT – RIZIN.

- KYLOW to BOBAA: 238°, 11 NM
- BOBAA to FIXIT: 302°, 18 NM
- FIXIT to RIZIN: 302°, 38 NM

- Total about 67 NM, top altitude FL230.

**Q66. What is the route for the DINTY transition?**

KYLOW – BOBAA – FIXIT – RIZIN – OINGO – DINTY.

- RIZIN to OINGO: 253°, 47 NM
- OINGO to DINTY: 233°, 118 NM
- Total about 232 NM, top altitude FL230, GPS required.

**Q67. What is the route for the MCKEY transition?**

KYLOW – BOBAA – FIXIT – RIZIN – MCKEY.

- RIZIN to MCKEY: 298°, 100 NM
- Total about 167 NM, top altitude FL230, radar required for non-GPS aircraft.

**Q68. What is the route for the STOKD transition?**

KYLOW – BOBAA – FIXIT – RIZIN – SPLNT – STOKD.

- RIZIN to SPLNT: 303°, 58 NM
- SPLNT to STOKD: 305°, 32 NM
- Total about 157 NM, top altitude FL230, radar required for non-GPS aircraft.

**Q69. What is the route for the SCTRR transition?**

KYLOW – BOBAA – FIXIT – RIZIN – TESEA – SCTRR.

- RIZIN to TESEA: 310°, 58 NM
- TESEA to SCTRR: 311°, 27 NM
- Total about 152 NM, top altitude FL230, radar required for non-GPS aircraft.

**Q70. What is the route for the FICKY transition?**

KYLOW – ZILLI – LAUER – PLLUT – DUNZL – IGOTT – FICKY.

- KYLOW to ZILLI: 207°, 11 NM
- ZILLI to LAUER: 175°, 16 NM
- LAUER to PLLUT: 214°, 19 NM
- PLLUT to DUNZL: 214°, 53 NM
- DUNZL to IGOTT: 213°, 47 NM
- IGOTT to FICKY: 229°, 59 NM
- Total about 205 NM, top altitude 14,000 ft, GPS required.

**Q71. What is the route for the GROGU transition?**

KYLOW – ZILLI – BEAUT – RIFFT – MALIT – GROGU.

- KYLOW to ZILLI: 207°, 11 NM
- ZILLI to BEAUT: 147°, 24 NM
- BEAUT to RIFFT: 216°, 22 NM
- RIFFT to MALIT: 216°, 53 NM
- MALIT to GROGU: 225°, 46 NM
- Total about 156 NM, top altitude 14,000 ft, GPS required, ATC assigned only.

**Q72. Where do the north and west transitions split?**

At RIZIN. From there OINGO and DINTY go west, and MCKEY, SPLNT/STOKD and TESEA/SCTRR fan out to the northwest.

**Q73. Where do the FICKY and GROGU transitions split?**

At ZILLI. FICKY continues on 175° to LAUER, and GROGU goes on 147° to BEAUT.

**Q74. Which transition is the shortest, and which is the longest?**

RIZIN is the shortest, about 67 NM from KYLOW. DINTY is the longest, about 232 NM. These are sums of the leg distances printed on the chart.

**Q75. Which fixes are shared by every north and west transition?**

BOBAA, FIXIT and RIZIN.

**Q76. What is the course and distance from KYLOW to BOBAA?**

238°, 11 NM.

**Q77. What is W292E, and how does it affect the FICKY route?**

W292E is a warning area over the water, drawn as a hatched area on the chart. The FICKY route passes through it between LAUER and DUNZL. When W292E is in use, expect ATC to give you the GROGU transition instead.

**Q78. What do the distances in parentheses mean?**

The leg length in nautical miles. For example, KYLOW to BOBAA is (11) NM. The chart is not to scale, so use the numbers.

**Q79. Are the courses on the chart magnetic or true?**

Magnetic, as on FAA procedure charts, unless the chart marks them otherwise.

**Q80. What do the arrows on the lines mean?**

The direction of flight along each leg.

**Q81. Why do IGOTT, GROGU and FICKY look so close together on the chart?**

The chart is not to scale. They are separate routes, and each leg has its own course and distance.

## Section 9 – Altitudes & the Numbers on the Legs

**Q82. What do the numbers on a leg, like "6500 *1700 233° (118)", mean?**

Each leg shows four things:

- 6500: the MEA, the minimum en route altitude
- *1700: the MOCA, marked with an asterisk
- 233°: the magnetic course
- (118): the distance in NM

**Q83. What is an MEA?**

Minimum En route Altitude. It gives obstacle clearance and acceptable navigation signal coverage.

**Q84. What is a MOCA?**

Minimum Obstruction Clearance Altitude. It gives obstacle clearance but not necessarily full navigation coverage. The chart marks it with an asterisk.

**Q85. What is the highest MEA printed on the chart?**

8,800 ft, on the legs from RIZIN to MCKEY, SPLNT and TESEA, and on to STOKD and SCTRR.

**Q86. What is the lowest MEA printed on the chart?**

3,000 ft, on KYLOW to BOBAA, KYLOW to ZILLI and ZILLI to LAUER.

**Q87. Which legs show an MEA of 5500?**

The legs of the south transitions: ZILLI to BEAUT, LAUER to PLLUT, PLLUT to DUNZL, DUNZL to IGOTT, IGOTT to FICKY, BEAUT to RIFFT, RIFFT to MALIT and MALIT to GROGU.

**Q88. What is the highest MOCA printed on the chart?**

5,700 ft, on the TESEA to SCTRR leg (the SPLNT to STOKD leg shows 5,100).

**Q89. What do the altitudes under BOBAA, LAUER and FIXIT mean?**

They are crossing altitudes at the fix. BOBAA 8000 and LAUER 9000 are underlined: at or above. FIXIT 12000 has a line over it: at or below. An underlined altitude is a minimum, and an overlined altitude is a maximum.

**Q90. Which fixes apply to which transitions?**

BOBAA and FIXIT are on the north and west transitions (RIZIN, DINTY, MCKEY, STOKD and SCTRR). LAUER is on the FICKY transition. GROGU has no crossing altitude on the chart.

**Q91. What does the number 7800 in the box mean?**

The MSA, the Minimum Sector Altitude within 25 NM of the LAX VOR, 7,800 ft.

**Q92. What is the difference between the star symbols and the triangle symbols on the fixes?**

Most fixes are drawn as four-point stars. MCKEY, DINTY and MALIT are drawn as triangles. Check the symbol legend of the chart booklet for the exact meaning.

## Section 10 – Calculations & Planning

**Q93. What is the total distance to the end of the FICKY transition?**

About 205 NM (11 + 16 + 19 + 53 + 47 + 59).

**Q94. What is the total distance to the end of the GROGU transition?**

About 156 NM (11 + 24 + 22 + 53 + 46).

**Q95. What is the total distance to the end of the DINTY transition?**

About 232 NM (11 + 18 + 38 + 47 + 118).

**Q96. How long does the FICKY transition take at 400 kt average groundspeed?**

205 ÷ 400 = 0.51 hours, about 31 minutes.

**Q97. What is the course change at BOBAA?**

From 238° to 302° is a 64° right turn.

**Q98. What is the course change at ZILLI for FICKY? For GROGU?**

FICKY: from 207° to 175°, a 32° left turn. GROGU: from 207° to 147°, a 60° left turn.

**Q99. What is the course change at LAUER?**

From 175° to 214° is a 39° right turn.

**Q100. What is the course change at BEAUT?**

From 147° to 216° is a 69° right turn.

**Q101. What is the course change at RIZIN for DINTY?**

From 302° to 253° is a 49° left turn.

**Q102. What is the course change at OINGO?**

From 253° to 233° is a 20° left turn.

**Q103. What is the course change at IGOTT?**

From 213° to 229° is a 16° right turn.

**Q104. What is the course change at SPLNT and at TESEA?**

At SPLNT from 303° to 305°, 2° right. At TESEA from 310° to 311°, 1° right.

**Q105. How many NM from KYLOW to RIZIN?**

About 67 NM (11 + 18 + 38).

**Q106. How far is RIZIN to MCKEY compared with RIZIN to OINGO?**

100 NM versus 47 NM, so MCKEY is 53 NM farther.

## Section 11 – Communications & Coordination

**Q107. Which frequencies are on the chart?**

- SoCal Departure Control: 124.3 / 363.2
- D-ATIS Departure: 135.65
- Clearance Delivery: 120.35 / 327.0 (and CPDLC)
- Ground: North 121.65, South 121.75, West 121.4 (all with 327.0 UHF)
- Tower: North 133.9 / 239.3, South 120.95 / 379.1

**Q108. Who do you talk to after takeoff?**

The tower hands you off to SoCal Departure. The chart shows 124.3 (UHF 363.2), but use the frequency the tower gives you.

**Q109. What do you say on initial contact?**

For example: "SoCal Departure, [callsign], KYLOW ONE departure, passing [altitude] for [assigned altitude]." Include any information ATC asked you to report.

**Q110. Give an example of a clearance for this SID.**

"[Callsign], cleared to [destination] via the KYLOW ONE departure, FICKY transition, then as filed. Departure frequency 124.3, squawk [code]."

**Q111. How do you read back that clearance?**

Read back the SID name, the transition, the altitude instruction, the frequency and the squawk code.

**Q112. ATC says "Cleared KYLOW ONE departure, RIZIN transition, runway 25R." What is your initial climb?**

Climb on heading 251°, intercept course 196° to cross EWOKK at or below 3,000 ft and at or below 230 kt, then track 196° to KYLOW, then fly the RIZIN transition.

**Q113. What do you do if the clearance does not match what you filed?**

Read back the clearance you received, verify that the FMS matches it, and tell ATC if you cannot comply.

**Q114. What should the pilot monitoring check on the FMS?**

The SID and transition name, the runway, the fixes, courses and distances against the chart, the altitude and speed restrictions, and the top altitude.

## Section 12 – Scenarios & Decision Making

**Q115. You are departing 24L and the required gradient is 280 ft/NM. How do you check you can make it?**

Convert it to a rate of climb with your expected groundspeed (700 ft/min at 150 kt, 933 ft/min at 200 kt). Check your performance data for the weight and temperature, and consider tailwind.

**Q116. You cannot meet the required gradient. What are your options?**

Reduce weight, use a different runway, or ask ATC for a different departure. Tell ATC if you cannot comply.

**Q117. ATC changes your runway from 24L to 25L before takeoff. What do you do?**

Re-brief. The initial climb changes from DSTAR (course 190°) to EWOKK (course 196°), the minimum gradient changes from 280 to 230 ft/NM, and the FMS must match.

**Q118. ATC assigns you the GROGU transition. What do you check?**

That your aircraft has GPS, that the FMS has the GROGU transition, and that the top altitude is 14,000 ft. GROGU is ATC assigned only.

**Q119. The ATIS says W292E is active. What do you expect?**

Expect the GROGU transition.

**Q120. You are flying at night and want to go west. Which transition do you file?**

Between 2100 and 0700 local, file the RIZIN transition in lieu of the LADYJ departure.

**Q121. ATC clears you to a speed above 230 kt before DSTAR or EWOKK. What do you do?**

Cross the fix at 230 kt or less and tell ATC if you cannot comply with the clearance.

**Q122. You are above 3,000 ft before DSTAR. Is that a problem?**

Yes, if you will be above 3,000 ft when you cross the fix. The restriction is at or below 3,000 at the fix. Adjust the climb and tell ATC if needed.

**Q123. You get a TCAS RA during the initial climb. What do you do?**

Follow the RA, disconnect the autopilot as required, and tell ATC when you can. Return to the clearance when clear of conflict.

**Q124. The FMS route does not match the chart. What do you do?**

Stop and verify. Check the SID and transition, the fixes, courses and distances against the chart, and correct the FMS or ask ATC for clarification before departure.

**Q125. You are a turboprop and ATC assigns this SID. What do you do?**

Tell ATC that you are unable because the SID is for jets only and request another departure.

**Q126. Weather is IFR and you are departing 25R. Which parts of the chart do you brief?**

Take-off minimums and the 260 ft/NM gradient, the initial climb to EWOKK, the top altitude, the transition, the frequency, the engine-out plan and the lost communications plan.

**Q127. What is your best defense against a mistake on a complex RNAV departure?**

Brief the chart before taxi, cross-check the FMS with the chart, use both pilots to monitor, follow the clearance when in doubt, and ask ATC if unsure.

## Section 13 – Rapid-Fire Review

**Q128. What is the SoCal Departure frequency?**
124.3.

**Q129. Which runways does this SID serve?**
24L, 24R, 25L and 25R.

**Q130. What is the maximum speed at DSTAR and EWOKK?**
230 kt.

**Q131. At what altitude do you cross DSTAR and EWOKK?**
At or below 3,000 ft.

**Q132. What heading do you fly on the initial climb?**
251°.

**Q133. Which course do you intercept for runways 24L/R? For 25L/R?**
190° and 196°.

**Q134. Which transitions require GPS?**
FICKY, GROGU and DINTY.

**Q135. Which transitions have a 14,000 ft top altitude?**
FICKY and GROGU.

**Q136. Which transition is ATC assigned only, always?**
GROGU.

**Q137. How many transitions does the SID have?**
Seven.

**Q138. What is the minimum climb gradient on 24L?**
280 ft/NM to 628.

**Q139. What is the MSA shown on the chart?**
7,800 ft, within 25 NM of LAX.

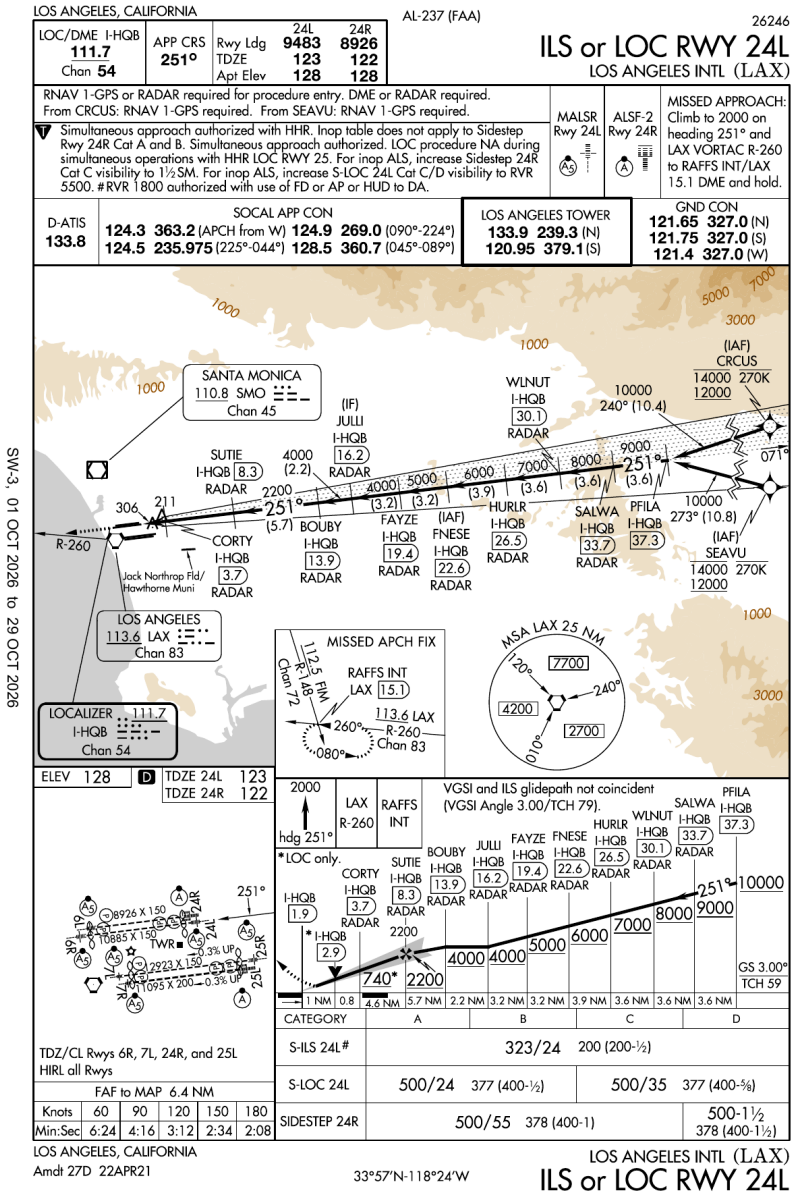
**Q140. What is the first fix after KYLOW to the north? To the south?**
BOBAA to the north and west, ZILLI to the south.

**Q141. What are the courses from KYLOW to BOBAA and to ZILLI?**
238° and 207°.

**Q142. Which transition goes through the hatched area W292E?**
FICKY, between LAUER and DUNZL.

**Q143. What is the highest MEA on the chart?**
8,800 ft.

# Los Angeles (KLAX) – ILS or LOC Rwy 24L




*FAA chart (public domain), cycle 01 OCT to 29 OCT 2026. For study only; not for navigation.*

## At a Glance – Fixes and Altitudes on the Final Approach Course

Altitudes and distances are as printed on the chart profile.

| Fix | DME IHQB | Altitude | To next fix (NM) | Note |
|---|---|---|---|---|
| CRCUS | – | 12,000–14,000, then 10,000 | 10.4 to PFILA | IAF, GPS |
| SEAVU | – | 12,000–14,000, then 10,000 | 10.8 to PFILA | IAF, GPS |
| PFILA | D37.3 | 10,000 | 3.6 | |
| SALWA | D33.7 | 9,000 | 3.6 | Radar fix |
| WLNUT | D30.1 | 8,000 | 3.6 | Radar fix |
| HURLR | D26.5 | 7,000 | 3.9 | Radar fix |
| FNESE | D22.6 | 6,000 | 3.2 | IAF, radar fix |
| FAYZE | D19.4 | 5,000 | 3.2 | Radar fix |
| JULLI | D16.2 | 4,000 | 2.2 | IF, radar fix |
| BOUBY | D13.9 | 4,000 | 5.7 | Radar fix |
| SUTIE | D8.3 | 2,200 (GS intercept) | 4.6 | FAF, radar fix |
| CORTY | D3.7 | – | 1.8 | Radar fix |
| MAP | D1.9 | DA 323 / MDA 500 | – | LOC MAP |

## Extended 3° Glide Path vs Charted Altitudes

Computed from the glide slope altitude of 2,200 ft at D8.3 and about 318 ft/NM (3.00°). For study only; the chart does not publish these values.

| Fix | DME IHQB | Charted altitude (ft) | Extended 3° path (ft) | Path above chart (ft) |
|---|---|---|---|---|
| PFILA | D37.3 | 10,000 | 11,400 | 1,400 |
| SALWA | D33.7 | 9,000 | 10,300 | 1,300 |
| WLNUT | D30.1 | 8,000 | 9,100 | 1,100 |
| HURLR | D26.5 | 7,000 | 8,000 | 1,000 |
| FNESE | D22.6 | 6,000 | 6,800 | 800 |
| FAYZE | D19.4 | 5,000 | 5,700 | 700 |
| JULLI | D16.2 | 4,000 | 4,700 | 700 |
| BOUBY | D13.9 | 4,000 | 4,000 | 0 |
| SUTIE | D8.3 | 2,200 | 2,200 | 0 |

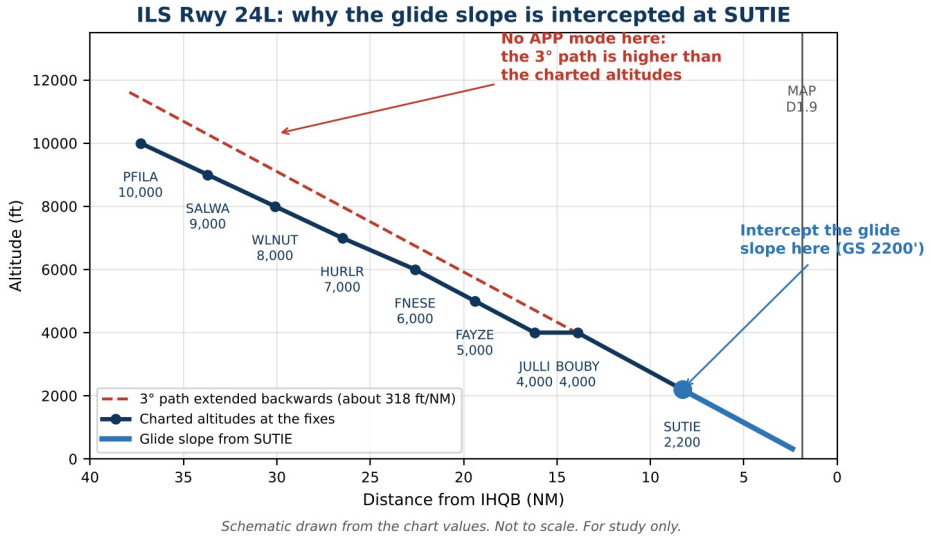



*Figure: charted altitudes against the extended 3° path.*

## Section 1 – Chart Identification & Header

### Q1. What chart is this?

The ILS or LOC Rwy 24L approach at Los Angeles Intl (KLAX/LAX). It is an FAA chart, AL-237, procedure ID 26246.

### Q2. What are the localizer frequency and identifier?

111.7, identifier I-HQB, Channel 54. The LOC/DME box in the top left corner shows "LOC/DME I-HQB 111.7 Chan 54", and the localizer box near the airport repeats it.

### Q3. What is the final approach course?

251° magnetic.

### Q4. What are the airport elevation and the touchdown zone elevation?

Airport elevation is 128 ft. The TDZE for Runway 24L is 123 ft (the TDZE for 24R is 122 ft).

### Q5. What is the ILS decision altitude?

DA(H) 323 ft, which is 200 ft above the TDZE (323 − 123 = 200).

### Q6. What does "Amdt 27D 22APR21" mean?

It is printed at the bottom left. The procedure is Amendment 27D, dated 22 April 2021. The amendment letter changes when the procedure itself is revised.

### Q7. What are the runway lengths and landing distances for 24L and 24R?

Rwy Ldg 9,483 ft on 24L and 8,926 ft on 24R (the landing length shown in the header box).

## Section 2 – Header, Frequencies & Missed Approach

### Q8. What is the D-ATIS arrival frequency?

133.8.

### Q9. What are the SoCal Approach frequencies?

The chart lists four, split by sector:

- 124.3 (363.2): APCH from W

- 124.9 (269.0): 090°–224°
- 124.5 (235.975): 225°–044°
- 128.5 (360.7): 045°–089°
- The sector bearings refer to your arrival direction. The first number is VHF, the second UHF.

**Q10. What are the Tower frequencies, and which do you use for 24L?**

North Complex 133.9 and South Complex 120.95. Runway 24L is in the north complex, so 133.9.

**Q11. What are the Ground frequencies?**

- West: 121.4
- North Complex: 121.65
- South Complex: 121.75

**Q12. What is SUTIE, and what is 2200 next to it?**

SUTIE is the final approach fix, the point where the glide slope is intercepted. It is drawn with the Maltese cross on the profile, at D8.3 I-HQB and 2,200 ft. That is 2,077 ft above the touchdown zone (2,200 − 123).

**Q13. What is the missed approach procedure?**

The chart says: Climb to 2000 on heading 251° and LAX VORTAC R-260 to RAFFS INT / LAX 15.1 DME and hold.

**Q14. What does the MSA box show?**

The Minimum Sector Altitude within 25 NM of the LAX VOR: three sectors, 7,700 ft, 4,200 ft and 2,700 ft, separated by the bearings 120°, 240° and 010° that are drawn on the circle. It gives obstacle clearance, not guaranteed navigation coverage, and is for emergencies and orientation.

## Section 3 – Equipment Requirements & Notes

**Q15. What are the equipment requirements at the top of the chart?**

- RNAV 1-GPS or RADAR required for procedure entry.
- DME or RADAR required.
- From CRCUS: RNAV 1-GPS required.
- From SEAVU: RNAV 1-GPS required.

**Q16. What does "RNAV 1" mean?**

Area navigation with a total system error of no more than 1 NM for at least 95% of the flight time.

**Q17. List the notes at the top of the chart.**

- RNAV 1-GPS or RADAR required for procedure entry. DME or RADAR required. From CRCUS and from SEAVU: RNAV 1-GPS required.
- Simultaneous approach authorized with HHR.
- Inop table does not apply to Sidestep Rwy 24R Cat A and B.
- Simultaneous approach authorized. LOC procedure NA during simultaneous operations with HHR LOC RWY 25.
- For inop ALS, increase Sidestep 24R Cat C visibility to 1½ SM. For inop ALS, increase S-LOC 24L Cat C/D visibility to RVR 5500.
- # RVR 1800 authorized with use of FD or AP or HUD to DA.

### Q18. What does "DME or RADAR required" mean?

The fixes on the final approach course are defined by DME from the I-HQB ILS DME. You need DME on board or ATC radar to identify them.

### Q19. What does the word RADAR under SALWA, WLNUT, HURLR, FNESE, FAYZE, JULLI, BOUBY, SUTIE and CORTY mean?

These fixes are defined by a DME distance from I-HQB and can also be identified by ATC radar. If you do not have DME, radar can identify them. PFILA, the first fix on the final approach course, is not labeled RADAR.

### Q20. What does "Simultaneous approach authorized with HHR" mean?

Simultaneous approaches are authorized with HHR, Hawthorne Municipal (Jack Northrop Field), which is just south of the final approach course. The chart names it in the notes. The LOC procedure is not authorized during simultaneous operations with HHR LOC RWY 25.

### Q21. What does "VGSI and ILS glidepath not coincident" mean?

The VGSI (visual glide slope indicator, the PAPI) and the ILS glide path are not the same. The VGSI is set at 3.00° with a threshold crossing height of 79 ft, and the ILS TCH is 59 ft (shown at the end of the profile). Flying the PAPI puts you about 20 ft higher over the threshold than flying the ILS.

### Q22. What does the lighting box on the chart tell you about the two runways?

Runway 24L has MALSR and Runway 24R has ALSF-2. The sketch also shows the runway lighting: TDZ/CL on Rwys 6R, 7L, 24R and 25L, and HIRL on all runways.

### Q23. What do the altitudes and 270K next to CRCUS and SEAVU mean?

14000 has a line over it and 12000 has a line under it: at or below 14,000 and at or above 12,000. The 270K is overlined, so it is a maximum speed of 270 kt. They apply at CRCUS and SEAVU, and then the legs to PFILA are at 10,000 ft.

### Q24. What do (IAF) and (IF) mean?

IAF is Initial Approach Fix, where the approach begins. IF is Intermediate Fix, the point where the intermediate segment begins. The chart shows IAF at CRCUS, SEAVU and FNESE, and IF at JULLI.

### Q25. What is the black triangle with a T beside the notes?

It tells you that IFR takeoff minimums and/or a departure procedure are published for this airport. Check them before departure. It has no effect on the approach.

## Section 4 – Radar Vectors, APP Mode & the Altitudes

### Q26. If you are under radar vectors and ATC gives you a vector to intercept the ILS between SALWA and PFILA, can you press the APP mode?

No. Even if you are cleared for the approach, you must respect the altitudes over the waypoints. You can intercept the glide slope just over SUTIE. Why:

- SALWA and PFILA are 33.7 and 37.3 DME from the localizer antenna (IHQB). The glide slope is only dependable close to the airport, about 10 NM.
- The chart does not publish a glide path there. It publishes altitudes: 10,000 ft at PFILA, 9,000 at SALWA, 8,000 at WLNUT, 7,000 at HURLR, 6,000 at FNESE, 5,000 at FAYZE, 4,000 at JULLI and BOUBY, and 2,200 at SUTIE.
- The glide slope intercept is published at SUTIE (D8.3 IHQB) at 2,200 ft.

- APP mode arms the localizer and the glide slope. The autopilot could capture the extended 3° path (about 10,300 ft at SALWA and 11,400 ft at PFILA, see the tables at the start of Part 3) or a false glide slope, instead of flying the charted altitudes.
- A clearance for the approach does not cancel the charted altitudes. Meet the altitude at each fix.

**Q27. How do you fly it, then?**

Intercept the localizer with the heading or lateral mode your company SOP calls for, then descend to the charted altitude at each fix with FLCH, V/S or VNAV. Arm APP per your SOP and the aircraft FCOM so that the glide slope captures from below at SUTIE, at 2,200 ft.

**Q28. Where is the glide slope intercept point?**

SUTIE, D8.3 IHQB, at 2,200 ft. The chart prints "GS 2200"' at SUTIE and the 3.00° glide slope.

**Q29. What altitudes do you meet between PFILA and SUTIE?**

- PFILA D37.3: 10,000
- SALWA D33.7: 9,000
- WLNUT D30.1: 8,000
- HURLR D26.5: 7,000
- FNESE D22.6: 6,000
- FAYZE D19.4: 5,000
- JULLI D16.2: 4,000
- BOUBY D13.9: 4,000
- SUTIE D8.3: 2,200

**Q30. Are the profile altitudes "at" or "at or above"?**

They are underlined, and an underlined altitude on an FAA chart is a minimum altitude, so "at or above". You may be higher, but you may not be lower until the next fix. The glide slope intercept altitude at SUTIE (2,200 ft) is also shown underlined.

**Q31. Where is the extended 3° glide path compared with the charted altitudes?**

Higher at every fix before BOUBY. Computed from GS 2,200 ft at D8.3 and about 318 ft/NM: PFILA 11,400 vs 10,000, SALWA 10,300 vs 9,000, WLNUT 9,100 vs 8,000, HURLR 8,000 vs 7,000, FNESE 6,800 vs 6,000, FAYZE 5,700 vs 5,000, JULLI 4,700 vs 4,000. At BOUBY they match (about 4,000), and at SUTIE both are 2,200. See the second table at the start of Part 3.

**Q32. Why does the glide slope matter at SUTIE and not before?**

Because the glide slope is intercepted from below at SUTIE at 2,200 ft. Before that, you fly level segments and step-downs. The published profile is a step-down profile, not a 3° descent from PFILA.

**Q33. You are at 10,000 ft near PFILA and ATC says "cleared ILS 24L approach". What do you do?**

Establish on the localizer, cross PFILA at 10,000, then descend to each charted altitude at SALWA (9,000), WLNUT (8,000), HURLR (7,000), FNESE (6,000), FAYZE (5,000), JULLI (4,000), BOUBY (4,000) and SUTIE (2,200). If ATC assigned a different altitude, follow it.

**Q34. ATC says "maintain 8,000 until established on the localizer, cleared ILS 24L". When do you descend?**

Once you are established on the localizer you may descend to the published altitude for your segment, and then follow the charted altitudes at each fix.

**Q35. You press APP too early and the aircraft begins to descend. What do you do?**

Stop the descent, return to the altitude required at your position (select altitude hold or a vertical mode), tell ATC if you deviated, and re-establish the approach.

**Q36. Can you intercept the glide slope from above?**

It is not recommended. Do not dive for the glide slope. If you are high at SUTIE, tell ATC and do not rely on the autopilot to capture the glide slope from above.

**Q37. When can you rely on the glide slope?**

Close to the airport, inside its coverage of about 10 NM, and only after being established on the localizer. On this chart that means at or near SUTIE (D8.3).

**Q38. What is a false glide slope?**

A false signal that can exist at steeper angles above the true 3° path. It is one more reason not to capture the glide slope far from the airport or high above the charted altitudes.

**Q39. FNESE is the IAF. What are SALWA and PFILA, then?**

Radar fixes on the final approach course extended, farther out than the IAF. PFILA is also where the CRCUS and SEAVU legs join the final approach course. The altitudes along them are on the profile, and you meet them.

## Section 5 – Initial Approach & Entry

**Q40. How can you enter this approach?**

By RNAV from CRCUS or SEAVU to PFILA, or by radar vectors to the final approach course. Procedure entry needs RNAV 1-GPS or radar.

**Q41. What is the route from CRCUS?**

CRCUS to PFILA on course 240°, 10.4 NM, 10,000 ft, MAX 270 KT. Then 251° on the final approach course.

**Q42. What is the route from SEAVU?**

SEAVU to PFILA on course 273°, 10.8 NM, 10,000 ft, MAX 270 KT. Then 251° on the final approach course.

**Q43. Is GPS required to enter from CRCUS or SEAVU?**

Yes. "From CRCUS RNAV 1-GPS required. From SEAVU: RNAV 1-GPS required."

**Q44. What is the course change at PFILA from CRCUS and from SEAVU?**

From CRCUS: 240° to 251°, an 11° right turn. From SEAVU: 273° to 251°, a 22° left turn.

**Q45. What altitude do you cross CRCUS and SEAVU?**

Between 12,000 and 14,000 ft (at or above 12000, at or below 14000), then 10,000 ft on the legs to PFILA.

**Q46. How far is it from CRCUS to SUTIE?**

About 39.4 NM (10.4 to PFILA, then 29.0 NM along the final approach course).

**Q47. How far is it from PFILA to SUTIE?**

29.0 NM (3.6 + 3.6 + 3.6 + 3.9 + 3.2 + 3.2 + 2.2 + 5.7).

**Q48. How far is it from PFILA to the MAP at D1.9?**

35.4 NM (D37.3 − D1.9).

**Q49. What fixes can ATC use for vectoring?**

PFILA is the farthest out. After it come SALWA, WLNUT, HURLR, FNESE, FAYZE, JULLI, BOUBY, SUTIE and CORTY, each identified by its DME from I-HQB and by radar.

## Section 6 – Fixes on the Final Approach Course

**Q50. Give the DME and altitude of each fix.**

- PFILA: D37.3 IHQB, 10,000
- SALWA: D33.7, 9,000
- WLNUT: D30.1, 8,000
- HURLR: D26.5, 7,000
- FNESE (IAF): D22.6, 6,000
- FAYZE: D19.4, 5,000
- JULLI (IF): D16.2, 4,000
- BOUBY: D13.9, 4,000
- SUTIE: D8.3, 2,200
- CORTY: D3.7
- D2.9 IHQB
- MAP: D1.9 IHQB

**Q51. What are the distances between the fixes?**

Printed on the profile, from PFILA toward the airport:

- PFILA to SALWA 3.6
- SALWA to WLNUT 3.6
- WLNUT to HURLR 3.6
- HURLR to FNESE 3.9
- FNESE to FAYZE 3.2
- FAYZE to JULLI 3.2
- JULLI to BOUBY 2.2
- BOUBY to SUTIE 5.7
- SUTIE to CORTY 4.6
- CORTY to D2.9 0.8
- D2.9 to D1.9 1.0

**Q52. What is the longest segment?**

BOUBY to SUTIE, 5.7 NM.

**Q53. What is the shortest segment?**

CORTY to D2.9, 0.8 NM.

**Q54. Which two fixes share the same altitude?**

JULLI and BOUBY, both at 4,000 ft.

**Q55. What is the altitude at WLNUT?**

8,000 ft.

**Q56. What is the altitude at FNESE, and what is FNESE?**

6,000 ft. FNESE is an IAF, at D22.6 IHQB.

**Q57. What is the altitude at JULLI, and what is JULLI?**
4,000 ft. JULLI is the intermediate fix (IF), at D16.2 IHQB.

**Q58. What do you do between BOUBY and SUTIE?**
Descend from 4,000 ft to 2,200 ft and be level at 2,200 ft before SUTIE, then intercept the glide slope from below at SUTIE.

**Q59. How many NM from SUTIE to the MAP?**
6.4 NM (D8.3 − D1.9).

**Q60. How far is CORTY from the MAP?**
1.8 NM (D3.7 − D1.9).

**Q61. What is the drop in altitude from PFILA to SUTIE?**
7,800 ft (10,000 − 2,200) in 29.0 NM, about 269 ft/NM on average, well below the 318 ft/NM of a 3° path.

## Section 7 – Glide Slope, FAF & Profile

**Q62. What is the glide slope angle?**
3.00°.

**Q63. What is the threshold crossing height of the ILS?**
59 ft (TCH 59').

**Q64. What is the height of the glide slope above the TDZE at SUTIE?**
2,077 ft (2,200 − 123).

**Q65. What is the rate of descent for a 3° glide slope at each groundspeed?**
The FAA chart prints a time table, not a rate-of-descent table. Computed with 318 ft/NM:

- 60 kt: 318 ft/min
- 90 kt: 477
- 120 kt: 636
- 150 kt: 796
- 180 kt: 955

**Q66. What is the formula behind the table?**
Rate of descent (ft/min) ≈ 318 ft/NM × groundspeed (kt) ÷ 60. A rule of thumb is groundspeed × 5.

**Q67. How long does it take from the FAF to the MAP?**
The chart says FAF to MAP 6.4 NM and gives the time:

- 60 kt: 6:24
- 90 kt: 4:16
- 120 kt: 3:12
- 150 kt: 2:34
- 180 kt: 2:08

**Q68. What is the required rate of descent at 150 kt groundspeed (not in the table)?**
About 796 ft/min (318.4 × 150 ÷ 60).

**Q69. What is the time from SUTIE to the MAP at 130 kt (not in the table)?**
About 2:57 (6.4 ÷ 130 × 60 = 2.95 minutes).

**Q70. Where do you reach the DA on the glide slope?**

About D2.4 IHQB, 5.9 NM after SUTIE (computed: (2,200 − 323) ÷ 318). This is a calculation, not a chart value.

**Q71. What does "MAP at D1.9 IHQB" mean?**

The missed approach point for the localizer-only (GS out) approach is at D1.9 IHQB. On the ILS you go missed at the DA if you do not have the visual references.

**Q72. What do the V symbol and D2.9 mean?**

The black V at D2.9 I-HQB is a Visual Descent Point (VDP) for the LOC-only approach: the point from which you can start the normal descent to the runway from the MDA, if you have the runway environment in sight. The footnote "*LOC only" applies to it.

**Q73. What is the 740 shown near CORTY?**

The 740 has an asterisk, so it falls under the footnote "*LOC only". It is an altitude on the LOC-only profile at CORTY (D3.7), close to the altitude of the 3° path there (about 735 ft).

**Q74. What happens if the glide slope fails?**

You fly the S-LOC 24L approach. The MDA is 500 ft (377 ft above the TDZE). The LOC procedure is not authorized during simultaneous operations with HHR LOC RWY 25.

## Section 8 – Minimums

**Q75. What are the ILS minimums?**

S-ILS 24L: DA(H) 323 ft (200 ft), RVR 24 (2,400 ft) or 1/2 SM, for all categories. The # footnote says RVR 1800 is authorized with a flight director, autopilot or HUD to the DA.

**Q76. What does "# RVR 1800" mean?**

RVR 1800 ft is authorized with use of a flight director (FD), autopilot (AP) or HUD down to the DA.

**Q77. What are the LOC (GS out) minimums?**

S-LOC 24L, MDA 500 ft (377 ft above the TDZE):

- Cat A and B: 500/24, 377 (400-1/2)
- Cat C and D: 500/35, 377 (400-5/8)
- If the ALS is inoperative, Cat C/D visibility increases to RVR 5500.

**Q78. What does the note about HHR LOC RWY 25 mean?**

The LOC procedure is not authorized during simultaneous operations with HHR LOC RWY 25.

**Q79. What are the sidestep minimums for landing Rwy 24R?**

MDA 500 ft (378 ft above the TDZE of 24R):

- Cat A to C: 500/55, 378 (400-1)
- Cat D: 500-1½, 378 (400-1½)
- With the ALS inoperative, Cat C visibility increases to 1½ SM. The inop table does not apply to Cat A and B.

**Q80. Why is the HAT 377 on the LOC and 378 for the sidestep?**

The HAT is the MDA minus the TDZE of the runway you land on. 500 − 123 = 377 for 24L, and 500 − 122 = 378 for 24R.

**Q81. What does RVR 24 mean?**

Runway Visual Range of 2,400 ft. The chart prints RVR in hundreds of feet.

**Q82. What is the difference between DA and MDA?**

DA(H) is used on an approach with vertical guidance. You go missed at the DA if you do not see the references. MDA is the lowest altitude on a non-precision approach, and you may level at the MDA until the MAP.

**Q83. Which category applies to your aircraft?**

The category is based on the approach speed (1.3 Vso at max landing weight): A, B, C or D. Use the column for your aircraft category or higher if you approach faster.

## Section 9 – Missed Approach

**Q84. What is the missed approach?**

Climb to 2000' on heading 251° and outbound on LAX VOR R-260 to RAFFS / D15.1 LAX and hold.

**Q85. What is the missed approach fix?**

RAFFS, at D15.1 LAX, on R-260.

**Q86. Which VOR is used on the missed approach, and what is its frequency?**

The LAX VOR, 113.6.

**Q87. Which heading do you fly on the initial missed approach climb?**

251°.

**Q88. Which radial do you track outbound?**

R-260 from the LAX VOR.

**Q89. What altitude do you climb to initially?**

2,000 ft.

**Q90. What is the Santa Monica VOR frequency shown near the missed approach?**

110.8 SMO.

**Q91. What do you do on the missed approach, step by step?**

Go-around thrust, follow the flight director and configure per your SOP, climb to 2,000 ft on 251°, then track R-260 outbound to RAFFS, tell ATC, and hold if you do not get other instructions.

## Section 10 – Lighting & Visual Aids

**Q92. What approach lighting does Rwy 24L have?**

MALSR, as shown in the lighting box at the top right.

**Q93. What lighting does Rwy 24R have?**

ALSF-2.

**Q94. What are the VGSI and the ILS TCH?**

VGSI angle 3.00° with TCH 79 ft, and the ILS TCH is 59 ft (the glidepath note and the GS 3.00° / TCH 59 box on the profile).

**Q95. What does MALSR stand for?**

Medium-intensity Approach Lighting System with Runway alignment indicator lights.

**Q96. What does PAPI stand for?**

Precision Approach Path Indicator.

**Q97. What do you need to see to continue below the DA?**

The required visual references for the runway environment, such as the approach lights, threshold, touchdown zone or runway markings, as 14 CFR 91.175 requires.

## Section 11 – Calculations

**Q98. What is the total distance from CRCUS to the MAP?**

About 45.8 NM (10.4 + 35.4).

**Q99. How long does it take from PFILA to SUTIE at 200 kt groundspeed?**

29.0 ÷ 200 × 60 = 8.7 minutes.

**Q100. How long does it take from PFILA to the MAP at 160 kt?**

35.4 ÷ 160 × 60 = 13.3 minutes.

**Q101. What gradient do you need to go from 10,000 ft at PFILA to 2,200 ft at SUTIE?**

7,800 ft in 29.0 NM = about 269 ft/NM.

**Q102. What rate of descent is that at 250 kt groundspeed?**

About 1,121 ft/min (269 × 250 ÷ 60).

**Q103. How far above the 3° path is the 10,000 ft altitude at PFILA?**

The extended path is about 11,400 ft at PFILA, so 10,000 ft is about 1,400 ft below it.

**Q104. What is the glide slope altitude at BOUBY?**

About 3,980 ft, which matches the 4,000 ft charted altitude.

**Q105. How far from the threshold is the DA?**

About 0.6 NM before the threshold along the glide slope (200 ft HAT ÷ 318 ft/NM). This is a calculation.

**Q106. How much do you descend from BOUBY to SUTIE?**

1,800 ft (4,000 − 2,200) in 5.7 NM, about 316 ft/NM.

## Section 12 – Plan View & Chart Features

**Q107. What does the thick line with 251° on the plan view show?**

The final approach course extended, with the localizer course 251°.

**Q108. What is the localizer box near the airport?**

It shows the localizer: 111.7, identifier I-HQB, Channel 54. The DME is paired with the localizer.

**Q109. Why does the chart show Hawthorne so close to the final approach course?**

HHR has its own LOC approach to Rwy 25, and the chart authorizes simultaneous approaches with it. The LOC procedure is not authorized during those simultaneous operations.

**Q110. What are the brown numbers and the shading on the map?**

The brown numbers (1000, 3000, 5000, 7000) are terrain contour elevations in feet, and the shading gets darker as the terrain gets higher.

## Section 13 – Communications

**Q111. Who do you talk to on the approach?**

SoCal Approach on the sector frequency (124.5, 128.5 or 124.9), then the Tower, 133.9 for 24L.

**Q112. What do you do with the D-ATIS?**

Copy the arrival ATIS (133.8), check the runway in use, approach in use, weather and NOTAMs, and tell ATC the information letter on initial contact.

**Q113. What is the read back for an approach clearance?**

"Cleared ILS runway 24 left approach, [callsign]." Add any altitude or speed restriction ATC gave you.

**Q114. ATC says "cleared ILS runway 24L approach" and you are not on the localizer yet. What do you do?**

Intercept the localizer, stay at your last assigned altitude until established on a published segment, then follow the charted altitudes.

**Q115. What do you report?**

Only what ATC requests, and any problem such as a missed approach or an inability to comply with a restriction.

## Section 14 – Scenarios & Decision Making

**Q116. ATC vectors you to intercept the localizer between SALWA and PFILA at 10,000 ft. What do you do?**

Do not press APP. Intercept the localizer with the lateral mode your SOP calls for, and fly the charted altitudes down to SUTIE. Capture the glide slope at SUTIE (2,200 ft).

**Q117. ATC gives you a vector that intercepts the localizer inside FNESE. What do you check?**

Your altitude against the charted altitude for that segment (6,000 ft at FNESE, 5,000 at FAYZE), your speed, and that you can still be at 2,200 ft by SUTIE.

**Q118. You are high at JULLI, at 5,500 ft instead of 4,000 ft. What do you do?**

Descend to meet the 4,000 ft altitude. If you cannot reach 2,200 ft at SUTIE, tell ATC and do not try to capture the glide slope from above.

**Q119. You are cleared for the approach and the glide slope flag appears. What do you do?**

Continue as the LOC (GS out) approach, using the MDA 500 ft and the LOC minimums. Tell ATC. Check the HHR note (simultaneous operations).

**Q120. The localizer fails during the approach. What do you do?**

Go missed and tell ATC.

**Q121. You do not see the runway at the DA. What do you do?**

Go missed. Climb to 2,000 ft on heading 251°, then outbound on R-260 to RAFFS.

**Q122. ATC assigns a speed above 270 kt at CRCUS. What do you do?**

Cross CRCUS at 270 kt or less, and tell ATC if you cannot comply.

**Q123. You use the PAPI for the last part of the approach. What is different from the ILS?**

The PAPI is set for a threshold crossing height of 79 ft versus the ILS TCH of 59 ft (Note 4). You cross the threshold higher on the PAPI.

**Q124. Weather is below the ILS minimums. What do you do?**

Do not start the approach below the required visibility (RVR). Hold, divert or go to the alternate, per your operations specifications.

**Q125. How do you brief this approach?**

- Approach name, runway and chart date.
- Frequencies, localizer 111.7 IHQB and final course 251°.
- Entry: CRCUS or SEAVU to PFILA, or radar vectors.
- Altitudes at each fix and the glide slope intercept at SUTIE (2,200 ft).
- DA 323 ft, minimums and lighting.
- Missed approach: 2,000 ft, 251°, R-260 to RAFFS and hold.
- The notes at the top: equipment, simultaneous approaches with HHR and the inop-lighting changes.
- Mode usage per SOP, including when to arm APP.

**Q126. What is your best defense against a mistake on this approach?**

Brief the altitudes, stay disciplined with them until SUTIE, use APP per your SOP and not before, cross-check with the DME, and use both pilots to monitor.

## Section 15 – Rapid-Fire Review

**Q127. What is the localizer frequency?**

111.7.

**Q128. What is the final approach course?**

251°.

**Q129. Where do you intercept the glide slope?**

SUTIE, D8.3, at 2,200 ft.

**Q130. What is the DA?**

323 ft (200 ft HAT).

**Q131. What is the LOC MDA?**

500 ft (377 ft).

**Q132. What is the TDZE for 24L?**

123 ft.

**Q133. What is the airport elevation?**

128 ft.

**Q134. Which fix is the IF?**

JULLI.

**Q135. Which fixes are IAFs?**

CRCUS, SEAVU and FNESE.

**Q136. What is the altitude at PFILA?**

10,000 ft.

**Q137. What is the altitude at SALWA?**

9,000 ft.

**Q138. What is the altitude at HURLR?**

7,000 ft.

**Q139. What is the altitude at FAYZE?**
5,000 ft.

**Q140. What is the MAP for the LOC procedure?**
D1.9 IHQB.

**Q141. What is the missed approach fix?**
RAFFS, D15.1 LAX.

**Q142. What is the glide slope angle and TCH?**
3.00° and 59 ft.

**Q143. Can you press APP between SALWA and PFILA?**
No.

**Q144. What is the max speed at CRCUS and SEAVU?**
270 kt.

**Q145. How long from SUTIE to the MAP at 140 kt?**
2:45.

# Chapter 13

## Scenario Question Bank (WWYD)

WWYD means What Would You Do. In a scenario question the interviewer wants to see your thinking more than the final answer: your priorities, your use of the crew and the resources, and a safe decision.

The questions are grouped by type. Each group has one model way to answer. Scenarios already covered in Chapter 10 are not repeated.

### Systems & Powerplant Failures

**Use for engine, generator, hydraulic, flight control, landing gear, fuel or any other system failure.**

**Q1. WWYD: A generator fails in cruise.**
**Q2. WWYD: You lose two hydraulic systems.**
**Q3. WWYD: You have a flight control degradation in the middle of the flight.**
**Q4. WWYD: You have an elevator jam while in IMC at cruise.**
**Q5. WWYD: You get a landing gear indication failure during the approach.**
**Q6. WWYD: You see a fuel leak from the wing during cruise.**

**How to answer. Use this order:**

1. Aviate first: fly the airplane, keep control, use the autopilot if it helps, and do not change anything until you are stable.

2. Identify and confirm the failure with the other pilot. Do the memory items only if they are required, then use the checklist (QRH).

3. Assess the situation: fuel, weather, terrain, aircraft performance and the nearest suitable airport. Decide whether to continue or divert.

4. Communicate: tell ATC (declare an emergency if needed), call dispatch and maintenance, brief the cabin crew and the passengers.

5. Plan the approach and landing: configuration, landing distance, runway, emergency equipment. Brief it, fly it, and be ready to go around.

6. After landing: stop where it is safe, decide whether an evacuation is needed, report, and debrief.

**Key points:** Say the order: Aviate, Navigate, Communicate. Use the crew, the checklist and the company. Show calm, a plan and a conservative decision.

### Fire, Smoke & Cabin Altitude

**Use for cargo fire, smoke, or any loss of cabin pressure.**

**Q7. WWYD: You get a cargo fire warning at cruise.**

**Q8. WWYD: You have smoke in the cabin after takeoff.**
**Q9. WWYD: You lose cabin pressure during the takeoff climb.**

**How to answer. Use this order:**

1. Protect yourselves first: oxygen masks and smoke goggles if needed, and establish communication between the pilots.

2. Fly the airplane: keep control, then start the memory items and the checklist. For smoke or fire, plan to land as soon as possible.

3. For a cabin altitude problem that cannot be controlled: start an emergency descent according to your procedures, taking terrain and the minimum safe altitude into account, and inform ATC.

4. Tell the cabin crew and the passengers, declare an emergency and choose the nearest suitable airport.

5. Prepare for an evacuation if the fire or smoke is confirmed, and brief the cabin crew.

**Key points:** Masks and communication first, then control, checklist and nearest suitable airport. Show that you do not wait with a fire.

## Incapacitation, Medical & Communication Failures

**Use for a pilot who becomes unresponsive, a medical emergency, or the loss of all communication.**

**Q10. WWYD: The captain becomes unresponsive on a transcontinental flight.**
**Q11. WWYD: There is a medical emergency in flight over the ocean.**
**Q12. WWYD: You lose all communication while in IMC.**

**How to answer. Use this order:**

1. Incapacitated pilot: say 'I have control', engage the autopilot, make sure the other pilot is secured and out of the controls, and call the cabin crew for help (a medical professional on board, if possible).

2. Declare an emergency, tell ATC that you are a single-pilot operation, and divert to the nearest suitable airport. Ask for medical services on arrival and brief the approach carefully.

3. Medical emergency over the ocean: collect the facts from the cabin crew, ask dispatch or the company medical service for advice, and decide using the time to the nearest suitable airport, fuel and weather.

4. Loss of communication in IMC: squawk 7600, try the other radios and frequencies, try the emergency frequency, and continue as required by the lost communication rules (route, altitude and time to leave a holding fix), then shoot the approach.

5. Keep the crew informed and prepare the cabin crew and the passengers.

**Key points:** Control, safe aircraft, help, divert. For lost communication, show you know the rules and the squawk.

## Approach, Landing & Terrain

**Use for difficult approaches, wet or short runways, birds, TCAS and runway excursions.**

**Q13. WWYD: You are arriving at a high-elevation airport with terrain, in IMC, down to minimums. How do you brief the arrival?**

**Q14. WWYD: You lose the visual references on a short runway approach and the runway is now wet.**

**Q15. WWYD: You are landing with a 200 ft and 1/2 mile minimum on a wet runway with a quartering tailwind.**

**Q16. WWYD: You get a TCAS RA at 800 ft AGL.**

**Q17. WWYD: You have a runway excursion caused by hydroplaning.**

**Q18. WWYD: You have a bird strike on short final.**

**How to answer. Use this order:**

1. Brief it well before the descent: the threats, the minimums, the runway condition, the missed approach and the divert plan.

2. Make the decision to continue only if the approach is stabilized and the landing distance is safe for the actual conditions (wet runway, tailwind or crosswind, and braking).

3. If the references are lost, or the approach is not stable, go around. A go-around is always a good decision.

4. TCAS RA: follow the RA, tell ATC and fly the profile according to your company procedure. Bird strike: land if the aircraft is flyable, check the controls and the engines, and tell ATC.

5. After a runway excursion or hydroplaning: stop, shut down as needed, decide on an evacuation, and report.

**Key points:** Prepare early, stay stabilized, and go around when in doubt. Show that you know the numbers for a wet runway and a tailwind.

## Fuel, Weather & Alternates

**Use for fuel, weather below forecast, no alternate, and diversion decisions.**

**Q19. WWYD: On a flight from LGA to DFW you are holding over LIT with an expected further clearance in one hour, and your alternate is LBB.**

**Q20. WWYD: The destination forecast was VFR, it is now 700 ft and 1 mile, no alternate was filed and you have 45 minutes of reserve.**

**Q21. WWYD: The fuel burn is higher than expected and there is no en route alternate.**

**Q22. WWYD: You see a discrepancy between the dispatch weather and the observed METAR.**

**Q23. WWYD: The runway you need is closed when you arrive.**

**Q24. WWYD: You have to divert to an airport with no ground support.**

**Q25. WWYD: You are holding for a gate at an outstation that has only one approach light.**

**How to answer. Use this order:**

1. Get the facts: current weather, trend, fuel on board, fuel burn, the alternates and the time to decide.
2. Compare them with the required fuel (alternate plus final reserve) and the regulations. A legal plan is the minimum, a safe plan keeps options.
3. Decide early. Divert before the fuel forces you. Choose an airport with good weather, enough runway, and the services you need.
4. Coordinate with dispatch and ATC. Use the correct fuel words: 'minimum fuel' when you cannot accept delay, and an emergency declaration if the fuel gets below the final reserve.
5. Tell the crew and passengers, and document the event afterward.

**Key points:** Early, conservative decision with numbers. Never 'continue and hope'. Show you know the difference between minimum fuel and an emergency.

## Duty, Rest, Maintenance & Performance Pressure

**Use when the company, the schedule or a technical item puts pressure on safety.**

**Q26. WWYD: The preflight shows a problem and fixing it would make the crew time out.**
**Q27. WWYD: A de-icing delay pushes the crew near the duty limit.**
**Q28. WWYD: Operations wants you to go with less than the minimum rest.**
**Q29. WWYD: You are asked to depart with an MEL item that you are not sure about.**
**Q30. WWYD: The performance numbers do not match before takeoff.**

**How to answer. Use this order:**

1. Stop and slow down. Say it out loud: 'Let us make sure we do this right.'
2. Get the facts: the regulation or limit, the MEL or the performance data, and the real status of the airplane and the crew.
3. Use the right resources: dispatch, maintenance, crew scheduling, the chief pilot and the manuals.
4. Choose safety over schedule. If the fatigue, the performance or the MEL is not clear, do not go until it is clear.
5. Document the decision and make a safety report so the company can learn.

**Key points:** Safety is not negotiable. Calm, factual, supported by regulations and the manuals.

## Crew Interaction & Authority

**Use when the problem is a person: a new First Officer, a jumpseater, or a crew that is not comfortable.**

**Q31. WWYD: A new First Officer omits a checklist item.**
**Q32. WWYD: A jumpseater challenges your decision.**
**Q33. WWYD: A crew member refuses to fly with a certain captain.**
**Q34. WWYD: The First Officer is new and overly cautious, causing a delay.**

**How to answer. Use this order:**

1. Listen first. Ask the person to explain the concern or the action.
2. Separate the safety issue from the personal one. A real safety concern is always handled right away.
3. Coach in a respectful way: say what you saw, why it matters, and agree on what will happen next.
4. Make the decision as Captain when needed, explain the reason and thank the person for speaking up.
5. Escalate to the chief pilot or the proper channel if the problem continues, and debrief after the flight.

**Key points:** Respect, clarity and authority without ego. Show that you keep the door open to input.

## ATC, Company Pressure & Unfamiliar Operations

**Use for ATC instructions you cannot accept, terrain alerts, or a company request for something you are not comfortable with.**

**Q35. WWYD: The company wants a visual approach at an airport that you do not know.**

**Q36. WWYD: ATC vectors cause a terrain proximity alert.**

**How to answer. Use this order:**

1. Remember that the Captain is responsible for the safety of the flight. You can say 'Unable' to ATC and ask for another option.
2. For a terrain warning, respond as trained at once (it is not the time for discussion), then tell ATC and recover.
3. For a request in an unfamiliar environment, review the charts and the weather, brief the threats and use all the available guidance (ILS or RNAV) as a backup.
4. Accept the request only if it is safe. Otherwise, explain the reason and offer an alternative.

**Key points:** Clear, calm and respectful communication. 'Unable' is a professional word.

# Chapter 14

## HR & Culture Question Bank

These questions test your motivation, your values and your fit with the company. Answer in your own words and with real examples.

Two questions are not repeated here because they already appear in Key Words and in the book: 'Tell me about yourself' and 'Why this airline?'.

### Motivation & Career Vision

**Use for 'why do you fly', 'what excites you' and 'what sets you apart'.**

**Q1. What motivates you to fly?**
**Q2. If you were not a pilot, what would you be doing?**
**Q3. What excites you about joining us now?**
**Q4. What do you bring that sets you apart?**
**Q5. What legacy do you want to leave in the company?**
**Q6. How do you manage work-life balance?**

**Model answer. Make it yours (60 to 90 seconds).**

I fly because I enjoy the combination of responsibility, teamwork and precision that this job requires. Over my career I have learned to stay calm under pressure and to bring the crew together, and I want to use these skills in an operation where I can grow and help others grow. What sets me apart is a habit of preparation and of learning from every flight. In the long term I see myself as a Captain who mentors the next generation of pilots.

**Key points:** Be personal and specific. Link your story to the company and use real examples. 60 to 90 seconds.

### Values, Professionalism & Culture

**Use for 'what is professionalism', 'what do others say about you' and 'how do you contribute'.**

**Q7. What do you value in a flight department?**
**Q8. How do you define professionalism?**
**Q9. How would you contribute to our culture?**
**Q10. What does inclusion mean to you?**
**Q11. What would others say is your biggest strength?**
**Q12. What does good CRM look like?**
**Q13. What does good customer service look like from a pilot?**
**Q14. How do you deal with stress?**

**Model answer. Make it yours (60 to 90 seconds).**

For me professionalism means being prepared, following procedures, treating everyone with respect, and taking responsibility for my decisions, even when no one is watching. People who fly with me say that I am calm, prepared and easy to work with. I would contribute to the culture by speaking up when something is not right, supporting my crew, and keeping a positive attitude, especially on difficult days. Good CRM is open communication, shared workload and a crew where everyone can speak up.

**Key points:** Define the word, then give a short real example. Avoid generic phrases and do not say 'I am a perfectionist'.

## Industry Awareness

**Use for questions about the airline industry, the future and the business.**

**Q15. Where do you see the airline industry in the next 5 to 10 years?**
**Q16. How can a pilot help the company to be more profitable?**
**Q17. How do you stay current with industry trends?**

**Model answer. Make it yours (60 to 90 seconds).**

I follow the industry through professional publications and the news, and I see three trends: the pilot shortage and the importance of training, the move to more efficient and sustainable aircraft, and more use of technology and data for safety. A pilot can help the company's profitability by flying efficiently (fuel, on-time performance), taking care of the equipment and the customers, and following the procedures that prevent costly events.

**Key points:** Show real knowledge and a positive view. Connect it to what the pilot can influence.

# Chapter 15

## Sample Technical Questions & the 13 Briefing Items

Some interviews include a short technical and briefing part. These are sample questions that pilots reported, with model answers. Always replace the general numbers with the ones from your own aircraft and your own company procedures.

### The 13 Briefing Items

A common structure for a departure briefing. Start with the threats and keep recognizing new threats as they come up during the flight.

| # | Item | What to cover |
|---|---|---|
| 1 | Threats | Everything that can hurt us today: weather, terrain, NOTAMs, MEL items, traffic, fatigue and anything unusual. Start here, and keep naming new threats during the flight. |
| 2 | Fuel plan | Block fuel, alternate, reserves and extra fuel. Decide when we would divert. |
| 3 | Maintenance status | Open items, MEL or CDL, and how they affect the flight. |
| 4 | NOTAMs | Closed runways or taxiways, navaid outages, airspace and restrictions that matter today. |
| 5 | Taxi plan / hot spots | The route, runway crossings, hold-short lines and the hot spots. Say who is heads-down. |
| 6 | Rejected takeoff criteria | The speed thresholds, what we reject for, who calls it and who flies the abort. Use your SOP. |
| 7 | Evacuation duties | Who decides, who gives the command, who does the checklist, and the cabin crew plan. |
| 8 | Departure clearance | Route, SID, initial altitude, frequency and transponder code. Confirm the read-back. |
| 9 | Departure SID review | Track, restrictions, top altitude, climb gradient and obstacles. |
| 10 | Weather / windshear | Weather on departure, en route and at the destination. Windshear precautions. |
| 11 | Takeoff performance | Flaps, thrust, V-speeds, runway length, weight and contamination. |
| 12 | Engine failure procedures | What we do after an engine failure: who flies, the profile, the acceleration altitude, the route and where we return. |
| 13 | Terrain / obstacles | The MSA, the high terrain and the obstacles on the departure path. |

## Sample Questions and Model Answers

**Q1. WWYD: You are a new Captain with a new First Officer just off IOE. Give me your briefing (13 items, threats first).**

Start with a short welcome and set the tone: 'If you see something, say it. I would rather hear it twice.' Then brief the 13 items in order, with threats first (see the list above). During the flight, keep naming the threats as they show up. Close with 'any questions or concerns?'. The goal is to show organization, leadership and an open crew.

**Q2. A flight is planned from a hub to a destination with a VFR forecast and no alternate required. What do you check?**

Check that the forecast really meets the no-alternate rule. Under Part 91 (14 CFR 91.169) it is the '1-2-3' rule: from one hour before to one hour after the ETA, a ceiling at least 2,000 ft above the airport elevation and visibility at least 3 SM. Part 121 uses 14 CFR 121.619, which has similar criteria. Even when the rule allows it, check the trend, the fuel and a backup plan.

**Q3. What is your abort criteria in your current aircraft?**

Answer with your own aircraft's SOP. A typical structure is: below a low-speed threshold (about 80 kt), reject for any malfunction or warning; between that speed and V1, reject only for serious problems (engine failure or fire, loss of control, or something that makes the aircraft unsafe or unable to fly); at V1, continue. Say who makes the call and who flies the abort.

**Q4. What does a number with a line above it mean on a SID? (For example 10000 with a line above.)**

A line above the altitude means 'at or above' that altitude. A line below means 'at or below', and a line above and below means 'exactly at' the altitude.

**Q5. What is the top altitude on a SID?**

It is the highest altitude published in the SID, where the procedure ends vertically. You normally climb to it unless ATC gives something different or another altitude is assigned in the clearance.

**Q6. The yaw damper fails at FL360. What are you thinking?**

Run the QRH and check the MEL and the aircraft limitations. At high altitude the yaw damper protects against Dutch roll, so the aircraft can be less stable. The procedure may require a lower altitude or a speed limit, depending on the type. Use the autopilot as the aircraft allows, avoid abrupt rudder inputs, tell ATC if you need a lower altitude and inform maintenance.

**Q7. You pull up an ILS approach. What are your weather minimums for your approach category (for example Category C)?**

Find your approach category by the aircraft's approach speed (Category C is a reference speed of 121 to 140 kt). Then read the DA(H) and the visibility or RVR in the minimums box for that category and for the equipment that is working. Check the inoperative components table (if the approach lights are out, the minimums go up) and any landing limits for the weather and the runway condition.

**Q8. If you are outside the FAF, can you begin the approach if the weather is below minimums? What if you are inside the FAF?**

The answer depends on the operation. Under Part 121 (14 CFR 121.651) you cannot continue the approach past the final approach fix if the latest reported weather is below minimums. If you are already inside the FAF when the report drops below minimums, you can continue to the DA or MDA. At the DA or MDA you can land only if you have the required visual references (14 CFR 91.175). Check your company's operations specifications.

**Q9. Where is the FAF on this plate?**

On a precision approach (ILS) the FAF is where the glide slope is intercepted, shown with a lightning bolt on the profile (for example SUTIE on the Los Angeles ILS 24L chart in this book). On a non-precision approach the FAF is shown with a Maltese cross.

**Q10. What is your decision height?**

On an ILS the DA(H) is the minimum shown on the plate, as an altitude (MSL) and a height above the touchdown zone in parentheses. At the DA you must go around if you do not have the required visual references.

**Q11. You are on the runway and ATC tells you to hold because the destination is closed. You have 1,000 lb of extra fuel. What is your bingo fuel?**

Bingo fuel is the fuel needed to fly to the alternate and land with the final reserve. Calculate it from the distance, the altitude, the winds and the alternate weather, and add a contingency. When the fuel reaches that number, you leave for the alternate. With only 1,000 lb of extra fuel the options are limited, so decide early and talk to dispatch and ATC.

**Q12. How can you find a diversion airport?**

Dispatch (the best source), ACARS, ATC, the FMS nearest-airport page, the charts and the airport data, and the company information. Check the weather, the runway length and condition, the approaches, the services (fuel, handling, emergency services) and the NOTAMs.

**Q13. Dispatch gives you an airport with company services where the weather is close to minimums but other aircraft are getting in. Another airport is VFR, 10 minutes away, but has no company services. Where do you go?**

The safest choice is normally the VFR airport. Weather close to minimums with no margin is a threat, and the lack of company services is an inconvenience that can be handled (handling company, passengers and rebooking). Check the fuel, the runway and the emergency services. As Captain, make the decision, tell dispatch, and brief the crew.

**Q14. You landed safely at the diversion airport. How do you debrief?**

Do a short, structured debrief with the crew: what went well, what could be improved, the threats and the errors, the decisions and the communication. Ask the First Officer for his view first, thank the crew, and decide whether a safety report is needed. End with a positive and specific point.

# Chapter 16

## Cold Temperature Altitude Correction

In very cold weather your altimeter reads higher than your real height. This is a small error in mild cold and a serious one when the temperature is far below standard, because the real distance to terrain or obstacles is shorter than the altimeter shows. This chapter explains the problem, shows the table and how to make the correction step by step, with a worked example on a real approach.

### Why It Matters

- The altimeter is calibrated for standard atmosphere (ISA). In air colder than ISA the pressure levels are closer together, so the real altitude is lower than the indicated altitude.
- The error grows with the cold and with the height above the airport. At -30°C, 2,000 ft above the airport you are about 380 ft lower than you think.
- The risk is highest where the margin is smallest: step-downs on a non-precision approach, the minimums and the missed approach in mountain terrain.
- The old rule: "from high to low, from hot to cold, look out below" (when going from a high to a low pressure or from warm to cold air, the real altitude is lower than indicated).

### When to Apply It

The rules depend on the country and on your operator. Always follow your own company procedures and the current charts. In general:

- United States: on airports with a cold temperature restriction, the chart shows a snowflake symbol with a temperature (for example, -2°C). When the reported temperature is at or below that value, you must correct the altitudes that the chart lists.
- Other countries: many operators require corrections at any airport when the temperature is 0°C or colder, because there is no list of airports.
- Some aircraft have an FMS function that applies the correction. Know how your system works and what it does not correct.

### What to Correct and What Not to Correct

| Correct | Do not correct |
|---|---|
| Altitudes of the approach segments you expect to fly: initial, intermediate, final approach fix and step-downs. | Altitudes on SIDs, STARs and obstacle departure procedures. |
| Minimums: DA(H) or MDA(H). | Altitudes assigned by ATC, or radar vectors, unless ATC clears the correction. |
| The missed approach altitude and the holding altitude. | The glide slope: it is a beam and is not affected by temperature. Use the corrected altitude as a check at the final approach fix. |

| Correct | Do not correct |
|---|---|
| The segments of the approach you anticipate flying, even if you are not there yet. | Anything when the temperature is above the charted or required temperature. |

## The Cold Temperature Correction Table

The table below is the ICAO table of the error to add to the charted altitude. It gives the correction in feet by temperature and by height above the airport. It is the standard ICAO table, also published in FAA material. Use the table in your own approved documents for real operations.

Use the height above the airport, not the altitude. For heights above 5,000 ft use the 5,000 ft column (no extrapolation).

| Temp \ Height (ft) | 200 | 300 | 400 | 500 | 600 | 700 | 800 | 900 | 1000 | 1500 | 2000 | 3000 | 4000 | 5000 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0°C** | 20 | 20 | 30 | 30 | 40 | 40 | 50 | 50 | 60 | 90 | 120 | 170 | 230 | 280 |
| **-10°C** | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 | 150 | 200 | 290 | 390 | 490 |
| **-20°C** | 30 | 50 | 60 | 70 | 90 | 100 | 120 | 130 | 140 | 210 | 280 | 420 | 570 | 710 |
| **-30°C** | 40 | 60 | 80 | 100 | 120 | 140 | 150 | 170 | 190 | 280 | 380 | 570 | 760 | 950 |
| **-40°C** | 50 | 80 | 100 | 120 | 150 | 170 | 190 | 220 | 240 | 360 | 480 | 720 | 970 | 1210 |
| **-50°C** | 60 | 90 | 120 | 150 | 180 | 210 | 240 | 270 | 300 | 450 | 590 | 890 | 1190 | 1500 |

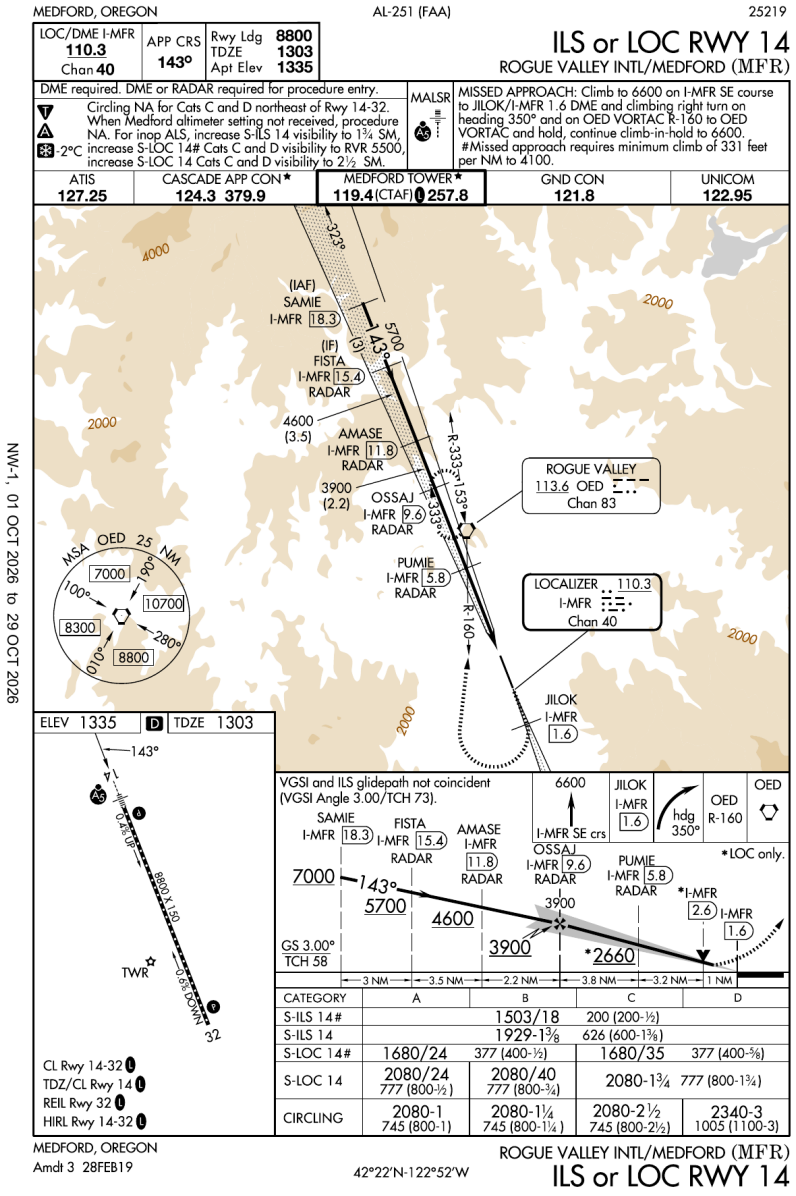
## The Method in Six Steps

1. Find out if a correction is required: the charted temperature, or your operator's rule for the airport.
2. Take the reported temperature (ATIS or the best source you have).
3. For each altitude to correct, subtract the airport elevation. This is the height above the airport.
4. Enter the table with the temperature (row) and the height (column). Interpolate between columns if needed.
5. Add the correction to the charted altitude. Round as your operator requires (usually up on the final segment).
6. Fly the corrected altitude, and tell ATC about the corrections when required.

## Worked Example: KMFR, ILS or LOC Rwy 14

Medford (Rogue Valley Intl, KMFR) is a cold temperature restricted airport: the chart shows the snowflake symbol with -2°C. The airport elevation is 1,335 ft and the TDZE is 1,303 ft. The ATIS reports -10°C, which is colder than -2°C, so the corrections are required. The crew expects radar vectors and the full approach, so it corrects the whole approach and the missed approach. The chart is on the next page.

# Rogue Valley Intl/Medford (KMFR) – ILS or LOC Rwy 14




*FAA chart (public domain), cycle 01 OCT to 29 OCT 2026. For study only; not for navigation.*

## Corrections for the Approach (Reported Temperature -10°C)

Heights are above the airport (elevation 1,335 ft). Corrections are from the table above and interpolated between columns. Rounding is only an example: follow your operator's rule.

| Point | Charted (ft) | Height above airport (ft) | Correction (ft) | Corrected (ft) | How to use it |
|---|---|---|---|---|---|
| SAMIE (IAF) | 7,000 | 5,665 | +490 | 7,490 → 7,500 | Altitude before SAMIE (above 5,000 ft, use the 5,000 ft column) |
| SAMIE to FISTA | 5,700 | 4,365 | +427 | 6,127 → 6,100 | Intermediate segment, nearest hundred |
| FISTA to AMASE | 4,600 | 3,265 | +317 | 4,917 → 4,900 | Intermediate segment, nearest hundred |
| OSSAJ (FAF) | 3,900 | 2,565 | +251 | 4,151 → 4,200 | Final approach fix, round up |
| DA (ILS) | 1,503 | 168 | +20 | 1,523 | Use the 200 ft column (table minimum) |
| Missed approach | 6,600 | 5,265 | +490 | 7,090 → 7,100 | Holding altitude. Above 5,000 ft, use the 5,000 ft column |
| PUMIE (LOC only) | 2,660 | 1,325 | +133 | 2,793 → 2,800 | Only if flying the LOC (GS out) |
| MDA (LOC only) | 1,680 | 345 | +35 | 1,715 | Only if flying the LOC (GS out) |

**Result: with -10°C, the final approach fix is flown at 4,200 ft instead of 3,900 ft, the DA is about 1,523 ft, and the missed approach is flown at 7,100 ft. Tell ATC the altitudes you require.**

## In Short

- Check whether a correction is required: the snowflake and temperature on the chart, or your operator’s rule.
- Use the reported temperature and the height above the airport (the charted altitude minus the airport elevation).
- Read the correction in the table and add it to the charted altitude. Round as your operator requires.
- Correct the approach segments you expect to fly, the minimums and the missed approach. Do not correct SIDs, STARs or altitudes assigned by ATC. The glide slope does not change with temperature.
- Tell ATC the altitudes you require, and have the other pilot check the numbers before the descent.

## Final Note

The questions in this guide represent real interview content from aviation companies around the world. Preparation, honesty, and professionalism are the keys to success. Good luck — you've got this.

Alcance Aviation

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# Your interview starts here.

Every pilot interview follows a pattern — and the pilots who prepare for that pattern are the ones who get the job. This guide condenses real questions from airline, cargo, fractional, and corporate interviews into sixteen focused chapters covering HR, behavioral, technical, and scenario-based content, plus key words and chart practice.

Whether you're walking into your first regional interview or your captain upgrade, use this book as your flight plan.

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*“The difference between a good pilot and a hired pilot is preparation. This book is that preparation.”*

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Pilot Career Preparation